Cystic Kidney Diseases

45

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CHAPTER OUTLINE

KEY POINTS

- Autosomal dominant polycystic kidney disease (ADPKD) is the most common hereditary kidney disease, characterized by progressive cyst development, renal enlargement, and eventual kidney function decline. The disease manifests with significant variability, influenced by genetic variants, environmental factors, and modifiers.
- Tolvaptan, a vasopressin V2 receptor antagonist, is the first U.S Food and Drug Administration–and EMA-approved treatment for ADPKD. It has been shown to slow disease progression by reducing cyst growth and preserving kidney function, particularly in patients at risk for rapid progression.
- ADPKD is predominantly caused by variants in *PKD1* and *PKD2*, but emerging data highlight the role of minor genes such as *DNAJB11*, *GANAB*, and *IFT140*. Advances in genetic testing have improved diagnostic accuracy, clarified disease variability, and enabled familial cascade screening.
- Autosomal Recessive Polycystic Kidney Disease (ARPKD), a rare but severe disorder, presents in infancy or early childhood with renal and hepatic involvement. Advances in understanding *PKHD1* variants and associated modifiers have expanded diagnostic capabilities and informed management strategies.
- The spectrum of cystic kidney diseases includes conditions such as autosomal dominant tubulointerstitial kidney disease (ADTKD), medullary sponge kidney, and acquired cystic kidney disease (ACKD). These conditions underscore the need for personalized diagnostic and therapeutic approaches.
- Breakthroughs in genetics have illuminated the pathogenesis of a wide array of cystic kidney diseases, enabling earlier diagnosis and the development of targeted treatments. Genetic insights now drive precision medicine and improve patient and family outcomes across these diseases.

Cystic kidney diseases include a group of disorders characterized by the formation of fluid-filled cysts within the kidneys. The burden of these diseases, marked by kidney dysfunction and associated complications, and their impact on patients' quality of life are significant. These diseases can be inherited or acquired and can affect a broad range of age groups. They include many types, each presenting unique nuances in diagnosis and management. Understanding the factors affecting severity and associated extrarenal manifestations is important for appropriate management.

PATHOPHYSIOLOGY OF KIDNEY CYST FORMATION

Nephrogenesis starts in the fifth week of in utero development, with reciprocal inductive signaling between the ureteric bud, an outgrowth of the mesonephric duct, and the metanephric blastema. This signaling is crucial for kidney formation and displays temporal and spatial coordination.[1] Simultaneously, the metanephric mesenchyme undergoes condensation and segregates into smaller forms, the S-shaped bodies, leading to the development of the renal epithelium.[1-3] Subsequently, blood vessels infiltrate the proximal part of the S-shaped bodies, forming vascularized and filtering glomeruli.[1] The initiation of glomerular filtration occurs around the 9th to 12th week of development. During this early stage, the kidney has a relatively low concentrating ability, and arginine vasopressin (AVP) plays a crucial role in regulating water balance. The role of AVP in fetal kidney water homeostasis is well documented, although the precise mechanisms are still not fully elucidated.[4] In human fetal kidneys, aquaporin-1 (AQP1) and aquaporin-2 (AQP2) channels are expressed starting from the 12th week of gestation. Notably, the distribution of both channels during embryogenesis is distinct. AQP1 is initially detected in the cortex and the descending thin loop of Henle, whereas AQP2 is restricted to ureteric bud structures.[3] This selective differentiation allows the proximal convoluted tubules (PCT) and thin descending limbs to become permeable to water early in development, while within the distal collecting duct, AQP2 expression co-occurs with the secretion of AVP by the neurohypophysis.[3,5]

Cyst formation is a complex process influenced by various molecular and cellular mechanisms during kidney development.[6,7] As the condensation of mesenchyme around the ureteric bud begins, the acquisition of polarity in the developing epithelial tubules becomes the second important step after mesenchyme condensation.[1,2] This includes the establishment of distinct apical, basal, and lateral borders, which are characteristic of epithelial tubules.[8] Proper cell-cell adhesion follows, mediated by several factors, including the cell adhesion molecule E-cadherin. In addition, abnormalities in the positioning of membrane proteins, such as the apical instead of basolateral placement of Na/K ATPase, can further lead to cyst formation.[1] Many of the genes involved in cyst development are expressed during embryogenesis and localize to the primary cilium-centrosome complex.[9] Alterations to the actin-integrin interactions impact the proliferation of cyst-binding epithelium. As such, the loss of β1 integrin, laminin, and *COL4A1* has been associated with altered in utero cyst development.[10,11]

The primary cilium of tubular epithelium also plays a crucial role in cyst formation. Renal epithelial cells converge via *Wnt9b*-regulated convergent extension, ensuring strict control of the division plane orientation, which must be parallel to dividing cells.[12] *PKHD1* and *TCF2/HNF1b* transcription regulators of ciliary protein genes are key factors in this process.[13] Ciliopathies can disrupt the centrosome position and the orientation of the mitotic spindle, contributing to cystic diseases.[6] The Notch1/2 signaling pathway maintains the alignment of a division plane perpendicular to the basement membrane in the renal proximal tubules during morphogenesis.[6] Disruption of the Notch signaling results in a partially randomized spindle orientation relative to the basement membrane and increased mitosis, leading to the stratification of some cells within the cysts.[6,14]

Exposure to hypoxia and subsequent reperfusion injury are conditions known to contribute to cystogenesis later in life by affecting kidney angiogenesis, apoptosis, and cell proliferation. This process is mainly controlled by hypoxia-inducible factor-1 alpha (HIF-1α) and vascular endothelial growth factor (VEGF).[15,16] Additionally, studies have shown that the germ-free environment and low M2 macrophages can influence cyst fluid dynamics. Depletion of macrophages using liposomal clodronate in a mouse model of cystic kidney disease has shown a reduction in cystic index and improved kidney function, suggesting that macrophages may promote cyst progression in cystic kidney disease.[17] Notably, alternatively activated M2-like macrophages expressing CD206 have been linked to kidney cyst formation.[18]

SIMPLE KIDNEY CYSTS

EPIDEMIOLOGY AND ETIOLOGY OF SIMPLE CYSTS

Simple kidney cysts are common, particularly among the elderly, and represent the most common kidney lesions.[19] Importantly, they are not neoplastic and remain asymptomatic in most cases, with only about 8% of individuals experiencing symptoms that necessitate medical intervention. In children, the incidence of kidney cysts is underestimated, ranging between 0.22% and 0.55%, but this rises with age,[20] increasing to more than 10% in patients aged at least 50 and more than 30% in patients above 70. Furthermore, simple renal cysts can exhibit significant growth, potentially doubling in size over a decade.[21,22]

Simple kidney cysts are acquired and not inherited.[23] Several nongenetic risk factors have been associated with the development of simple kidney cysts including age, male sex, hypertension, and renal insufficiency.[24,25] These simple cysts are most frequently identified incidentally during radiologic studies.[26,27]

PATHOPHYSIOLOGY OF SIMPLE CYSTS

Simple kidney cysts manifest as fluid-filled pockets in the renal cortex, typically starting unilaterally, but may become bilateral with age or declining kidney function. Simple renal cysts are typically filled with serous fluid and have a normal epithelial lining, which may consist of a single layer of cuboidal or flattened epithelial cells. Some cysts may exhibit signs of epithelial atrophy.[28,29]

CLINICAL MANIFESTATIONS OF SIMPLE CYSTS

Typically, sporadic kidney cysts are asymptomatic. However, when symptoms do manifest, individuals may experience flank pain, the presence of a palpable mass, hypertension (HTN), hematuria, intracystic hemorrhage, infection, fever, hydronephrosis, or gastrointestinal symptoms. Acquired cystic disease is observed in patients with advanced chronic kidney disease (CKD), particularly patients on dialysis.[24,30]

DIAGNOSIS OF SIMPLE CYSTS

Simple renal cysts maybe diagnosed by ultrasound (US), computed tomography (CT), or magnetic resonance imaging (MRI).[31] The evaluation of complex renal cysts usually requires

Stage	Cyst Wall	Septa	Calcified	Enhanced?	Work Up
1	Hairline Thin Wall	No	No	No	No F/U Needed
2	Minimal Regular Wall Thickening	Thin, Smooth, Few Septa	Minimal	No	No F/U Needed
2F	Minimal Regular Wall Thickening	Multiple Minimal Smooth Thickening	Thick Nodular	No	F/U at 6 month, 12 month, then annually for 5 years
3	Thick Irregular on Smooth Wall	Thick and Irregular	Thick Nodular and Irregular	Yes	Partial Nephrectomy or Radioablation in poor surgical candidates
4	Irregular Gross Thickening	Irregular Gross Thickening	Thick Nodular and Irregular	Yes (Tissue and Cyst)	Partial or Total Nephrectomy

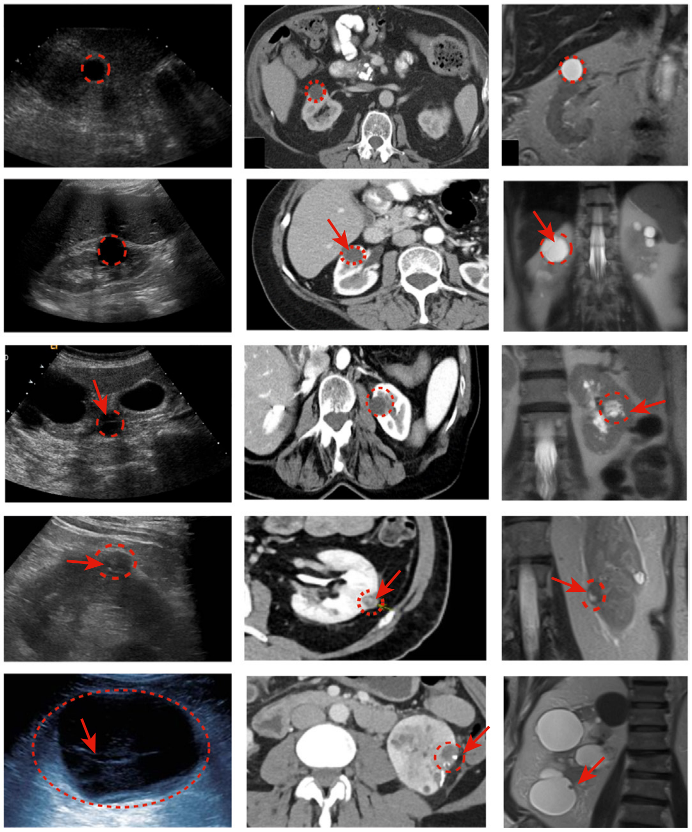


Fig. 45.1 Characteristic findings on various abdominal imaging modalities and the criteria for the Bosniak Classification. *CECT,* Contrast-enhanced computed tomography scan; *F/U,* follow-up.

CT scans or MRIs.[32,33] Scans before and after intravenous contrast administration are needed as contrast enhancement is a feature of some malignancies.[31] CT scans may be cheaper but involve radiation exposure and the risk of kidney injury (AKI) due to iodinated contrast administration in presence of advanced CKD or other risk factors. While costly, MRIs do not involve radiation and may be able to characterize the cyst without the need for gadolinium contrast. The newer gadolinium preparations are safe in CKD with no reports of nephrogenic systemic fibrosis.[31,34] On US, simple cysts appear round and anechoic with posterior enhancement. By CT, imaging appears consistent with water content with a Hounsfield unit of 0 to 20.[35] On MRI, cysts show similar signal characteristics to water, displaying uniform hyperintensity on T2-weighted images.[31,36] A simple cyst can be confidently diagnosed on an unenhanced CT scan when it exhibits simple fluid characteristics, features a thin wall, and lacks septations, central or peripheral calcifications. In contrast, cysts with features of wall thickening, septum partitioning, and calcification are named “complex cysts.”[31] The Bosniak classification system (Fig. 45.1) is designed to assess the risk of cystic renal masses and is instrumental to guide therapy. It is primarily based on contrast-enhanced CT findings and categorizes cysts into four classes based on morphology and enhancement characteristics, with simple cysts falling into the Bosniak 1 and 2 classes. A Bosniak 2F requires follow-up, and classes III and IV have an increasing likelihood of cancer. This differentiation between simple and complex cysts (Table 45.1) is vital to ensure accurate clinical assessment and management.[31,36] For further discussion on kidney imaging, see Chapter 24.

MANAGEMENT OF SIMPLE CYSTS

Management of kidney cysts depends on whether they are asymptomatic or symptomatic. For asymptomatic cysts, a conservative approach is typically employed. When symptomatic, various cyst interventions may be considered.[37] These include percutaneous aspiration with the use of a sclerosing agent, such as ethanol or sodium tetradecyl (Sotradecol). Laparoscopic decortication, whether conventional or laparoendoscopic single-site surgery (LESS), percutaneous ablation, and retrograde endoscopic marsupialization are other available choices.[38,39]

PRENATAL DIAGNOSIS OF KIDNEY CYSTS

US serves as the primary diagnostic tool for detecting, categorizing, and tracking prenatal kidney cystic diseases. It helps identify renal pathology and also plays a significant role in assessing amniotic fluid volume and detecting any associated anomalies.[40] Cystic kidney diseases can initially manifest as hyperechogenic (bright) kidneys without visible cysts. Therefore a comprehensive evaluation should collectively

Table 45.1 Differences Between Simple and Complex Cysts

	Simple Cyst	Complex Cyst
Characteristics	Most common renal lesion containing water-like fluid	Less common renal mass with thicker walls and solid-like material
Presentation	Typically, asymptomatic. Dull abdominal or back pain and urinary changes in case of enlargement	Typically, asymptomatic. May present with fever, pain, and hematuria
Imaging	Incidental finding. Oval anechoic mass	Poorly demarcated thick cyst walls with septations and internal echoes reflecting solid-like material inside.

Table 45.2 Comparison Between Autosomal Dominant Polycystic Kidney Disease (ADPKD), Multiple Simple Cysts, Unilateral Renal Cystic Disease (URCD), Multicystic Dysplastic Kidneys (MCDK), and Multiple Unilateral Subcapsular Hemorrhagic (MUCH)

Disease		Acquired vs. Inherited	Prevalence	Clinical Features
ADPKD	Typical (MIC 1A-1E)	Autosomal dominant	1/1000-1/2500	Bilateral cyst involvement with extrarenal manifestations. Leads to several complications including cyst hemorrhage, hypertension, and cerebral aneurysms
	Atypical (MIC 2A-2B)			Focal disease (MIC-2A) or patterns of parenchymal atrophy (MIC-2B). Milder form of the disease with a later onset of symptoms, slower cyst growth, and lower risk of complications
Multiple Simple Cysts		Acquired	Common	Usually unilateral but may increase and show up independently in both kidneys with age. Typically, asymptomatic.
Unilateral Renal Cystic Disease (URCD)		Acquired	Unknown	Looks like PKD but manifests unilaterally. Does not typically progress to ESKF and lacks extrarenal manifestations and complications.
Multicystic Dysplastic Kidney disease (MCDK)		Acquired	1/1000-1/4300	Group of cysts of different sizes. May lead to complications such as vesicoureteral reflux.
Multiple Unilateral Subcapsular Hemorrhagic (MUCH) Cysts		Acquired	Unknown	Multiple unilateral small kidney cysts displaying hemorrhagic contents (hyper-attenuated on unenhanced CT or extremely hypointense on T2-weighted magnetic resonance image

ESKF, End-stage kidney failure; *MIC,* Mayo imaging classification.

consider the differential diagnosis of these conditions.[41] Imaging studies could be complemented by genetic testing, which can identify monogenic diseases in children with two or more renal cysts and/or increased cortical echogenicity.[40,42] In cases where fetuses exhibit extrarenal anomalies, chromosomal rearrangements or aberrations are more common.[43] If there are indications of hyperechoic kidneys, whether with or without cysts and an absence of amniotic fluid, fetal MRI can offer a more in-depth characterization.[40] This highlights the importance of considering the broader clinical context and genetic factors when making prenatal diagnoses and treatment decisions related to renal cystic diseases, which may be required immediately after birth.[40]

LOCALIZED/UNILATERAL RENAL CYSTIC DISEASE

Unilateral renal cystic disease (URCD) is a relatively rare, typically nonfamilial and nonprogressive, kidney disease. URCD manifests as a cluster of simple cysts that are asymmetrically distributed within the affected kidney, diagnosed by abdominal imaging.[44,45] It is important to differentiate URCD from other conditions such as autosomal dominant polycystic kidney disease (ADPKD) or multilocular cystic renal neoplasms to ensure an accurate diagnosis and appropriate treatment.[44] Unlike ADPKD, which may show extrarenal manifestations and familial patterns, URCD is confined to the kidneys and does not follow a hereditary pattern. This distinction is particularly important in pediatric cases, where ADPKD might initially appear asymmetric.[44,45] The clinical course of URCD is generally benign, contrasting with the progressive worsening seen in ADPKD. URCD also differs from multicystic dysplastic kidney disease (MCDK), which is marked by extensive dysplasia and potential renal dysfunction or proteinuria (for further discussion see Chapter 71). The affected kidney in MCDK can appear small, atrophic, normal-sized, or even enlarged, depending on the degree of cystic involvement and dysplasia.[44,45] Clinical manifestations of URCD include the presence of an abdominal mass, flank pain, and possible gross hematuria.[45] Table 45.2 compares URCD, ADPKD (typical and atypical),

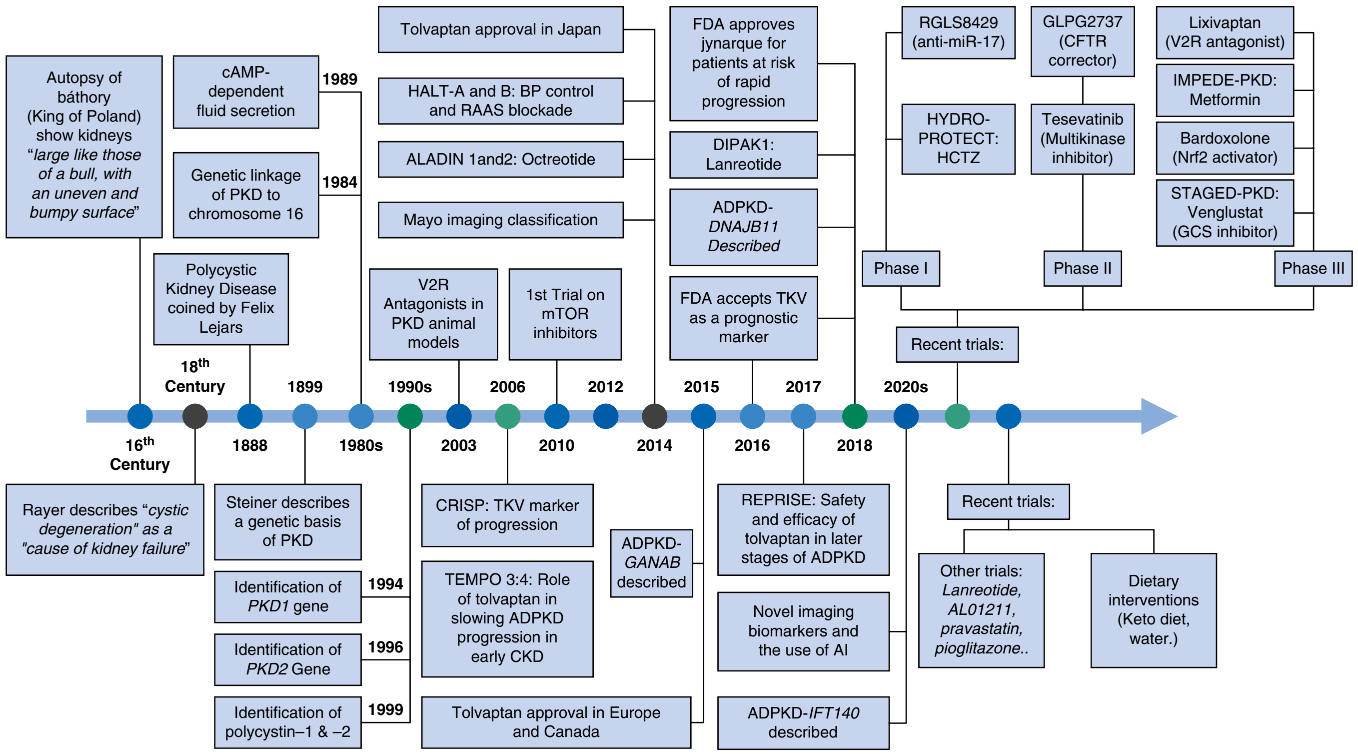


Fig. 45.2 Timeline of the major historical milestones and trials related to autosomal dominant polycystic kidney disease.

multicystic dysplastic kidney disease, and multiple unilateral subcapsular hemorrhagic cystic disease.

HEREDITARY CYSTIC KIDNEY DISEASES

AUTOSOMAL DOMINANT DISEASES

AUTOSOMAL DOMINANT POLYCYSTIC KIDNEY DISEASE

Epidemiology of ADPKD

ADPKD is the most prevalent hereditary kidney disease and the fourth most common cause of end-stage kidney failure (ESKF). This genetic disorder is characterized by the development of multiple kidney cysts, leading to kidney enlargement and structural and functional alterations that progress to ESKF in most patients.[46,47] ADPKD history dates back over 300 years. Fig. 45.2 represents the major milestones in ADPKD history. Both sexes and all racial and ethnic backgrounds are susceptible to this condition. Epidemiologic studies estimate the prevalence of ADPKD to range from 1 in 400 to 1 in 1000 live births with annual incidence of 2.75 per 100,000 per-years, affecting up to 12.5 million people globally.[46,48,49] Data from clinical registries suggest slightly lower prevalence rates due to varying factors such as geographic distribution, population size, epidemiologic measurement methods, and health care system characteristics (Table 45.3),[46,50] which could explain the variability in prevalence.

Prevalence rates for ADPKD have been estimated through worldwide population-based studies with specific studies reporting rates of 1 in 2459 in the United Kingdom, 1 in 1111 in France, and 1 in 542 for the Seychelles.[51-53] In Copenhagen, during the period 1920–1953, the incidence was recorded at 0.8 cases per 100,000 patient-years.[54] In the United States, ADPKD incidence is observed to be higher in males than in females, with rates of 8.2 versus 6.8 cases per million, respectively.[55] ADPKD is the cause of ESKF in about 5% of individuals who need kidney replacement therapy (KRT) each year in the United States, constituting 5% to 10% of all ESKF patients.[46] In most patients with ADPKD, kidney function remains relatively intact in the first 3 decades of life, after which the glomerular filtration rate (GFR) progressively declines, leading to ESKF. Patients reach ESKF at various ages depending on disease severity with approximately 50% of them reaching ESKF by age 62.[56] A French study found that 22% of patients with ADPKD reached ESKF by the age of 50, rising to 42% by the age of 58, and to a substantial 72% by the age of 73.[46,50] Lately, there has been a shift toward later-onset ESKF, potentially due to reduced cardiovascular mortality among older patients before reaching ESKF.[46,50]

Etiology and genetics of ADPKD

ADPKD is an autosomal dominant disorder. Most PKD1 and PKD2 disease-causing gene variants have complete penetrance and almost equal sex distribution. Within the same family, affected members may show varying degrees of severity, which may reflect the influence of environmental factors, epigenetic factors, or modifying genes on disease manifestation.[47,50] ADPKD is mainly caused by pathogenic variants in one of two genes: *PKD1,* on chromosome 16, encodes polycystin-1 (PC1) and is responsible for approximately 78% of ADPKD cases, while

Table 45.3 Table Representing National and International Prevalence of ADPKD

Location	Year	Prevalence
Globally	1957	Point prevalence ranges from 1 in 400 to 1 in 1000 live births, affecting approximately 12.5 million people globally.[49]
	2017	Lifetime prevalence of 9.3 per 10,000, out of which *PKD1* constitutes 6.8 per 10,000 and *PKD2* 2.6 per 10,000. *HNF1B* truncating variant detected in 1 per 28,770 individuals while TSC complex 1 in 2315 individuals. Minimal genetic prevalence 9.3 per 10,000 individuals.[48]
European Union	2012	Screening prevalence 3.96 in 10,000 Minimum point prevalence 3.29 in 10,000 91.1/million on RRT[410]
United States	2002-2018	Overall crude prevalence 4.26 per 10,000. When standardized to age, prevalence decreases to 4.15 per 10,000.[411]
	1935-1980	ADPKD prevalence of 100-250:100,000 inhabitants[412]
	1980-2016	2010 Prevalence: Definite ADPKD: 4.7 per 10,000 Definite, likely, or possible ADPKD: 12.4 per 10,000 2010 incidence: Definite ADPKD: 179 per 100,000 Definite, likely, or possible ADPKD: 9.44 per 100,000[413]
	2013-2015	Commercial and Medicare Database: 2013 Prevalence: 1.74 per 10,000 2014 Prevalence: 1.97 per 10,000 2015 Prevalence: 2.10 per 10,000 Managed Medicaid Database: 2013 prevalence: 2.26 per 10,000 2014 prevalence: 2.40 per 10,000 2015 prevalence: 2.20 per 10,000[414]
	2016-2017	2017 prevalence: 2.34 per 10,000 2-yr prevalence: 3.61 per 10,000[415]
Seychelles	1993-1995	3-yr prevalence 5.7 per 10,000 1 in 544 among the Caucasian population[51]
United Kingdom	1991	Prevalence: 1 in 2459[52]
	1999	Minimum prevalence: 3.9 per and screening prevalence: 4.6[416]
France	1996	Point prevalence rate 1 in 1111[53]
Germany	1999	Minimum prevalence 2.4 per 10,000, screening prevalence 3.3[416]
	2013	overall prevalence of ADPKD in southeast Germany was 3.27/10000: ranging from 1.73/10,000 in the third decade to 5.73/100 000 in the 6th decade of life[417]
Modena, Italy	1980-2017	Point prevalence: 3.63 per 10,000 and predicted prevalence 4.76 per 10,000[418]

15% are attributed to *PKD2,* located on chromosome 4 and encoding polycystin-2 (PC2).[57-59] Around 7% of genetic testing remains inconclusive and may account for rarer new pathogenic variants in few genes, namely *IFT140, DNAJB11, GANAB, ALG5, ALG8,* and *ALG9,* among others.[58,60] Some cohorts are showing a lesser percentage of *PKD1* and *PKD2* variants and an increased percentage of the newly discovered variants.[61,62] PC1 is a complex multidomain membrane protein, resembling a G-protein-coupled receptor (GPCR) (Fig. 45.3). It regulates protein-protein, cell-cell, and cell-matrix interactions and participates in intracellular signaling pathways governing cell proliferation and survival.[63] It is localized at critical sites of cyst formation including renal tubular epithelia, hepatic bile ductules, endothelial cells, and pancreatic ducts. PC2 is a member of the transient receptor potential (TRP) family of calcium-regulated cation channels. It is found in the distal tubules, collecting ducts, and thick ascending limb and is involved in cell calcium signaling.[64,65] PC1 and PC2 are on nonmotile primary cilia, which receive and relay external signals into the cell. Ample evidence underscores their role in inhibiting cystogenesis, with cyst formation occurring when polycystin levels dip below a certain threshold. The interaction between these two polycystins proves to be crucial for PC1 maturation, its trafficking to the primary cilia, and ultimately its stability.[66,67]

Lanktree and colleagues[48] conducted a study in which they examined the frequency of pathogenic variants in large, public whole-genome sequencing (WGS) and whole-exome sequencing (WES) databases. They found the lifetime genetic prevalence of ADPKD to be 9.3 per 10,000. This indicates that the true prevalence of ADPKD is often underestimated as large studies miss asymptomatic cases or variants that do not necessarily end up with a significant disease course.[48] Pathogenic variants in *PKD1* were detected in 6.8/10,000 individuals while those in *PKD2* were found in 2.6/10,000. This results in a *PKD1* to *PKD2* ratio of 2.6.[48] In another study by Chang and colleagues[57] examining WES of the Geisinger population in the United States, it was found

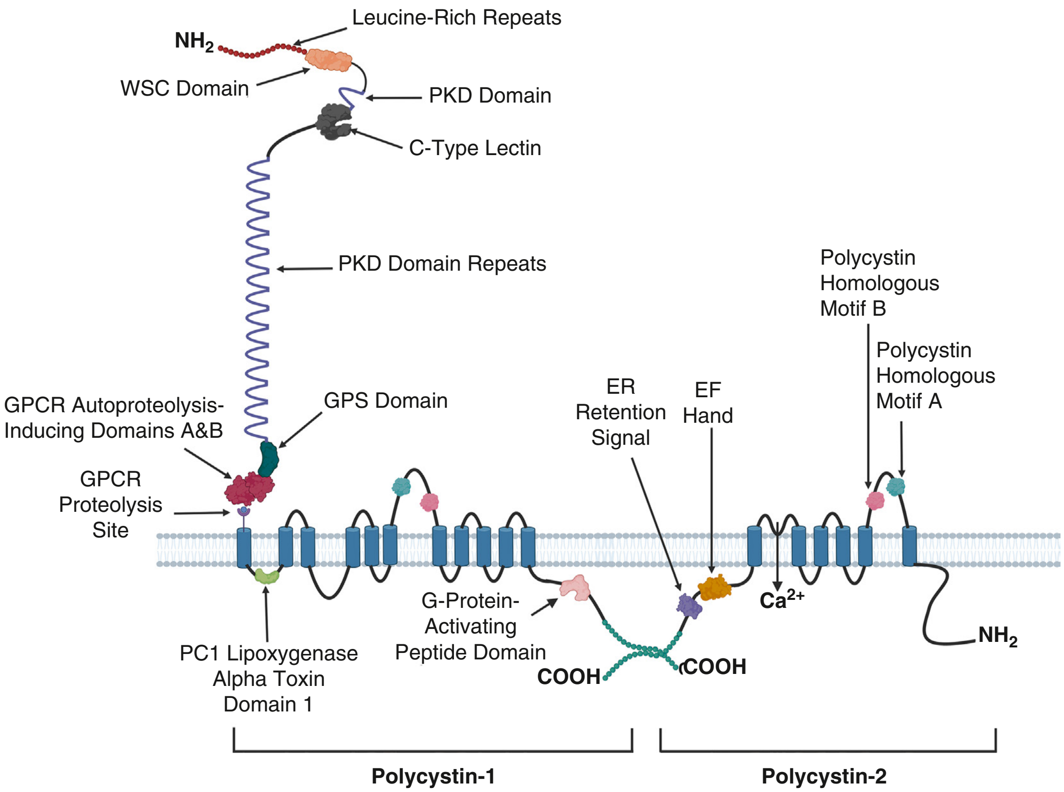


Fig. 45.3 Structure of polycystin proteins. Polycystin 1 (PC1) and polycystin 2 (PC2) form the polycystin complex.

that the loss-of-function variants in *PKD1* and *PKD2* were predominantly associated with ADPKD, with rare variants found in other genes including *IFT140, GANAB, PKHD1, HNF1B, ALG8,* and *ALG9*. To note, the estimated prevalence of ADPKD based on genotype (8.64/1000) was higher than the estimated prevalence based on phenotype (1.35–1.74/1000).[57] Table 45.4 and Fig. 45.4 show the differences between the ADPKD pathogenic variants.

One of the standout features of ADPKD is the remarkable phenotypic variability among individuals. This is due to the affected locus (*PKD1* vs. *PKD2*); nature of the allelic variant (truncating, nontruncating, or hypomorphic); timing of gene inactivation; presence of mosaicism; and an individual's unique genetic background. The ADPKD pathogenic variant is the primary determinant governing the progression rate in each patient. When compared with individuals with *PKD1* variants, those with *PKD2* variants tend to develop fewer cysts and experience a more gradual progression. The median onset age of ESKF is around 54 and 74 years for *PKD1* and *PKD2*, respectively.[46,66] In a rare but more severe presentation, <2% of patients exhibit deletions in both the *PKD1* gene and the tuberous sclerosis complex-2 (*TSC2*) gene, which gives rise to a more severe form of cystic kidney disease, the *TSC2/PKD1* contiguous gene syndrome, that typically manifests in childhood.[68] Protein-truncating variants of *PKD1* and *PKD2* exhibit 100% disease penetrance while missense variants have variable manifestations and incomplete penetrance.[57] Truncating variants in *PKD1* generally yield a more severe disease phenotype compared with nontruncating missense variants. Inheriting two inactivating variants in either *PKD1* or *PKD2* results in different outcomes. When such variants are present in both alleles, it leads to lethality in utero.[69] Family history can help predict whether a *PKD1* or *PKD2* variant is likely. The presence of at least one family member who developed ESKF before the age of 55 years strongly favors a *PKD1* variant, with a PPV of 100% and a sensitivity of 72%. Conversely, if a family member reaches 70 years of age without experiencing ESKF, it is suggestive of a *PKD2* variant, with a PPV of 100% and a sensitivity of 74%.[46,50,70]

Despite the germline defect being present in all cells, cysts form in less than 5% of tubules, with focal cystic dilation within each tubule. These observations, coupled with the discovery of somatic *PKD1* or *PKD2* mutations in kidney and liver cysts of patients with ADPKD, have led to the hypothesis of a recessive cellular mechanism for cystogenesis with inactivation of both copies of either *PKD1* or *PKD2* within a cell necessary for cyst formation.[46] There is considerable clinical overlap between ADPKD and autosomal dominant polycystic liver disease (ADPLD) caused by genes, such as *SEC63* and *PRKCSH*. PC1 localization in the primary cilia is influenced by disruptions in the endoplasmic reticulum (ER) endogenous proteins or the N-glycosylation process.[47] Proper coordination of those events involves a wide array of genes, namely *GANAB, PRKCSH, ALG8,* and *PMM2* for N-linked glycosylation, as well as *SEC63, SEC61A1,* and *SEC61B* for ER translocation.[47,71,72] Genes such as *SEC63, PRKCSH, GANAB, ALG5, ALG8, ALG9,* and *SEC61B,* which encode ER proteins, play a crucial role in the maturation of PC1 to its functional site on the primary cilium's surface membrane (Fig. 45.5).[73] However, the ADPKD genotype does not necessarily predict the growth rate or severity of

Table 45.4 Main Differences Among ADPKD-Spectrum Genes, Expression, and Manifestations

Gene	Disease	Chr.	Year	MOI	Protein	Protein Function	Proportion	Kidney Manifestations	Renal Outcomes	Liver Manifes-tations	ICA Risk
PKD1	ADPKD-*PKD1*	16p13.3	1994	AD	Polycystin-1	Receptor, Complex with PC2. Tubulogenesis (not well understood)	78%	Multiple bilateral cysts. Highest cystic burden among other pathologic variants	ESKF at mean age 58.1	Absent to severe	Increased risk (unclear rate)
PKD2	ADPKD-*PKD2*	4q21	1996	AD	Polycystin-2	Ca2+-permeable nonselective cation channel. Complex with PC1	15%	Multiple bilateral cysts	ESKF at mean age 79.7	Absent to severe	Increased risk (unclear rate)
IFT140	ADPKD-*IFT140*	16p13.3	2021	AD	IFT140 Protein (part of IFT complex A)	Retrograde ciliary transport. Well development and functioning of cilia	1-2%	Few, large bilateral cysts, Enlarged Kidneys	Preserved GFR until old age	Rare	Unclear
GANAB	ADPKD-*GANAB*	11q12.3	2016	AD	α-Subunit of Glucosidase- II	Catalytic subunit of glucosidase II. PC1&PC2 maturation and localization into the cilia and cell surface	>0.5%	Mild cystic burden	Limited CKD, no ESKF	Mild to severe	Unclear
DNAJB11	ADPKD-*DNAJB11*	3q27.3	2018	AD	DNAJ Heat Shock Protein 40 Subfamily B, member 11	Co-chaperone for HSPA5. Maturation and correct trafficking of PKD1	>0.5%	Bilateral, small cysts, mild enlargement	Limited early CKD, ESKF in 70s	Mild	Possible
ALG5	ADPKD-*ALG5*	13q13.3	2022	AD	Dolichyl-phosphate beta-glucosyl-transferase	Assembly of oligosaccharides in kidney epithelial cells. Proper PC1 glycosylation and maturation	<0.5%	Mild-Moderate cystic burden, mild enlargement	CKD, ESKF in older patients	Few, rarely	Unclear
ALG6	ADPKD-*ALG6*	1p31.3	-	AD	α -1,3-glucosyltransferase	Glycosylation of lipid-linked oligosaccharides. Maturation and localization of PC1 into the primary cilia	<0.5%	Mild cystic disease	Generally preserved GFR	Liver cysts can present with PLD	Unclear
ALG8	ADPKD-*ALG8*	11q14.1	2017	AD	α-3-glucosyl-transferase	Glycosylation of lipid-linked oligosaccharides. Maturation and localization of PC1 into the primary cilia	~1%	Mild cystic disease	Preserved GFR into old age	Mild to Severe	Unclear
ALG9	ADPKD-*ALG9*	11q23.1	2019	AD	α-1,2-mannosyl-transferase	Transfer of Mannose into lipid-linked oligosaccharides. Proper PC1 maturation	>0.5%	Mild to Moderate Cystic disease	Significant CKD in older individuals	Common	Unclear
PKHD1	ADPKD-*PKHD1*	6p12.3–p12.2	1997	AD	Fibrocystin	Ciliogenesis, Tubulogenesis. Cell-Cell and Cell-ECM interactions.	~1%	Generally, very mild cystic disease	Preserved function into old age.	Common	Unclear

Chr, Chromosome; *CKD,* chronic kidney disease; *ESKF,* end-stage kidney failure; *ICA,* intracranial aneurysm; *Manifest,* manifestation; *MOI,* mode of inheritance; *Prop,* proportion.

Fig. 45.4 Representative abdominal images of patients with autosomal dominant polycystic kidney disease seen on the large disease spectrum. *GFR,* Glomerular filtration rate; *Ht,* height; *MIC,* Mayo Imaging Classification; *NT1,* nontruncating tier 1; *NT2,* nontruncating tier 2; *T,* truncating; *TKV,* total kidney volume.

polycystic liver disease (PLD)[74]; therefore modifiers and estrogen might affect the phenotype.[74] Somatic mosaicism is another process implicated in ADPKD development when no overt variants are identified. Somatic mosaicism involves the presence of two distinct genetic cell populations within a single individual, stemming from a somatic variant that occurs during embryonic development.[72] The clinical diversity seen in somatic mosaicism is tied to the type of cell affected and the specific stage of development at which the variant arises.[75] The presence of atypical kidney imaging patterns (such as asymmetry, unilaterality, and lopsided pattern) in a patient with de novo PKD should raise suspicion of possible somatic mosaicism.[72,76] Deep intronic variants in *PKD1* and *PKD2* contribute significantly to the diagnostic challenge in ADPKD, accounting for approximately 7% of cases where standard genetic

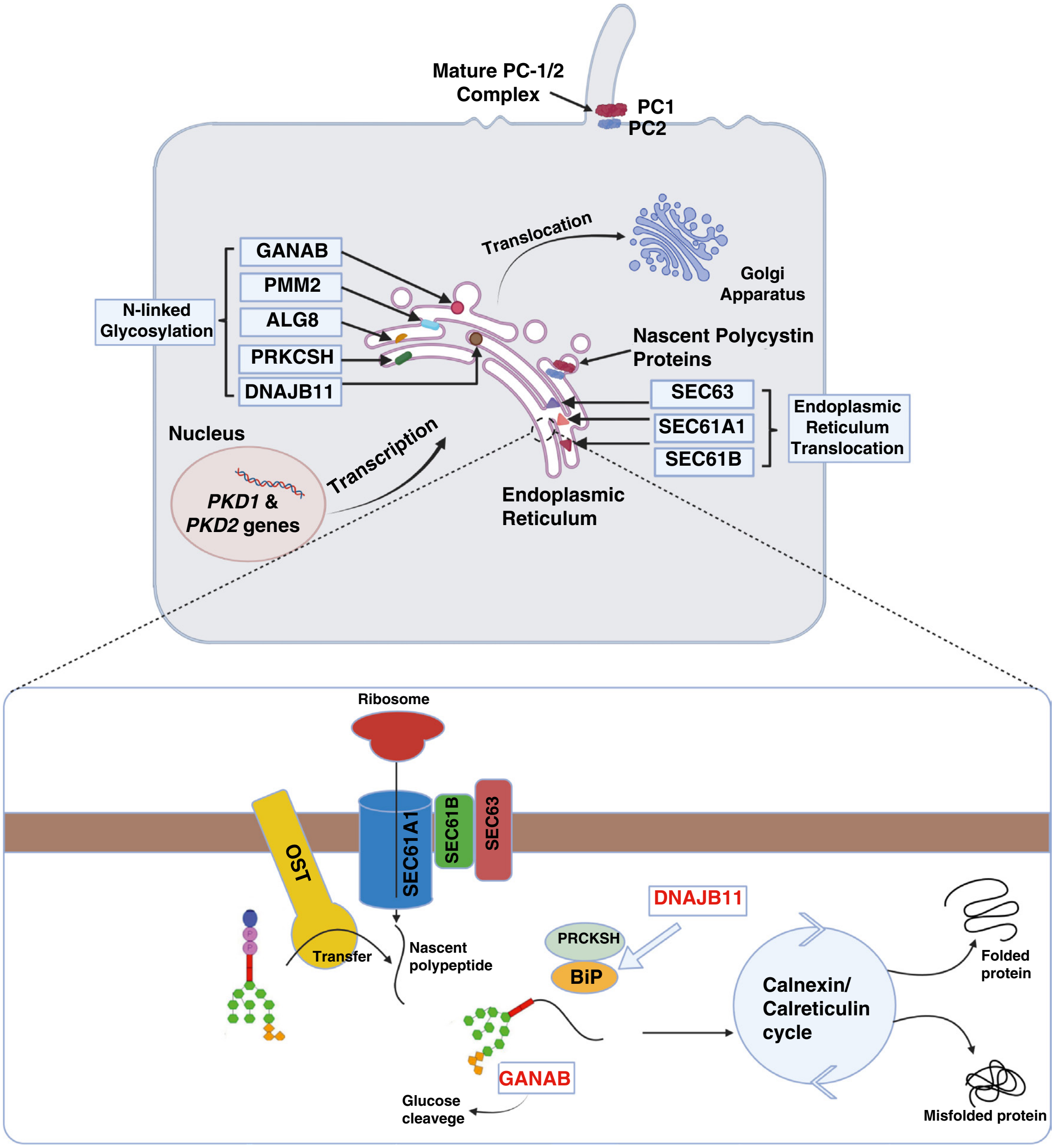


Fig. 45.5 Maturation process in the tubular epithelium. The maturation process of ADPKD-associated proteins, primarily polycystin 1 (PC1) and polycystin 2 (PC2), from gene expression to functional integration within the primary cilium. Initiated within the nucleus, genes encoding PC1 and PC2 undergo transcription, producing mRNA transcripts. Subsequently, these transcripts are transported to the endoplasmic reticulum (ER), where translation takes place, leading to the synthesis of PC1 and PC2 proteins. Crucial proteins in this maturation cascade, including *SEC61* and *SEC63* (involved in ER translocation) and *DNAJB11, ALG8, GANAB, PMM2,* and *PRKCSH* (involved in N-linked glycosylation), play important roles in guiding proper folding, post-translational modifications, and trafficking of PC1 and PC2. Notably, *DNAJB11* emerges as a key facilitator in ensuring the maturation and correct trafficking of *PKD1*. Additionally, the figure underscores the significance of various proteins in the accurate placement of PC1 and PC2 within the ciliary structure, where their functional roles are realized.

testing fails to yield a diagnosis.[77,78] These variants, often affecting gene splicing, necessitate more comprehensive sequencing approaches that include both protein-coding and noncoding regions. One common *PKD1* deep intronic variant, rs3874648G>A, plays a crucial role in altering gene expression by influencing splicing patterns.[79] This variant disrupts normal splicing by activating a cryptic splicing site, ultimately leading to a reduction in full-length *PKD1* mRNA levels. Specifically, the variant affects the binding affinity of a splicing regulator, Tra2-β, thereby affecting *PKD1* expression.[79]

Achieving a definitive genetic diagnosis of ADPKD is needed to inform clinical care and facilitate cascade testing within affected families. However, the classification of variants of uncertain significance (VUS) remains a challenge, particularly in cases where variants are unique to individual families.[77] Despite these challenges, expanding diagnostic approaches to include analysis of noncoding regions and advanced sequencing methodologies offer promising avenues for improving diagnostic rates in ADPKD.[77,80] For further discussion on genetics in kidney diseases, see Chapter 44.

Pathophysiology of ADPKD

During the past few years, there have been significant advancements in our understanding of the pathogenesis of ADPKD. However, the precise roles of polycystins and the molecular mechanisms governing the development of the disease remain areas of ongoing investigation. Polycystins are believed to play a crucial role in regulating intracellular calcium signaling. Initially, it was thought that the polycystin complex localized in the primary cilium transduced shear stress generated by extracellular fluid flow into a cellular calcium signal.[81] However, subsequent studies have indicated that while polycystin proteins can induce localized changes in ciliary calcium concentration, they do not substantially alter global cytoplasmic calcium levels.[82] Polycystin proteins are also functional at other subcellular locations as PC1 may repress signaling pathways that regulate the interaction between the extracellular matrix and integrins.[46,50,81,83] The precise mechanism underlying cystogenesis remains unclear. In the early stages, cysts originate as dilatations within intact tubules. However, as cysts grow by the secretion of fluid into the cysts, accompanied by hyperplasia of the cyst epithelium, they lose their connection to functional nephrons.[81] Cystogenesis occurs when functional polycystin levels drop below a certain threshold level, rather than requiring complete loss of function.[46,50,81,84]

Variants in *PKD1* or *PKD2* lead to reduced intracellular calcium levels, resulting in increased cAMP, and activation of protein kinase A (PKA). The pathogenesis of ADPKD is critically influenced by cyclic AMP (cAMP), which drives cyst initiation and expansion by stimulating cystic epithelial cell proliferation, fluid secretion, and downstream signaling pathways that impair tubulogenesis, promote interstitial inflammation, and enhance collecting duct principal cell sensitivity to AVP (Fig. 45.6).[85] In fact, vasopressin V2 receptor antagonists were shown to increase intracellular calcium and decrease cystogenesis.[85,86] One prominent hypothesis suggests that decreased calcium in PKD affects both the synthesis and hydrolysis of cAMP. Synthesis is enhanced through the activation of calcium-inhibitable Adenylyl cyclase 6 (AC6), while cAMP hydrolysis is diminished due to the inhibition of calcium-calmodulin-dependent phosphodiesterase 1 (PDE1).[87,88] PDE3 might also play a role in regulating cAMP signaling, as it is localized in the ER membrane and degrades pools of cAMP that control cell proliferation and chloride secretion mediated by the CFTR transporter.[88] Consequently, abnormal epithelial chloride secretion occurs via the cAMP-dependent transporter encoded by the CFTR gene, significantly contributing to the generation and maintenance of fluid-filled cysts in ADPKD. The upregulation of PKA is crucial in CFTR-mediated chloride-driven fluid secretion. The calcium-dependent chloride channel anoctamin-1 (ANO1), also known as TMEM16A, which is notably present in ADPKD kidneys, is also implicated in fluid secretion into cysts as inhibiting this channel halts cystogenesis.[89] The activation of PKA could result in a maladaptive response or “self-perpetuation of disruption” that further disrupts calcium signaling. With the progression of PKD, the molecular alterations in the kidneys increasingly result from secondary events that disrupt the network of signaling pathways. This sustained disruptive response eventually leads to IP3R and PC2 hyperphosphorylation, depletion of calcium stores, and the expression of the PKD phenotype.[90,91] Decreased Ca^{2+} entry, increased cAMP levels, and the irregular activation of RAS–RAF–ERK pathways within renal epithelial cells contribute to cyst growth. This process is amplified by enhanced signaling from growth factors. For example, the higher levels of the epidermal growth factor receptor and its mislocalization on the apical surface of cystic epithelial cells, coupled with the existence of ligands such as epidermal growth factor and transforming growth factor–α (TGF-α) in cystic fluid, are possible drivers of tubular epithelial cell proliferation.[50,85]

Another hypothesis sheds light on the role of decreased PC2-mediated calcium entry, which activates AC5 and/or AC6 while inhibiting PDE4C. This effect is secondary to a dysfunction in ciliary proteins within the A-kinase anchoring protein complex.[92,93] However, this theory has been challenged as studies have shown that primary cilia are largely impermeant to calcium.[94] Regardless of what is initiating calcium entry, polycystin loss or dysfunction has been consistently linked to disrupted calcium signaling.[66,95] A final hypothesis related to cAMP signaling involves STIM1 oligomerization and translocation to the membrane due to the depletion of ER calcium stores and subsequently AC6 activation.[92,96]

The dysregulation of purinergic signaling may also lead to altered cAMP signaling. Under normal conditions, renal tubular epithelial cells exhibit constitutive ATP release and spontaneous calcium variations. This is the result of ligand-gated P2X receptors and G-protein-coupled P2Y receptors that interact with ATP, tightly controlling intracellular calcium levels. In ADPKD cells, there is a significant reduction in flow-sensitive ATP release, disturbing intracellular calcium regulation. This reduction in function may be linked to decreased mRNA levels of the aforementioned receptors.[97,98] Cyst expansion is accompanied by changes in cellular metabolism including increased glucose consumption and impaired fatty acid oxidation. Anomalous activation of the mammalian (mechanistic) target of rapamycin (mTOR) pathway has been associated with the upregulation of aerobic glycolysis, increase in ATP levels, and further inhibition of 5’-AMP-activated protein kinase. Furthermore, cysts can induce mechanical stress on nearby nephrons, leading to apoptosis of renal epithelial cells and local cyst formation.[85,99,100]

Downstream factors also play a role. The Wnt-β-Catenin pathway becomes activated downstream of PKA and may be intensified in ADPKD due to the inhibition of glycogen synthase kinase-3β and stabilization of β-catenin. This leads to the phosphorylation of PKA. Consequently, the upregulated PKA enhances the production of several transcription factors including cAMP-response element-binding protein, paired box protein Pax-2, and STAT3.[85,99] Both innate and adaptive immunity have been implicated in the pathogenesis of ADPKD. Monocyte chemoattractant protein 1 (MCP1), also known as CCL2, accumulates in cystic kidneys promoting cyst growth.[85]

On the macroscopic level, ADPKD is characterized by a gradual enlargement of kidney cysts. In early stages, the kidney typically contains relatively few cysts, and most of the parenchyma remains intact.[101,102] The cyst epithelium secretes a significant quantity of chemokines and cytokines, which in turn trigger an inflammatory response in the surrounding tissue. In advanced stages, marked kidney enlargement is

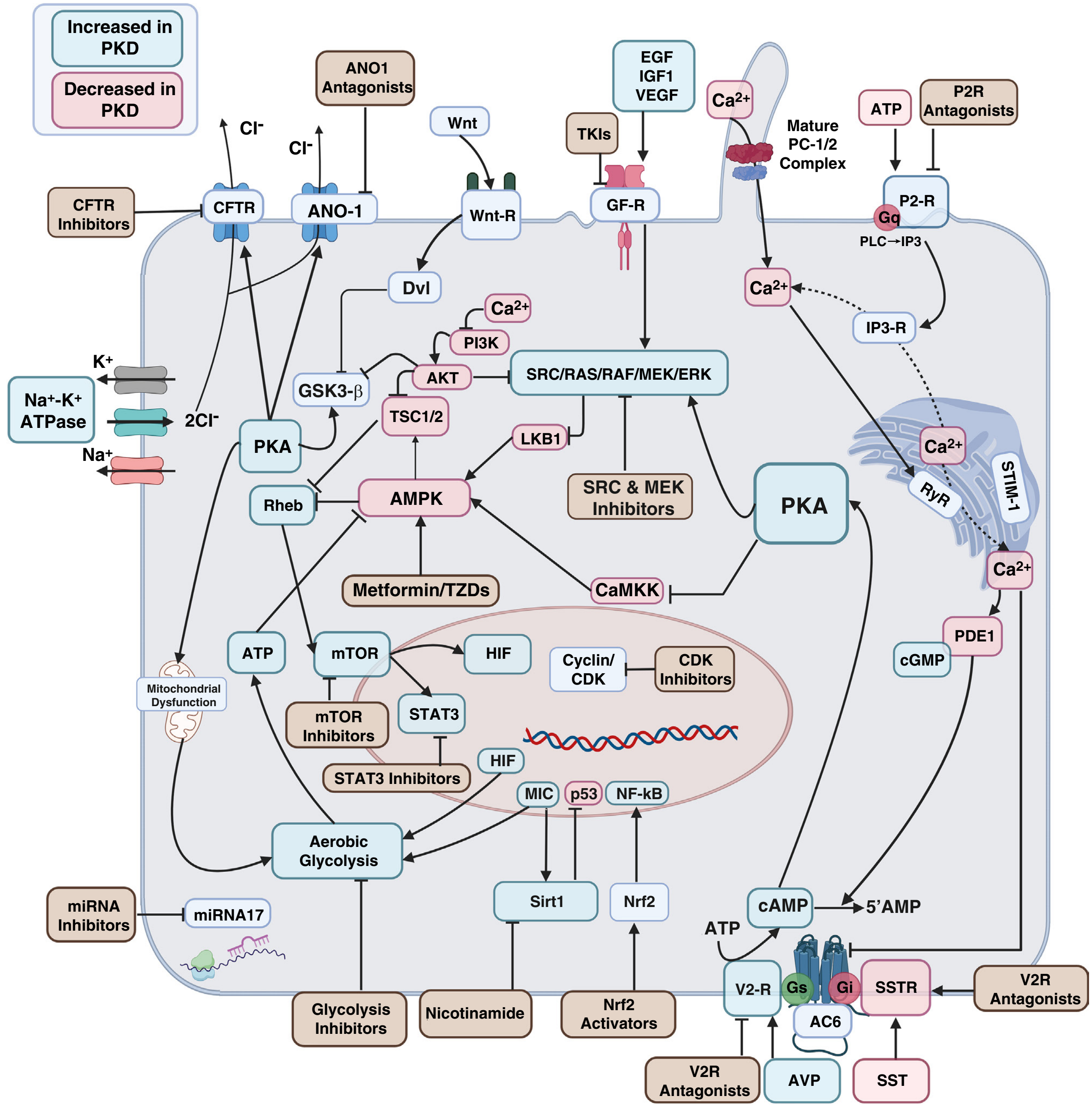


Fig. 45.6 Hypothetical pathways upregulated (green) or downregulated (red) in PKD and possible treatment targets. The molecular cascade driving the progression of autosomal dominant polycystic kidney disease (ADPKD). Initiated by *PKD* variants, aberrant crosstalk between calcium (Ca) and cyclic AMP (cAMP) signaling pathways disrupts cellular homeostasis. This disruption enhances cAMP and protein kinase A (PKA) signaling through the activation of Ca-inhibitable adenylyl cyclases and inhibition of calcium-dependent phosphodiesterase 9. The ensuing PKA-induced effects include the perturbation of intracellular calcium balance, phosphorylation of endoplasmic reticulum calcium cycling proteins, and the activation of CFTR, leading to Cl and water secretion, thus enhancing fluid secretion. PKA activation exhibits contrasting effects on cell proliferation in wild-type and PKD cells. Calcium modulation, through deprivation or delivery, reverses these effects, proposing a mechanism involving PI3K inhibition and the regulation of transcription factors governing cell progression and energy metabolism. Disease progression in ADPKD is further fueled by factors such as mislocalized EGFR, overexpressed growth factors, cytokines, chemokines, and their receptors. Abnormal miRNA expression is linked to disease advancement, with miR-17 controlling various target genes in proliferative signaling pathways including PPARA and mTOR and directly repressing *PKD1, PKD2,* and *HNF-1β* transcripts, thereby contributing to reduced gene expression and cyst formation. This comprehensive illustration provides insights into the interconnected molecular events shaping the complex pathogenesis of ADPKD. *AC6,* Adenylate cyclase 6; *AMPK,* AMP-activated kinase; *ATP,* adenosine triphosphate; *AVP,* vasopressin; *CaMKK,* calcium/calmodulin-dependent protein kinase; *CDK,* cyclin-dependent kinase; *EGF,* epidermal growth factor; *ERK,* extracellular signal-regulated kinase; *Gi,* inhibitory G protein; *Gq,* a G-protein subunit; *Gs,* stimulatory G protein; *GSK3β,* glycogen synthase kinase 3β; *HIF,* hypoxia-inducible factor; *IGF1,* insulin growth factor 1; *IP3R,* inositol 1,4,5 triphosphate (IP3) receptor; *LKB1,* liver kinase B1; *MEK,* mitogen-activated protein kinase kinase; *mTOR,* mammalian target of rapamycin; *PC1,* polycystin-1; *PC2,* polycystin-2; *P2R,* purinergic 2 receptor; *PLC,* phospholipase C; *Rheb,* Ras Homolog enriched in brain; *RSK,* ribosomal s6 kinase; *RYR,* ryanodine receptor; *Sirt1,* sirtuin 1; *SOC,* store operated channel; *STAT3,* signal transducer and activator of transcription 3; *STIM1,* stromal interaction molecule 1; *SSTR,* somatostatin receptor; *TKIs,* tyrosine kinase inhibitors; *TSC,* tuberous sclerosis proteins tuberin (TSC2) and hamartin (TSC1); *TZDs,* thiazolidinediones; *V2R,* vasopressin V2 receptor.

evident, accompanied by notable vascular remodeling and interstitial fibrosis.[46] Cystic epithelial cells are characterized by increased proliferation, abnormal fluid secretion, and excessive deposition of extracellular matrix. These alterations are closely linked to changes in the blood and lymphatic microvasculature around the cysts. Most cysts are eventually independent of the originating nephron. The expansion of these cysts also exerts pressure on nearby tissue and vasculature. Obstructed nephrons progress toward forming atubular glomeruli, and proximal tubule cells undergo apoptosis.[46,50,85]

Diagnosis of ADPKD

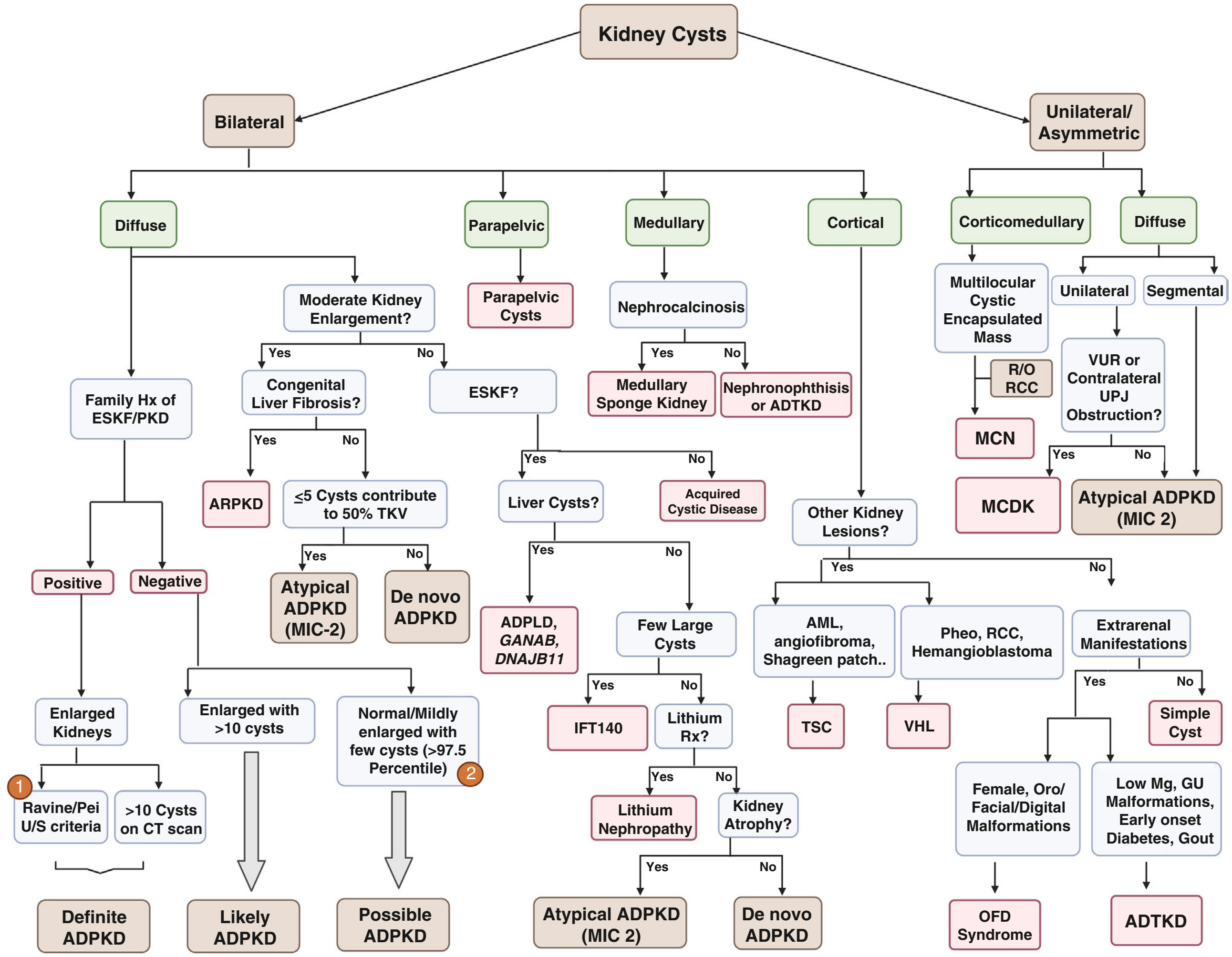
ADPKD is diagnosed clinically when both kidneys are enlarged and uniformly distributed cysts are noted on imaging. An individual's medical history, clinical presentation, physical examination, and, in some cases, genetic studies can provide valuable insights. Additional factors that complement imaging studies include a family history of the disease, the age of presentation, and the number of kidney cysts.[46] Therefore ADPKD should be considered in patients with a relevant medical history, such as a history of hypertension, multiorgan involvement, recurrent urinary infections, gross hematuria, kidney stones, or a family history of ADPKD or kidney failure. Physical examination findings may include palpation of masses in the upper to midabdomen or elevated blood pressure (BP). Extrarenal manifestations such as a mitral valve prolapse, murmur on auscultation, or evidence of hepatomegaly can also raise suspicion for ADPKD. Laboratory tests, primarily involving a renal profile, and urinalysis are also helpful.[46,81] Fig. 45.7 shows a diagnostic algorithm for cystic kidney diseases. The diagnostic approach can be categorized on the basis of the patient's presentation:

1. *In asymptomatic patients with normal kidney function and a family history of ADPKD:*
   US is the imaging method of choice for presymptomatic individuals due to its accessibility and cost-effectiveness. A diagnosis of ADPKD can be established if at-risk individuals aged 15 to 39 years have three or more kidney cysts and those aged 40 to 59 years have two or more cysts in each kidney (Table 45.5).[81] If a definite diagnosis is not achieved, CT or MRI studies become valuable given their sensitivity of detecting smaller cysts. In individuals older than 40 with no cysts detected by US, the diagnosis of ADPKD can be excluded. For those younger than 40, MRI is superior to US for exclusion, as the detection of fewer than five cysts by MRI is sufficient to rule out ADPKD. MRI or CT with intravenous contrast is recommended when masses or complex cysts are incidentally discovered.[103,104] Asymptomatic patients can be clinically monitored by assessing BP and kidney function. Once diagnosis is confirmed, a baseline contrast-enhanced CT (CCT) or MRI, if not obtained yet, is recommended to assess the risk of rapid progression through measuring total kidney volume (TKV).[105,106] For individuals younger than 18, current recommendations advise against screening due to the absence of disease-modifying agents for this age group. However, it is recommended that pediatricians be informed of the family history, monitor BP regularly, and promote a healthy lifestyle within the household.
2. *In patients with clinical findings suggestive of ADPKD and a positive family history of ADPKD:*
   When ADPKD is highly suspected in an individual at risk of ADPKD, A CCT or MRI is preferred over an initial US (Fig. 45.8). These advanced imaging methods confirm the diagnosis and also offer a baseline for future reference, help identify extrarenal complications of ADPKD, and enable the calculation of htTKV for prognosis and treatment planning.[105] The choice between CCT and MRI depends on the patient's kidney function, considering the risk of contrast exposure with CT. Patients with preserved kidney function (estimated glomerular filtration rate [eGFR] ≥60 mL/min/1.73 m^2) may undergo a CCT for TKV calculation, to distinguish between cystic and noncystic tissue and assess cyst burden. Noncontrast CT helps assess stones in the collecting system. For patients with an eGFR <60 mL/min/1.73 m^2, MRI is the preferred choice due to lack of iodinated contrast and radiation exposure. MRI is effective in distinguishing between cystic and noncystic tissue but may not reliably detect intrarenal factors like kidney stones or parenchymal calcifications.[46,50,107]

Utilizing genetic testing in ADPKD

The application of genetic testing becomes essential when the diagnosis is inconclusive (e.g., lack of family history or atypical imaging findings) or when the goal is to rule out the disease at early stages and at a young age (e.g., preimplantation diagnosis or assessing potential living donors). Comprehensive pregenetic and postgenetic counseling should be provided to all individuals undergoing genetic testing.[108,109] With the advancements in next-generation sequencing (NGS), it is now possible to simultaneously test for multiple cystic diseases. Consequently, high-throughput sequencing methodologies have been established to investigate atypical presentations of ADPKD.[110] Advanced genetic testing has explored possible manifestations, such as the contiguous *TSC2/PKD1* deletion syndrome, "compound heterozygosity," which involves having one *PKD1* variant alongside a second variant in a non-*PKD1* cystic gene (i.e., *PKD2, COL4A1,* or *HNF1B*), or having a truncating *PKD1* variant in trans, associated with a second nontruncating *PKD1* variant.[68,108,111,112] Genetic testing is also insightful in cases of significant intrafamilial disease variability, given the role of genetic mosaicism and biallelic disease, which can affect patients with ADPKD from the same family differently. Applications of genetic testing have been observed in patients with no apparent family history and atypical imaging findings that reflect new gene variants, such as *GANAB, IFT140, DNAJB11, ALG9, ALG5, ALG8, COL4A3, COL4A4,* and *APOL1* variants in black patients with proteinuria.[50,72,108]

To standardize the process of diagnosis and medical documentation, the diagnosis of ADPKD could be also stratified on the basis of family history and kidney cysts findings (Fig. 45.8). "Definite ADPKD" is established either clinically, with a family history of ADPKD and multiple renal cysts according to the Ravine/Pei criteria, or genetically through the detection of pathogenic *PKD1* or *PKD2* variants.[107,113] "Likely ADPKD" is assigned to individuals who exhibit typical ADPKD features, such as enlarged kidneys with more than 10 cysts per kidney, but who lack a family history; in these instances, genetic testing is



1

ADPKD Diagnosis				
Positive FHx				
Age	US-Based	PPV	CT-Based	PPV
15-29	≥3 cysts (Total)	97.3	>10 cysts (Total)	100
30-39	≥3 cysts (Total)	94.7	>10 cysts (Total)	100
40-59	≥2 cysts/ kidney	100	Not determined	
>60	≥4 cysts/ kidney	100	Not determined	
Negative FHx				
Any Age	≥10 cysts/ kidney (Bilateral kidney enlargement)	N/A	Not determined	

Ruling out ADPKD				
Positive FHx				
Age	US-Based	NPV	CT-Based	NPV
15-29	No Cysts	98	<5 Cysts	100
30-39	No Cysts	98.3	<5 Cysts	100
40-59	No Cysts	100	Not determined	
>60	No Cysts	100	Not determined	

2

Age (Years)	Number of Cysts (Diameter ≥5 mm)	
	Males	Females
≤29	≥2	≥2
30-39	≥3	≥3
40-49	≥4	≥3
50-59	≥6	≥4
60-69	≥11	≥5
≥70	≥11 (≥5 in each kidney)	≥10 (≥5 in each kidney)

Fig. 45.7 Algorithm involved in diagnosing autosomal dominant polycystic kidney disease *(ADPKD)* based on the location of the kidney cysts, family history, extrarenal manifestations, and congruency between cystic burden and kidney function. Among the large differential, we emphasize how to diagnose definite, likely, and possible ADPKD. *ADPLD,* Autosomal dominant polycystic liver disease; *ADTKD,* autosomal dominant tubulointerstitial disease; *AML,* angiomyolipoma; *ESKD,* end-stage kidney disease; *Hx,* history; *MCDK,* multicystic dysplastic kidneys; *MCN,* multilocular cystic nephroma; *MIC,* Mayo imaging classification; *NPV,* negative predictive value; *OFD,* oral facial digital; *Pheo,* pheochromocytoma; *PPV,* positive predictive value; *RCC,* renal cell carcinoma; *Rx,* treatment; *TKV,* total kidney volume; *TSC,* tuberous sclerosis complex; *UPJ,* ureteropelvic junction; *VHL,* von Hippel-Lindau syndrome; *VUR,* vesicoureteral junction.

recommended. A diagnosis of "possible ADPKD" is considered if the individual has normal-sized or mildly enlarged kidneys with cysts each measuring ≥5 mm (excluding parapelvic cysts) that exceed the 97.5th percentile of their demographic group, with no severe CKD or other cystic diseases. The specific cyst count thresholds used for this assessment are based on age and sex, derived from data on healthy potential kidney donors from contrast-enhanced abdominal CT scans (Table 45.6).[107]

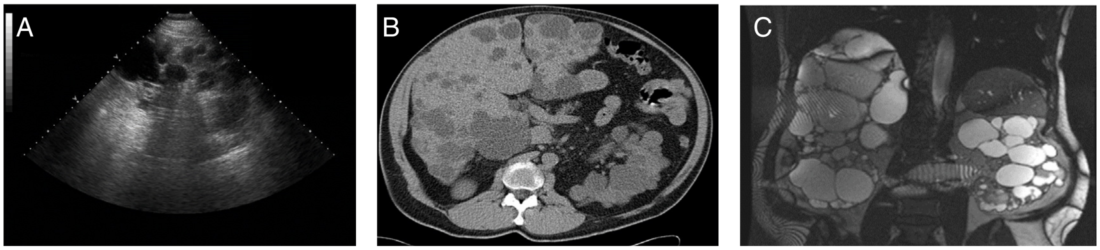
Clinical manifestations, complications and management of ADPKD

ADPKD is characterized by a broad range of renal and extrarenal clinical manifestations. Typically, patients remain asymptomatic early in life, with the age of symptom onset varying widely. While the clinical manifestations may overlap in patients with ADPKD who have *PKD1* or *PKD2* pathogenic variants, those with *PKD1* variants generally experience more severe symptoms, related to larger kidney size and earlier

Table 45.5 Performance of Ultrasound-Based Unified Criteria for Diagnosis or Exclusion of ADPKD in Patients (*PKD1* or *PKD2* pathogenic variants) With Positive Family History

Age (yr)	Imaging Findings	*PKD1*	*PKD2*	Unknown Gene Type
For Diagnostic Confirmation				
15-29	A total of ≥3 cysts*	PPV = 100% Sensitivity = 94.3%	PPV = 100% Sensitivity = 69.5%	PPV = 100% Sensitivity = 81.7%
30-39	A total of ≥3 cysts*	PPV = 100% Sensitivity = 96.6%	PPV = 100% Sensitivity = 94.9%	PPV = 100% Sensitivity = 95.5%
40-59	≥2 cysts in each kidney	PPV = 100% Sensitivity = 92.6%	PPV = 100% Sensitivity = 88.8%	PPV = 100% Sensitivity = 90%
For Diagnostic Exclusion				
15-29	No kidney cyst	NPV = 99.1% Specificity = 97.6%	NPV = 83.5% Specificity = 96.6%	NPV = 90.8% Specificity = 97.1%
30-39	No kidney cyst	NPV = 100% Specificity = 96.0%	NPV = 96.8% Specificity = 93.8%	NPV = 98.3% Specificity = 94.8%
40-59	No kidney cyst	NPV = 100% Specificity = 93.9%	NPV = 100% Specificity = 93.7%	NPV = 100% Specificity = 93.9%

ADPKD, Autosomal dominant polycystic kidney disease; *PPV,* positive predictive value; *NPV,* negative predictive value; *, unilateral or bilateral.
Adapted from Chapman AB, Devuyst O, Eckardt KU, et al. Autosomal-dominant kidney disease: executive summary from a kidney conference. *Kidney Int*. 2015;88:17–27).



	Ultrasound	CT Scan	MRI
Use	Screening (Cyst size detection 7-10 mm)	Prognostication and follow-up (Cyst size detection 2 mm)	Prognostication and follow-up (Cyst size detection 2 mm)
Pros	Cost-effective No exposure to radiation Noninvasive	High resolution, quick and excellent visualisation of cysts	Excellent soft tissue contrast and cyst assessment Multiparametric imaging
Cons	Operator-dependent Limited characterization, visualization and sensitivity	Exposure to contrast and radiation Less suitable in impaired kidney function	Longer imaging time Availability and cost Contraindicated in patients with metallic implants

Fig. 45.8 Abdominal Imaging representation of a 64-year-old man with autosomal dominant polycystic kidney disease. (A) Ultrasound. (B) Computed tomography scan transverse section. (C) Magnetic resonance imaging coronal section.

progression to ESKF. A summary of the major renal and extrarenal manifestations of ADPKD is provided in Fig. 45.9.

Clinical Manifestations in ADPKD

Kidney cysts and GFR. Cyst formation, though initiated in utero, generally remains asymptomatic until the third to fourth decade of life. This early clinical quiescence is attributed to compensatory hyperfiltration by intact nephrons.[108] Variants in *PKD1* or *PKD2* disrupt ciliary cellular flow sensation and intracellular calcium signaling, impairing renal tubular epithelium function. This results in significant cellular proliferation and abnormal ECM deposition. As cysts grow in size and number, they begin to detach and compress surrounding vasculature, leading to vascular damage, inflammation, and fibrosis, and eventually ESKF.[81,108] Cyst size starts from a few millimeters with cystic kidneys potentially growing over 40 cm in length and weighing up to 8 kg in severe cases.[114] Baseline TKV is associated with the rate of increase in kidney volume; larger kidneys tend to enlarge faster and more significantly, with increases in kidney size predicting declines in kidney function. A htTKV of >600 mL/m is predictive of CKD stage 3 within 8 years.[56] Early manifestations of ADPKD include defects in urinary concentration and ammonia excretion.[115] The decreased concentration ability and elevation in vasopressin levels contribute to cystogenesis.[116,117] Typically, GFR is stable in the first 3 decades with the decline in kidney function seen as a later manifestation declining at 2 to 5 mL/min per year. Notably, renal blood flow significantly impacts kidney function, highlighting the role of vascular remodeling in exacerbating the function decline beyond the extent suggested by cystic burden alone.[56,118] Proteinuria, defined by urinary protein excretion >300 mg/day and observed in 25% of adults with ADPKD, seldom surpasses 1 g/day but holds prognostic significance, correlating with disease severity. Higher levels of proteinuria are associated with increased kidney volume, more rapid GFR decline, and earlier onset of ESKF.[46]

Table 45.6 Specific Values Above 97.5th Percentile by Age and Sex for Possible ADPKD Diagnosis; Cysts Must Be ≥5 mm in Diameter

	Number of Cysts (Diameter ≥5 mm)	
Age (Years)	**Males**	**Females**
≤29	≥2	≥2
30-39	≥3	≥3
40-49	≥4	≥3
50-59	≥6	≥4
60-69	≥11	≥5
≥70	≥11 (≥5 in each kidney)	≥10 (≥5 in each kidney)

Hypertension. Hypertension, defined as BP readings higher than 130/80 mm Hg, is the most prevalent clinical manifestation of ADPKD, affecting 50% to 70% of patients by an average age of 30 years, often before significant renal function decline.[47,119] Notably, hypertension is also found in approximately 20% of pediatric patients with ADPKD.[120,121] The development of hypertension in ADPKD is multifactorial, characterized by increased renal vascular resistance, altered pressure-natriuresis relationship, and heightened salt sensitivity. Primarily, it is linked to the activation of the renin-angiotensin-aldosterone system (RAAS), compounded by factors such as increased sympathetic activity and the

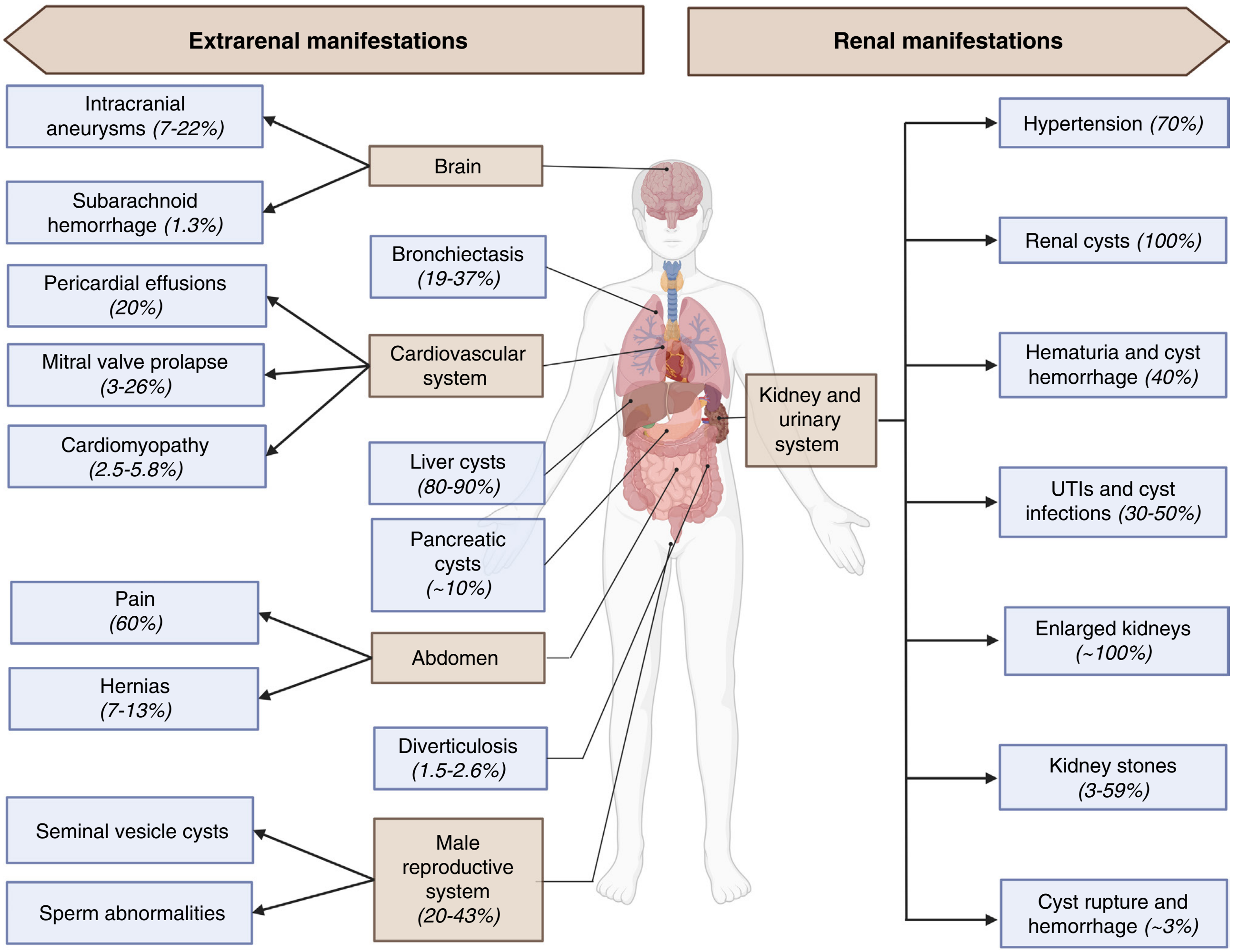


Fig. 45.9 Summary of the renal and extrarenal manifestations seen in autosomal dominant polycystic kidney disease.

effects of endothelin and nitric oxide (NO) on the vascular tone.[114] Cyst expansion and renal structural abnormalities in ADPKD lead to vascular compression and ischemia, triggering the intrarenal RAAS. This activation is evidenced by elevated renin levels in serum or cystic fluid, significantly contributing to arterial hypertension. ACEIs significantly decreased mean arterial pressure, renal vascular resistance, and filtration fraction in hypertensive patients with ADPKD.[122]

Clinical trials including the Halt Progression of Polycystic Kidney Disease (HALT-PKD) study—a double-blinded, randomized, placebo-controlled trial—have underscored the effectiveness of ACE inhibitors in managing hypertension in ADPKD. Monotherapy with lisinopril was shown to provide effective BP control and slow TKV increase, but it resulted in an initial steeper decline in eGFR, followed by a slower decline.[116] Interestingly, additional therapy with telmisartan did not yield further benefits over lisinopril monotherapy. Furthermore, other studies demonstrated that despite similar high plasma levels of angiotensinogen, renin, and aldosterone between hypertensive ADPKD and CKD patients, urinary levels were much more elevated in patients with ADPKD, likely due to cyst ischemia and differential tubular reabsorption.[123] Additionally, RAAS is implicated in promoting angiogenesis and cytogenesis, exacerbating ADPKD progression.[121] Sympathetic nerve activity at the juxtaglomerular apparatus is also heightened in ADPKD, enhancing RAAS activity and thus playing a crucial role in modulating renin secretion, renal vascular resistance, and blood flow. Chronic sympathetic activation correlates with hypertension and its systemic impacts.[124] Another intriguing aspect involves ciliary dysfunction in ADPKD, where abnormal renal primary cilia disrupt NO biosynthesis, typically induced by shear stress within the tubules.[125] Current guidelines recommend a BP target of <120/80 mm Hg for patients older than 50 years, with a lower target of <110/75 mm Hg for individuals aged 18 to 49 years with eGFR >60 mL/min/1.73 m^2.[126]

Early detection and diagnosis of hypertension are crucial as cardiovascular disease is the number one cause of death in ADPKD.[46] Both lifestyle modifications and pharmacologic interventions are important to maintain an adequate BP. Nonpharmacologic interventions include sodium restriction to <100 mmol per day, maintaining a healthy body weight (BMI <25 kg/m^2), physical exercise as tolerated (typically 30 minutes per day for 5 days a week), and smoking cessation.[121] The preferred first-line pharmaceutic agents are usually ACEI or ARB if ACEIs are not well tolerated. Dual RAAS blockade is not recommended due to the risk of hyperkalemia. RAAS blockers should be used in caution in pregnant patients due to their teratogenic effects. Most patients might need two antihypertensive agents. The choice of the second-line agent depends on patients' characteristics and comorbidities.[121]

Pain. Pain is a common complaint among ADPKD patients, affecting approximately 50% to 60% of individuals.[127] It arises from various causes including UTIs such as cystitis and pyelonephritis, cyst infections, nephrolithiasis, or complications like acute abscess formation. Pain may also result from cyst hemorrhage or the physical stress of enlarged kidneys compressing adjacent structures. Additionally, diffuse or right upper quadrant abdominal pain often stems from enlarged liver cysts, while diffuse unilateral nonradiating flank pain may indicate kidney cyst infection, and point tenderness may suggest cyst hemorrhage.[127,128] Chronic pain in ADPKD is primarily caused by cyst enlargement, which leads to the stretching of the kidney capsule and increased kidney weight. This added weight strains the back muscles, contributing significantly to discomfort and pain. This chronic pain can significantly impact mental health, leading to depression, sleep disturbances, and a reduced quality of life.[47,114,129] Furthermore, chronic pain may exacerbate hypertension, either by directly stimulating sympathetic activity or indirectly affecting BP control by altering the nocturnal dip and increasing morning BP levels.[121] The ADPKD pain and discomfort scale (ADPKD-PDS) is the first systematic tool designed to measure the pain and discomfort experienced by ADPKD patients across various stages of CKD. This scale assesses two primary dimensions: pain severity and pain interference, providing insights into the impact of pain on daily activities.[130] Pain management in kidney disease is discussed further in Chapter 61.

Among the multiple strategies used for pain management, ice massages or heating pads, support garments, the Alexander technique, and physical therapy are some of the major noninvasive ones. Pain medications can be added, mainly nonopioid analgesics such as acetaminophen and clonidine. NSAIDs should be used cautiously for no longer than a week in patients with adequate renal function and, if needed, opioids may be considered after dose adjustments according to eGFR.[131,132] Before moving to more invasive procedures, percutaneous aspiration can help diagnose and alleviate the symptoms of cyst expansion in select patients with relatively larger-sized cysts. Other more invasive procedures that were shown to be effective include celiac plexus blockade, radiofrequency ablation, spinal cord stimulation, laparoscopic, or transluminal catheter-based kidney denervation. As such, cyst aspiration and fenestration alleviate pain related to cyst expansion, celiac and/or splanchnic nerve blocks, and interventions cater to pains related to the celiac plexus while renal denervation targets pain secondary to aortorenal plexus.[127,131,132] Nephrectomy is a last resort tool to address pain in ADPKD. It must be avoided, unless really necessary, as it leads to an earlier need for KRT.[127] In the case of acute abdominal pain, it is crucial to rule out cyst infection and identify the source, considering the potential need for a prolonged antibiotic regimen and occasionally cyst drainage.

Cyst Infections and Urinary Tract Infections. UTIs occur in 30% to 50% of patients with ADPKD, with female patients being particularly susceptible to urethritis and cystitis, commonly caused by gram-negative enteric organisms.[46] The treatment of lower tract infections aligns with that of the general population and should not be delayed. Upper tract infections, such as pyelonephritis and kidney cyst infections, are indicated by abdominal or flank pain accompanied by fever and elevated inflammatory markers, C-reactive protein, or erythrocyte sedimentation rate.[133] Diagnostic workup should include urinalysis, urine culture with antibiotic susceptibilities, and blood cultures. Abdominal imaging, essential for accurate diagnosis, typically involves CT or MRI, though fludeoxyglucose positron emission tomography (FDG-PET) is helpful,[134] particularly when distinguishing between kidney and liver cyst infections is required.[134,135] Liver cyst infections often present with fever, malaise, and right upper quadrant abdominal pain. These infections generally require prolonged antibiotic treatment. Initial antibiotic therapy starts with broad-spectrum agents, later tailored based on susceptibilities to narrower oral agents with effective cyst penetration, such as trimethoprim-sulfamethoxazole and fluoroquinolones.[136] The duration of the antibiotic therapy varies; isolated acute pyelonephritis requires 10 to 14 days, whereas an infected cyst may need 4 to 6 weeks. If no clinical improvement is observed with antibiotics, CT-guided aspiration may be necessary, with the aspirate sent for microbiology analysis.[46] For larger cysts (>5 cm), a combination of antibiotics and cyst aspiration is often essential. Should there be no improvement after the first 72 hours of adequate management, further investigation of complications such as abscesses, emphysematous pyelonephritis, or severe urinary tract obstruction is warranted, which in some rare cases may necessitate nephrectomy.[133,136,137]

Cyst Hemorrhage and Hematuria. Cyst hemorrhage in ADPKD is frequently seen on imaging as a high-density cyst, similar in appearance to proteinaceous cysts. The extent of hemorrhagic cysts is linked to TKV.[46,138] The cysts, through their formation and associated marked angiogenesis, compress nearby vasculature, which can lead to internal or external cyst hemorrhage, resulting in hematuria. Not limited to the urinary tract, cysts may also rupture into the subcapsular compartment, causing retroperitoneal bleeding.[138] Hematuria may appear either microscopically or gross hematuria, with the latter more frequently observed in patients with larger kidneys and advanced CKD. While hemorrhage may spontaneously resolve within a week, it necessitates further investigation to exclude more serious underlying causes. Gross hematuria at a younger age (<35 years old) may indicate a prognosis of poorer renal outcomes.[81,129,139]

Management is largely conservative. In addition to rest and hydration, it is advised to avoid or temporarily discontinue anticoagulants. In severe cases, holding ACEIs or ARBs may help prevent AKIs. When needed, blood transfusions should be used with caution to prevent sensitization in potential future kidney recipients. After management of shock, massive cyst bleeding may require more invasive interventions such as interventional radiology-guided embolization. When all the conservative therapies fail, tranexamic acid, an antifibrinolytic agent, was shown to be effective to control cyst hemorrhage.[140,141]

Nephrolithiasis. Kidney stones affect 20% to 35% of patients, with calcium oxalate and uric acid stones being the most common types. CT scans are more effective than ultrasound in differentiating between these stone types. Contributing factors to kidney stone formation include urinary stasis and reduced excretion of urinary ammonia.[139,142] Hypocitraturia is frequently observed in individuals with ADPKD and has been associated with stone formation.[143] Experimental studies further underscore the impact of low urinary citrate levels on disease progression, demonstrating an association with accelerated eGFR decline and increased cyst burden.[84,144]

Medical management of stones in individuals with ADPKD is similar to the general population. In addition to adequate hydration and symptomatic management, potassium citrate is the treatment of choice in most patients. Alkalinization of the urine with potassium citrate has been shown to be effective in reducing the formation of uric acid stones and decreasing calcium oxalate supersaturation, providing a beneficial strategy for managing nephrolithiasis in ADPKD.[145,146] In more complicated cases with obstructing kidney stones, extracorporeal shock wave lithotripsy, percutaneous nephrolithotomy, and flexible ureterorenoscopy are options to be considered. Among those, flexible ureteroscopic lithotripsy showed the highest safety and faster recovery rate.[46,147-149]

Renal Cell Carcinoma. RCC is uncommon in ADPKD, with an incidence of <1%, similar to that seen in other kidney diseases. However, RCC in ADPKD patients tends to present at a younger age and is often bilateral, multicentric, and sarcomatoid in type. Patients may exhibit systemic symptoms such as fever, anorexia, weight, loss, and fatigue. Diagnostically, rapid growth of a complex cyst on US or speckled calcifications may be evident on CT, often accompanied by local lymphadenopathy and thrombi on CT or MRI. Screening for RCC is not a routine practice in patients with ADPKD.[150]

Extrarenal Manifestations in ADPKD

Polycystic Liver Disease. Cysts in ADPKD can involve various extrarenal organs including the liver, pancreas, seminal vesicles, and arachnoid membrane in the brain. Among these, liver cysts are the most common extrarenal manifestation. Their prevalence increases with age, affecting 80% to 90% of patients older than 35 years old.[46,47] The development of hepatic cysts is influenced by several factors, particularly exposure to estrogen hormones such as those in estrogen-containing oral contraceptives, hormone replacement therapy, and pregnancy. Consequently, cysts typically appear at a younger age in women than men.[151] Severe PLD, defined as height-adjusted total liver volume (htTLV) of ≥1.8 L/m, occurs more often in women (80%) than men. This frequency appears independent of the presence of truncating or nontruncating variants in *PKD1* or *PKD2*.[46] Although liver cysts typically do not impair liver function, elevations in γ-glutamyl transferase (GGT) and alkaline phosphatase may occur.[152] Liver cyst epithelia produce carbohydrate antigen 19-9 (CA19-9), leading to elevated serum and hepatic cyst fluid levels in individuals with PLD. The most significant impact of PLD on quality of life arises from space-occupying symptoms including abdominal pain, distention, early satiety, and dyspnea due to diaphragmatic compression.[46,47] Complications of liver cysts include cyst hemorrhage, infection, and rupture. Cyst hemorrhage, often linked to rapid cyst growth or abdominal trauma, may result in complications such as cholecystitis and infection. Imaging techniques like CT or MRI are helpful. Cyst infections, which occur in approximately 1% of patients, are associated with a 2% mortality if not promptly treated.[153] Management of liver cysts depends on the severity of the symptoms and complications. Patients may require pain management (acetaminophen and narcotics; NSAIDs are usually discouraged), cyst aspiration with sclerotherapy, cyst fenestration, partial hepatectomy, and, in the most severe cases, liver transplantation.[81,153]

Other cysts. Pancreatic cysts are observed in 19% of individuals with ADPKD but are rarely symptomatic or associated with recurrent pancreatitis.[154] Seminal vesicle cysts occur in 40% of male individuals with ADPKD.[155] Infertility in ADPKD is more commonly due to sperm flagella dysmotility secondary to polycystin variants, rather than seminal vesicle cysts.[156]

Cardiac manifestations. Cardiovascular disease is the leading cause of mortality in patients with ADPKD. Hypertension, which develops early, is accompanied by structural abnormalities, such as left ventricular hypertrophy (LVH) and mitral valve defects.[121] LVH increases the risks of arythmias, heart failure, and premature death. A study by Arjune and colleagues[157] reported a higher prevalence of LVH in patients with ADPKD compared with controls (65% vs. 55%), although findings have been variable across studies.Variability may reflect earlier screening, detection, and the increased use of RAAS blockers.[158,159] Hypertension is the most important contributor to LVH, with contributing factors including borderline hypertension, masked hypertension, and loss of the nocturnal BP dipping.[160] Although rigorous BP control can slow LVH progression, it has not been consistently associated with reduced cardiovascular events. The HALT-PKD studies demonstrated no significant reduction in cardiovascular complications despite lower BP targets.[161] Mitral valve prolapse (MVP) is a valvular manifestation of ADPKD with a prevalence up to 26%. It arises from connective tissue defects and can occur in normotensive patients.[162,163] Complications include MVP mitral regurgitation, which warrants echocardiography when a murmur is detected on auscultation.[164]

Gastrointestinal manifestations. The association between ADPKD and diverticular disease remains unclear, though complications such as diverticulitis and colon perforation become more apparent in patients who reach ESKF or those with post-transplant immunosuppression.[165,166] Dysfunction in PC1 and PC2 in intestinal smooth muscle may contribute to diverticular disease pathogenesis.[166]

Pulmonary manifestations. Patients with ADPKD have a threefold increased risk of bronchiectasis compared with other CKD populations. Bronchiectasis is usually mild and may result from dysfunctional PC1 and PC2 in the motile epithelial airway cilia or bronchial smooth muscle cells.[167,168]

Intracranial aneurysms. Intracranial aneurysms (IAs) are a significant extrarenal manifestation of ADPKD, with a prevalence of 8%, which is notably higher than the 3.2% observed in the general population. This prevalence increases to 22% in patients with a family history of IAs or subarachnoid hemorrhage (SAH).[169-171] Most IAs in ADPKD are small, saccular, and often multifocal, primarily located in the anterior circulation of the brain including the internal carotid artery (ICA) and middle cerebral artery (MCA).[172] Risk factors for IAs in ADPKD include both nonmodifiable, such as age, female sex, and family history, and modifiable risk factors, such as hypertension, smoking, and alcohol consumption.[173] Genetic factors also contribute, with variants in *PKD1* and *PKD2* genes, particularly in the 5′ region of *PKD1,* associated with a higher risk of IA formation.[174-176]

Screening for IAs in ADPKD remains a debated topic, with strategies ranging from universal screening of all patients to targeted screening of high-risk individuals. Universal screening is supported by the significant prevalence of IAs in ADPKD and their potential for catastrophic complications like rupture, which occurs at higher rates in ADPKD compared with the general population. Studies have suggested that noninvasive imaging modalities, such as magnetic resonance angiography (MRA), can improve outcomes and life expectancy without neurologic disability.[177] However, universal screening raises concerns about cost-effectiveness, potential overtreatment, and management dilemmas.[178,179] Targeted screening focuses on individuals with additional risk factors, such as a family history of IAs, PKD genotype, hypertension, and smoking, to optimize resource use and minimize unnecessary interventions.[178,180] It is important to have a shared decision on screening strategy for patients aged 18 to 70 by discussing the risks and benefits. In patients with high risk of developing IAs, it would be reasonable to screen every 5 years; otherwise, every 10 years is the recommended interval.

Most IAs in ADPKD are asymptomatic and detected incidentally, but complications including ischemia, embolic events, and rupture leading to subarachnoid hemorrhage can occur.[181] Ruptured IAs carry a high risk of morbidity and mortality and typically present with a sudden, severe “thunderclap” headache, altered mental status, nausea, and neck pain.[182,183] Educating patients and their caregivers about the symptoms of thunderclap headaches and the need to present urgently to the emergency department is important. This headache is typically described as the most severe headache of a lifetime and peaks within a minute. Smaller aneurysms (<10 mm) carry a low annual rupture risk of 0.05%, but the risk increases significantly for aneurysms larger than 25 mm.[184] With the coordination with a multidisciplinary team of neurologists and neurosurgeons, management of unruptured IAs is tailored to patient age, aneurysm size, and symptoms, with larger (>5 mm in diameter) or symptomatic aneurysms in younger patients often requiring immediate intervention.[185,186] Options include surgical clipping, coil embolization, or active surveillance with serial imaging. Comprehensive management includes addressing modifiable risk factors, particularly hypertension and smoking, to reduce the risk of aneurysm formation and complications.

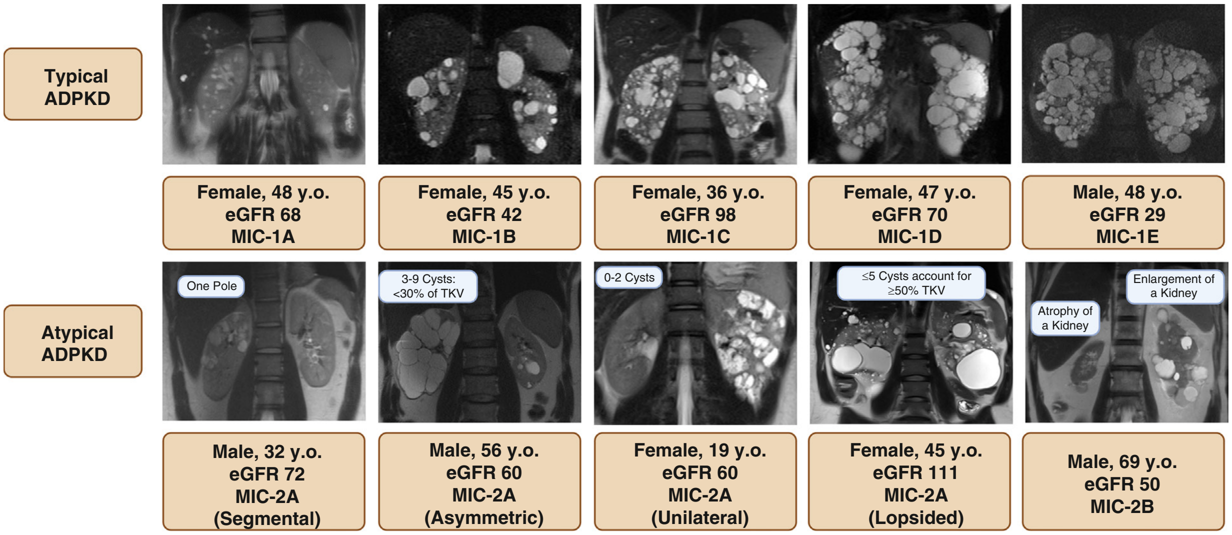


Fig. 45.10 Magnetic resonance imaging representation of typical (MIC-1A, 1B, 1C, 1D, and 1E) versus atypical (MIC-2A and 2B) autosomal dominant polycystic kidney disease.

Disease Stratification in ADPKD

ADPKD is a heterogeneous disease with a large phenotypic variability. As kidney cysts grow, kidney size and volume increase, leading to a gradual decline in GFR.[113] The onset and rate of GFR decline varies among patients with some patients reaching ESKF in their fourth decade while others in their eighth decade. Given that GFR remains relatively stable during the first 2 to 3 decades of ADPKD, alternative prognostic biomarkers are essential to predict the risk of rapid progression and differentiate disease severity. Biomarkers such as TKV, height-adjusted TKV (htTKV), and PROPKD score based on clinical manifestations and PKD genotyping are typically used.

The CRISP (Consortium for Radiologic Imaging Studies of Polycystic Kidney Disease) studies demonstrated the utility of TKV as a prognostic marker for kidney function decline. A baseline TKV >1500 mL was strongly associated with a GFR decline of 4.33 mL/min/year, outperforming serum creatinine and albuminuria.[187] TKV has been approved by the U.S. Food and Drug Administration as a prognostic biomarker in ADPKD. TKV measurements require precision imaging tools such as planimetry, but simplified techniques using MRI or CT can provide reliable estimates via the ellipsoid equation by measuring the sagittal, coronal, and axial kidney lengths. The Mayo Imaging Classification (MIC) is a practical, imaging-based model used to categorize ADPKD severity. Originally developed at the Mayo Clinic Translational PKD Center using data from a cohort of 590 ADPKD patients aged 15 to 80 years, the MIC has become a valuable tool for clinical trial selection and disease monitoring.[188] Patients are classified as either typical (Mayo Class 1) or atypical (Mayo Class 2) (Fig. 45.10). Typical ADPKD, representing 95% of cases, is characterized by bilateral kidney cysts that contribute nearly equally to TKV. Atypical ADPKD, accounting for 5% of cases, is further subdivided into Class 2A, where the disease is focal and asymmetric, and Class 2B, characterized by unilateral or bilateral atrophic kidneys. For MIC 1, patients are further subcategorized into five subclasses, MIC 1A through 1E, by adjusting TKV to height and age at time of imaging. The risk of progression to ESKF increases significantly from MIC 1A to 1E, making this classification an essential tool for identifying patients at higher risk who might benefit from disease-modifying treatments. Patients with MIC 1C, 1D, or 1E are considered at risk of rapid progression, defined as reaching ESKF before age 62. In contrast, those in lower-risk classes (1A and 2A) are less likely to benefit from clinical trials but still require regular monitoring every 3 to 5 years to reassess their progression risk.

PKD genotype is associated with prognostic insight. Patients with *PKD2* variants reach ESKF 20 years after those with *PKD1* variants. The Predicting Renal Outcomes in PKD (PROPKD) tool is a clinical scoring system designed to assess the risk of kidney function decline and progression to ESKF in ADPKD (Table 45.7).[189] It combines several clinical parameters (male, cyst infection, or bleeding before age 35, hypertension before age 35) and PKD genotype data (*PKD1* truncating, *PKD1* nontruncating, or *PKD2*) into a single score, where a score of ≤3 predicts no risk of ESKF before age 60, with a negative predictive value (NPV) of 81.4%. A score >6 indicates a high likelihood of rapid progression to ESKF before age 60, with a positive predictive value (PPV) of 90.9%. Intermediate scores (4–6) indicate an uncertain prognosis, requiring closer monitoring and individualized assessment. Table 45.8 highlights the strengths and limitations of various methods for assessing the risk of rapid progression in ADPKD.

Other biomarkers have been explored in ADPKD including advanced imaging biomarkers, such as imaging texture analysis and cyst segmentation, which offer innovative methods to predict disease progression in ADPKD. Texture analysis, using features like entropy, gradient, and correlation from T2-weighted MRIs, enhances predictive models when combined with traditional factors like age and eGFR, effectively distinguishing patients at risk for advanced CKD.[190] Cyst segmentation, demonstrated in the CRISP cohort, has shown strong correlations among markers like total cyst volume (TCV), renal parenchyma volume (RPV), and total cyst number (TCN) with kidney function decline over time. Among these, Cyst-Parenchyma

Table 45.7 Different Factors Constituting the PROPKD Score

PROPKD Score	Factor	Category	Point
	Sex	Female	0
		Male	1
	Hypertension before age 35	No	0
		Yes	2
	At least 1 urologic complication before age 35	No	0
		Yes	2
	Pathogenic variant	PKD2	0
		Nontruncating *PKD1*	2
		Truncating *PKD1*	4

Characteristics	Categories	Score	Median Age at KF	Risk of KF by age 60	KF by age 60
	Low Risk	0-3	70.6 (63.6-84.5)	19.3±2.7	NPV= 81.4%
	Intermediate Risk	4-6	56.9 (50.7-65.4)	60.8±3.0	
	High Risk	7-9	49 (43.8-55.6)	91.9±3.2	PPV= 90.9%

KF, Kidney failure.
From Cornec-Le Gall E, Audrezet M-P, Rousseua A, et al. The PROPKD score: a new algorithm to predict renal survival in autosomal dominant polycystic kidney disease. *JASN*. 2016;27.3:942–951.

Table 45.8 Strength and Limitations of Various Methods Used to Assess Rapid Progression Risk in ADPKD[202]

Prognostic Biomarker	Strengths	Limitations
eGFR (index by age)	Classic biomarker indicating rapid progression if congruent with cystic burden	Delay therapy in early stages. Does not exclude other factors affecting GFR
eGFR rate of decline	Measures actual rather than predicted progression	Delay therapy in young patients. Does not exclude non-ADPKD factors
htTKV (>600 mL/m)	Approved predictor of CKD 3	CT/MRI are costly, and US is less precise. Unreliable in atypical ADPKD
TKV rate of growth (>5%/yr)	Measures actual growth	Needs expertise for precise measurements and unreliable for atypical ADPKD
MIC (classes 1C, 1D, and 1E)	Predicts eGFR decline and practical for atypical ADPKD	Costly, small class separation at young age and underrepresents non-White population
Kidney length >16.5 cm in patients aged <45 yr	Less expensive, correlates with TKV >750 mL and predicts CKD stage 3	Denies treatment for some young patients, mislabel atypical ADPKD cases, and operator dependent
Genetic testing (*PKD1* variant)	Links variant to severity	High cost and limited availability. More accurate at the population rather than individual level
Family history	Predictive of *PKD1* variant	Variable accuracy and no individual precision
PROPKD (Score >6)	Combines clinical and genetic markers and high chance of reaching ESKF by age 60	Not helpful in patients <35 yr unless hypertensive with urologic complications

eGFR, Estimated glomerular filtration rate; *ESKF,* end-stage kidney failure; *htTKV,* height-adjusted TKV.
Adapted from Chebib FT, Torres VE. Assessing risk of rapid progression in autosomal dominant polycystic kidney disease and special considerations for disease-modifying therapy. *AJKD*. 2021;78(2):282–292.

Surface Area proved particularly predictive.[191] Biomarkers for ADPKD progression fall into four categories: clinical (e.g., eGFR and demographics), imaging (e.g., TKV, MIC, and AI tools), genetic (e.g., PKD variants), and molecular, though the latter require further validation (Fig. 45.11). While imaging biomarkers improve predictive accuracy, challenges remain in their routine clinical application, necessitating ongoing research and refinement. Fig. 45.12 represents the disease progression in ADPKD along with the associated symptoms.

Management of Kidney Disease

There have been significant advancements in improving kidney outcomes in ADPKD. The following section summarizes approaches to slow disease progression in ADPKD. Fig. 45.13 shows a detailed algorithm for the management of patients with ADPKD based on their risk to develop ESKF.

Water Prescription. Hydration has been studied as a potential therapeutic intervention in ADPKD. From a physiologic perspective, increased water intake theoretically attenuates

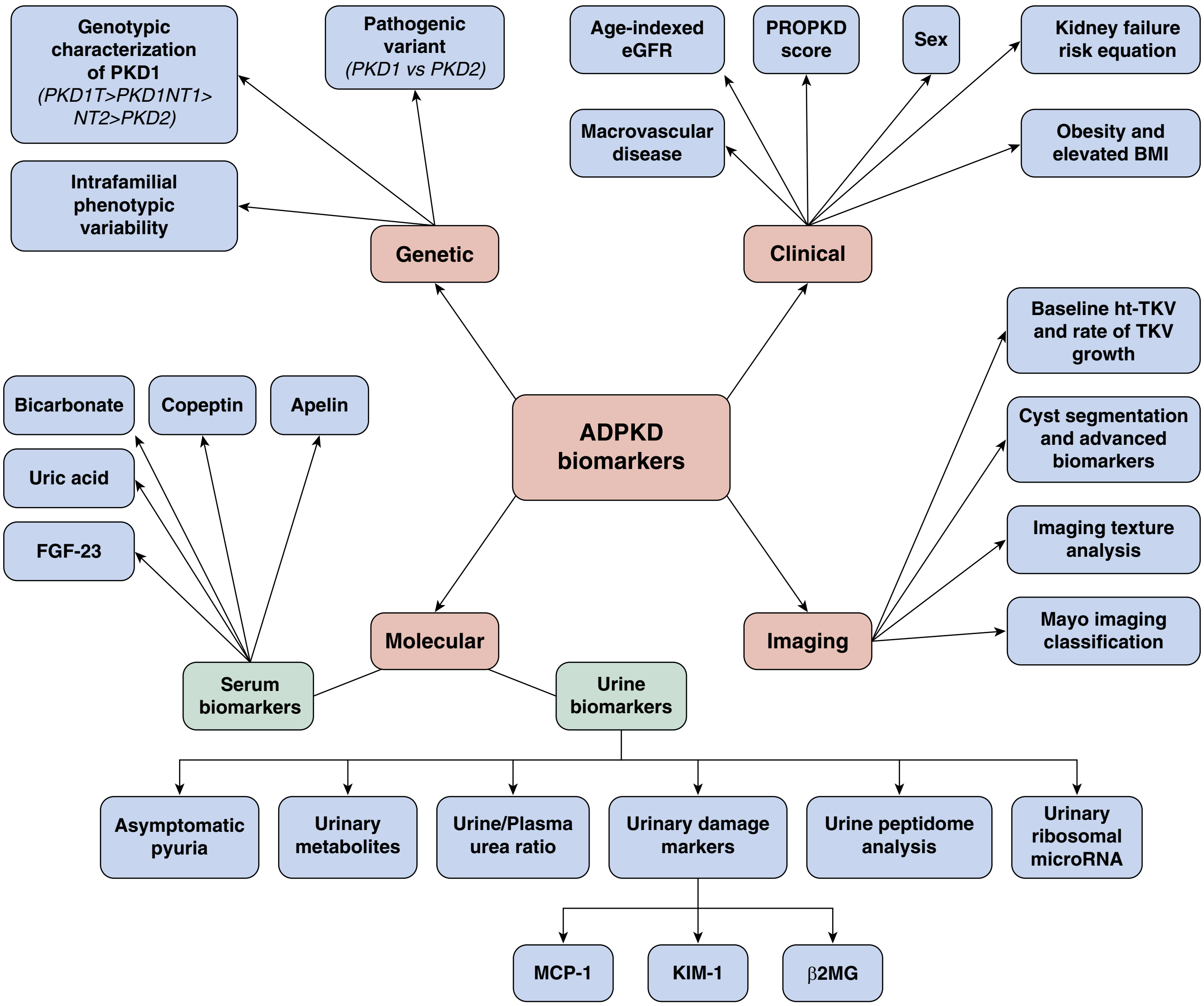


Fig. 45.11 The different biomarkers identified in autosomal dominant polycystic kidney disease (ADPKD) progression and prognosis. This detailed classification organizes ADPKD biomarkers into four distinct groups. Genetic biomarkers include the type of genetic variant, highlighting the role of *PKD1* pathogenic variants in severe presentations. Further delineation within *PKD1* pathogenic variants reveals genotypic and intrafamilial variations. Clinical markers including age-indexed eGFR, PROPKD scores, and considerations of macrovascular diseases provide insights into risk factors, with males experiencing more severe ADPKD. Imaging biomarkers such as ht-TKV and TKV growth rate are approved as prognostic biomarkers in ADPKD while advanced imaging techniques offer nuanced insights. Molecular markers, divided into serum and urinary categories, include apelin, copeptin, bicarbonate, uric acid, FGF23, asymptomatic pyuria, urine/plasma urea ratio, and more, offering systemic and renal-specific perspectives.

cyst growth by suppressing vasopressin central release. While no definitive target for daily fluid intake has been established, patients with ADPKD are at an increased risk for nephrolithiasis with current guidelines recommending a daily fluid intake of 3 L to mitigate this risk.[47] A multicenter, long-term randomized controlled trial evaluated the impact of prescribed water intake aimed at achieving hypotonic urine (<270 mOsm/kg) on kidney disease progression in ADPKD. Over a 3-year follow-up, the mean 24-hour urine volume was 2364 mL in the ad libitum water intake group and 2997 mL in the prescribed water intake group, with a difference of 633 mL (95% CI, 369–896). Despite the increased urine volume in the prescribed water intake group, no significant difference in TKV growth or GFR decline was observed compared with the ad libitum group.[192] Interestingly, only half of the patients in the prescribed prescription group achieved and sustained hypotonic urine, highlighting the inherent challenges of sustaining high fluid intake over the long term.

Disease-Modifying Treatments

Tolvaptan (V2 receptor antagonism). Tolvaptan, a selective vasopressin V2 receptor antagonist, represents the first EMA- and U.S. Food and Drug Administration–approved disease-modifying therapy for ADPKD. The rationale for its use lies in the pathophysiology of ADPKD, where elevated arginine

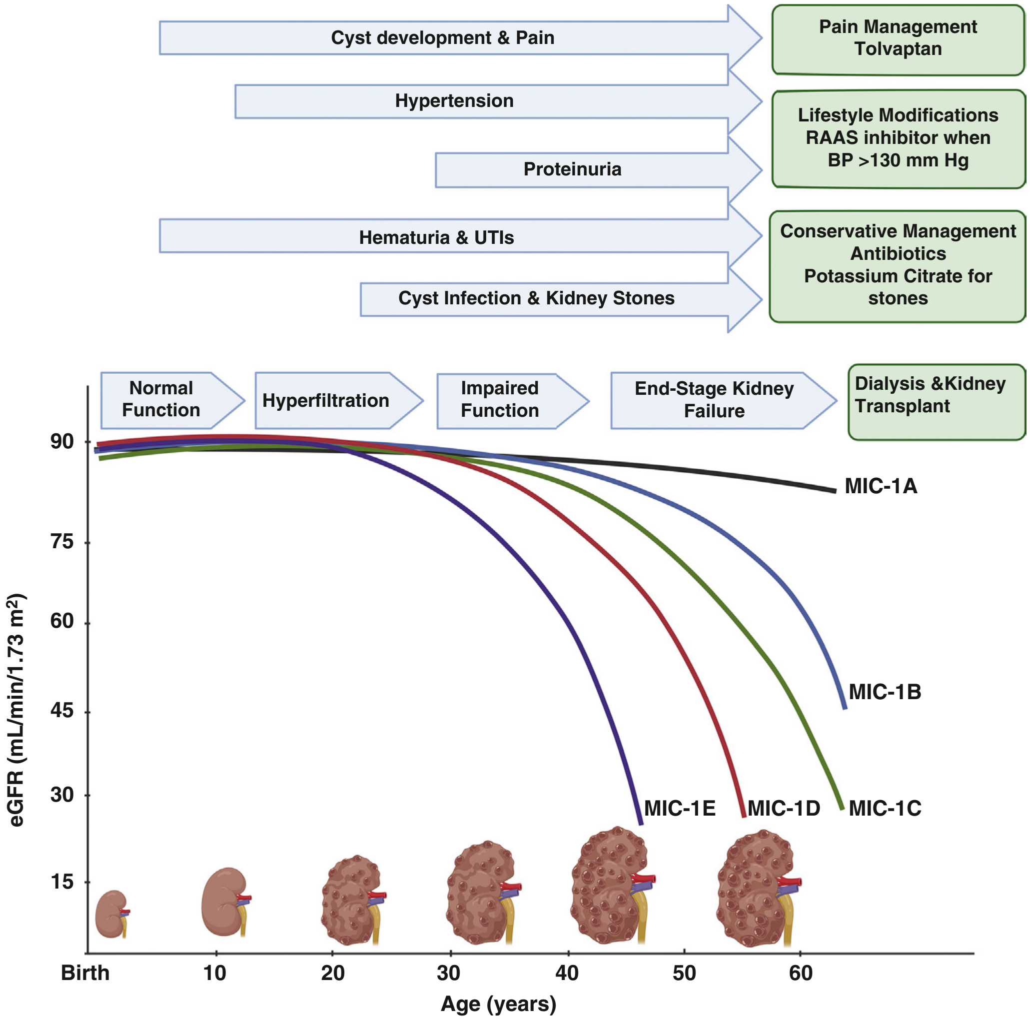


Fig. 45.12 Disease progression in autosomal dominant polycystic kidney disease (ADPKD). This representation outlines the progressive stages of ADPKD in the context of chronic kidney disease (CKD). Starting with normal kidney function, the trajectory advances through hyperfiltration and impaired function, ultimately leading to end-stage kidney failure (ESKF), mandating interventions like dialysis or transplantation. Noteworthy early manifestations include cyst development, pain, hematuria, urinary tract infections (UTIs), and hypertension, with later stages featuring proteinuria, cyst infections, and stone formation. Different interventions are shown for different manifestations: tolvaptan for limiting cystic and disease progression, renin-angiotensin-aldosterone system (RAAS) blockers for hypertension, potassium citrate for stones, and antibiotics for UTIs. The comprehensive approach of conservative management proves crucial in addressing various ADPKD-associated symptoms including hematuria, stones, and pain, contributing to comprehensive patient care. *MIC,* Mayo imaging classification.

vasopressin (AVP) levels play a critical role in cystogenesis and disease progression. AVP binds to V2 receptors (V2R) on the basolateral membrane of kidney collecting duct cells, triggering cyclic AMP (cAMP) accumulation.[193,194] This promotes cell proliferation, chloride-driven fluid secretion, and cyst expansion.[195] Elevated copeptin, a surrogate marker for vasopressin, correlates strongly with ADPKD severity and progression.[193] Tolvaptan is a significant advance in the treatment of ADPKD, offering meaningful reductions in cyst growth, eGFR decline, and symptom burden. Patient selection, adherence, and monitoring are important in optimizing its safe use.

Preclinical Evidence. Animal models of polycystic kidney disease have demonstrated that genetic elimination[196] or pharmacologic inhibition[87] of AVP signaling significantly attenuates cystogenesis. Tolvaptan has shown efficacy in slowing cyst growth and preserving kidney function across various rodent models, including cpk mice, PCK rats (autosomal recessive polycystic kidney disease [ARPKD]), and *Pkd1*$^{RC/RC}$ mice (ADPKD).[87,197-199] In vitro, Tolvaptan inhibits AVP-stimulated chloride secretion, highlighting its mechanism in reducing cyst expansion.[200]

Clinical Trials. In the TEMPO 3:4 trial, involving ADPKD patients with preserved kidney function (creatinine clearance >60 mL/min/1.73 m^2), Tolvaptan reduced the annual rate of kidney volume growth by 45% and slowed kidney function decline by 26% over 3 years.[193] Similarly, the REPRISE trial demonstrated a 35% reduction in eGFR decline among patients with more advanced ADPKD (creatinine clearance 25–65 mL/min/1.73 m^2) over 1 year.[201] Both studies affirm Tolvaptan's efficacy in delaying disease progression and support its use in patients at risk for rapid

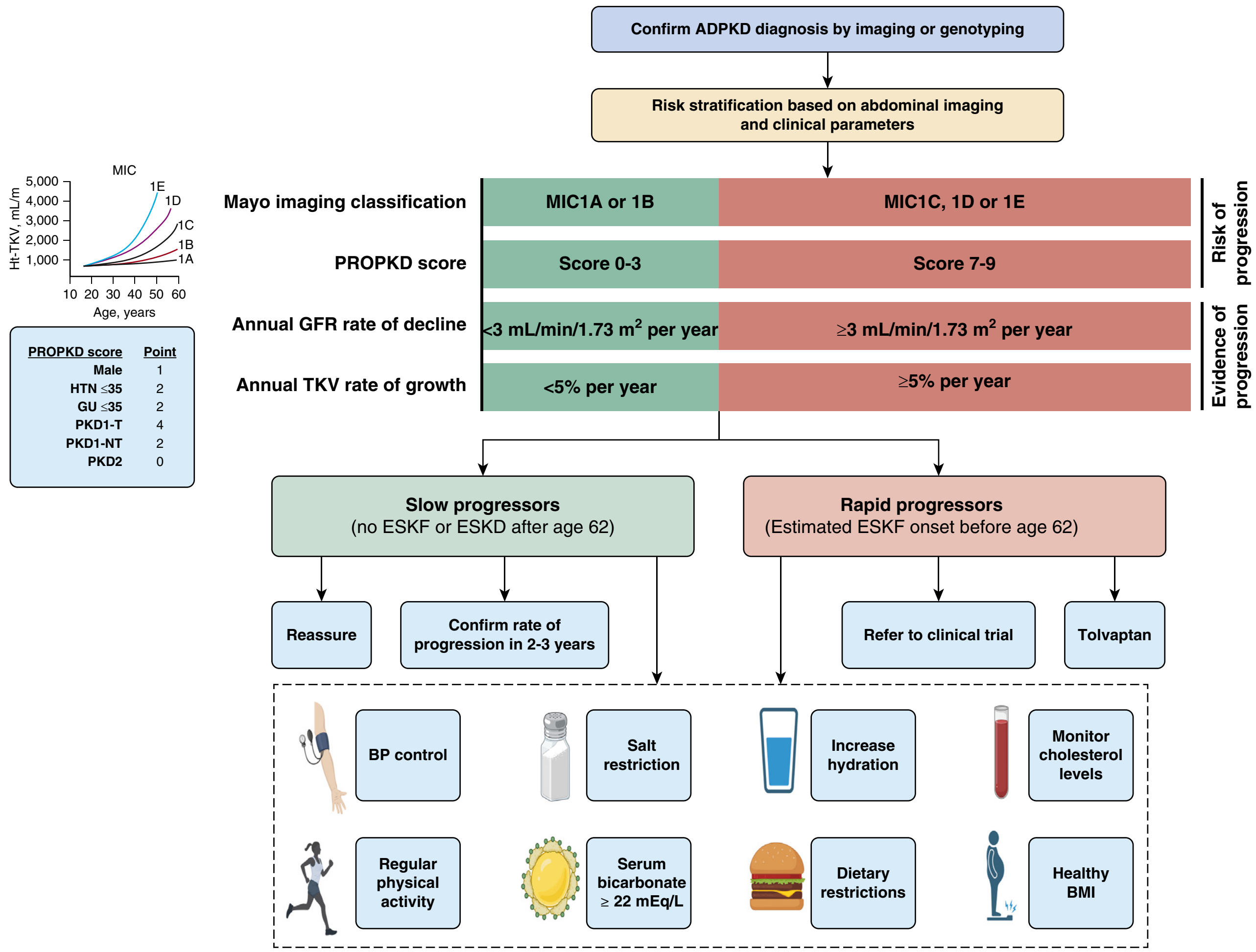


Fig. 45.13 Treatment diagram of patients with autosomal dominant polycystic kidney disease (ADPKD). ADPKD management starts with a confirmation of the diagnosis through imaging or genotype testing. Subsequently, patients are stratified based on abdominal imaging and clinical parameters. Slow progressors, classified as MIC 1A or 1B with a PROPKD score of 0 to 3, an annual GFR rate of decline <3 mL/min/SA per year, and an annual TKV rate of growth <5% per year, are identified. These individuals are expected to either not develop end-stage kidney disease (ESKD) or do so after the age of 62. Conversely, rapid progressors, categorized as MIC 1C, 1D, or 1E with a PROPKD score of 7 to 9, an EGFR rate of decline >3 mL/min/SA per year, and a TKV rate of growth >5% per year, require prompt intervention as ESKD is estimated before the age of 62. Management strategies differ accordingly: Slow progressors are reassured and their progression is reconfirmed in 2 to 3 years, while rapid progressors are referred to clinical trials and may be started on tolvaptan. Additionally, all patients are advised to adhere to lifestyle modifications including blood pressure control, regular physical activity, salt restriction, maintenance of serum bicarbonate levels >22, increased hydration, dietary adjustments, monitoring of cholesterol levels, and maintenance of a healthy body mass index (BMI).

progression. Tolvaptan was also shown to decrease symptoms, such as pain, hematuria, kidney stones, and UTIs. It also slightly reduces BP.

Treatment Guidelines

Tolvaptan is recommended for adults aged 18 to 55 with evidence of rapid disease progression based on imaging (e.g., MIC 1C, 1D, and 1E) or eGFR decline >3 mL/min/1.73 m^2 per year.[202-204] Post hoc analyses suggest benefits for patients aged 56 to 65 with rapid progression (historical GFR rate of decline >3 mL/min per year) and CKD stages 3–4.[202] Treatment is generally continued until KRT is required.[205]

The most common side effects of Tolvaptan include polyuria, excessive thirst, and nocturia due to aquaretic effects. Tolvaptan-induced reversible hepatic injury (5.6%) necessitates regular monitoring of liver function tests, monthly in the first 18 months of therapy and every 3 months thereafter.[201] The monitoring schedule involves biweekly testing for the first month, monthly testing for 18 months, and quarterly testing thereafter.

To mitigate nocturia and polyuria, dietary sodium restriction—particularly at the evening meal—can reduce osmolar load and aquaresis. Alternative strategies, such as adjunctive use of metformin or hydrochlorothiazide, have been explored to manage tolvaptan-induced polyuria, but long-term safety data remain insufficient to recommend these combinations.[206,207] Potential drug interactions with CYP3A inhibitors, such as certain antifungals, antibiotics, and antiretrovirals, must also be considered.[204] Additionally, patients should avoid factors that may increase the risk of liver injury including excessive alcohol consumption and high-dose acetaminophen. To minimize hepatic risks, it is recommended that patients temporarily discontinue

Tolvaptan during acute viral illnesses or events where significant alcohol intake is anticipated.

Patients should also be educated to temporarily withhold Tolvaptan in situations where they may be at risk for dehydration, such as during acute gastrointestinal illnesses (e.g., vomiting or diarrhea) or periods when fluid intake is restricted. Proper counseling ensures patients understand the importance of maintaining hydration and promptly resuming therapy once these conditions resolve.

Tolvaptan is contraindicated in pregnancy, lactation, uncorrected hyponatremia, history of liver injury, hypovolemia, and urinary obstruction.[205] When initiated early in adulthood with preserved GFR, Tolvaptan is projected to delay the onset of ESKF by approximately 3.1 years for every 13.7 years of therapy.[208] However, access to the medication may be influenced by regional health care policies and availability.

Management of Kidney Failure. Compared with other patients with ESKF, patients with ADPKD have a generally better prognosis. The preferred method of treatment is living donor kidney transplantation, ideally preemptive. Both forms of dialysis are used in ADPKD and should follow patient preference, if possible. Although less commonly used, peritoneal dialysis is not contraindicated in ADPKD. Treatment with hemodialysis is associated with better survival in patients with ADPKD when compared with other patients with other causes of ESKF.[209,210]

Indications for Nephrectomy. The timing and indications for nephrectomy vary among practices. The main indications include recurrent cyst infections or bleeding, severe and chronic pain, recurrent kidney stones, suspicious renal mass, and patient preference. Nephrectomy might be performed pretransplant for space considerations or at time of transplant with unilateral or bilateral nephrectomy. Otherwise, there is no absolute need to remove the cystic kidney as its volume tends to decline with time.[211] Nephrectomy is associated with perioperative morbidity, mortality, and risk for sensitization for future donors when blood transfusions are needed.[211,212] Nephrectomy can be either done before, at the time of, or after transplantation depending on local surgical expertise and preference.[105]

Dietary Interventions in ADPKD. Several dietary interventions, such as caloric restriction, intermittent fasting, time-restricted eating, and ketogenic diets, have emerged as potential strategies to slow the progression of ADPKD. These interventions aim to induce metabolic reprogramming, improve metabolic health, and reduce oxidative stress in ADPKD. However, their long-term effects and adverse events remain uncertain and require medical supervision.[213] To optimize the management of ADPKD, patients are advised to maintain a healthy body weight. This includes high fluid intake, low sodium, and limited concentrated sweets. Caloric restriction is particularly beneficial for overweight patients as it promotes weight loss and improves metabolic parameters.[213] Ketogenic dietary interventions, mild to moderate caloric restriction, and intermittent fasting have demonstrated potential in slowing disease progression and cyst growth in preclinical models by suppressing mTOR signaling and/or activating AMPK pathways, thereby triggering metabolic reprogramming.[213,214] Clinical studies on caloric restriction in patients with ADPKD are limited. However, obesity has been identified as an independent risk factor for ADPKD progression, especially in early stage patients. A pilot study comparing daily caloric restriction and intermittent fasting showed higher adherence and more pronounced weight loss with daily caloric restriction, potentially slowing ADPKD progression. Intermittent fasting, however, raised concerns regarding its tolerability and adverse effects.[214,215] While excess protein may contribute to osmolar load and vasopressin release, its association with annual changes in eGFR in ADPKD remains inconclusive.[216] Potassium-rich diets have shown associations with slower ADPKD progression, and caffeine consumption appears to have no significant impact on ADPKD outcomes.[217] While certain supplements like curcumin, ginkgolide B, and specific vitamins have shown potential in animal models, clinical evidence supporting their use in patients is limited. Clinical trials are needed to establish their benefits and safety.[213] The exploration of dietary interventions in children with ADPKD remains relatively understudied. Despite evidence from animal models indicating potential benefits of dietary protein restriction on kidney cyst volume and function, similar effects have not been observed in ADPKD adult humans, particularly in advanced disease stages.[218] However, caution is advised against excessive protein intake, although dietary protein restriction is not recommended for growing children including those with CKD.[219] The Dietary Approaches to Stop Hypertension program, emphasizing a balanced diet rich in fruits, vegetables, and low in sodium, may hold promise in promoting healthy eating habits and managing cardiovascular risks in children with ADPKD.[220]

Emerging Therapeutic Agents in ADPKD. Several therapies have been investigated in patients with ADPKD, with varying outcomes (Table 45.9). While most previously explored therapies showed positive results in preclinical studies, several clinical trials using those regimens showed nonsignificant or conflicting results. For instance, clinical trials on mTOR inhibitors (e.g., sirolimus and everolimus) failed to have significant effects on kidney function decline. Somatostatin analogs (e.g., lanreotide and octreotide) had conflicting results; while some trials (e.g., ALADIN) reported reductions in TKV growth, others (e.g., ALADIN 2 and DIPAK 1) did not have significant effects on eGFR decline. GCS inhibitors (e.g., Venglustat) lacked efficacy in clinical trials, leading to study termination. Nrf2 activators (e.g., bardoxolone) initially showed promising results in improving kidney function but were later discontinued (PHOENIX and FALCON trials). Currently, several emerging therapies are under investigation. CFTR modulators (e.g., GLPG2737) were studied in a phase 2 trial, assessing their safety and tolerability in patients with rapid disease progression. miRNA inhibitors (e.g., RGLS4326 and RGLS8429) are in early phase studies. Pioglitazone, though safe in nondiabetic ADPKD patients in phase 1 trials, requires further studies to determine its effects on TKV and kidney function. Lastly, tyrosine kinase inhibitors (e.g., tesevatinib and bosutinib) are being evaluated for their potential to reduce kidney growth, though their clinical use

Table 45.9 Emerging Therapeutics in Cystic Kidney Diseases: Mechanisms, Preclinical Evidence, and Clinical Insights

Categories	Mode of Action	ADPKD Preclinical Studies	Clinical Trials
mTOR Inhibitors[419] Everolimus, Sirolimus	Inhibition of MDM2 prevents TP53 degradation and increases p21 expression, decrease cell proliferation and activation of apoptosis	Induced disease regression by activating autophagy in ADPKD cells	The sirolimus study[420] did not show a significant change in TKV or eGFR. However, the urinary albumin excretion rate was higher in the sirolimus group Everolimus slows TKV growth in patients with ADPKD but does not prevent kidney impairment progression
CFTR Modulators VX-809 GLPG2737	CFTR chloride channel is responsible for driving net fluid secretion into the cysts, promoting cyst growth. Studies show that it is regulated by AVP[421]	An animal model of slowly progressing cyst formation typical of human ADPKD that VX-809 reduces the growth of already established cyst[422]	For VX809, a phase 2, double-blind, placebo-controlled followed by 1-year open-label phase but was terminated early due to lack of efficacy (ClinicalTrials.gov ID NCT04578548) Phase 2 trial to assess safety and tolerability of GLPG2737 in ADPKD patients at risk of fast progression
Biguanide Analogs	Metformin inhibits the mTOR and CFTR pathways and activates AMPK[100]	Metformin has been associated with reduced kidney cyst growth in an ADPKD mouse mode[100] Diminishes leukocyte infiltration and downregulates inflammatory markers and kidney injury markers as well as accumulation of ECM in rat kidney tubular epithelial cell line NRK-52E, thus decreasing fibrotic changes in ADPKD[423]	Phase 2 showed no significant impact on TKV growth rate or kidney function. In the present analysis, metformin also showed a favorable trend, but it was not significant, possibly because of a small sample size. IMPEDE-PKD phase 3 study aims to include a larger cohort[100]
miRNA Inhibitors Anti-miR-17 Anti-miR-21	miR-17 rewires cyst epithelial metabolism to enhance cyst proliferation	anti-miR-17 demonstrated cyst-reducing effects, but no overt toxicity, in a second 6-month, preclinical trial involving a slow cyst-growth mouse mode [424].	A phase 1, with anti-miR-21, clinical study showed some changes in polycystin 1 and -2 levels in urinary exosomes Current study is completed (ClinicalTrials.gov ID NCT04536688) Next phase study of Anti-mR-17 is under design at this point.
Tesevatinib (KD019)	Multi-kinase inhibitor targets multiple abnormal signal transduction events in PKD. Also targets abnormal angiogenesis necessary for cyst growth [425].	In vivo pharmacological inhibition of multiple kinase cascades with tesevatinib reduced phosphorylation of key mediators of cystogenesis: EGFR, ErbB2, c-Src, and KDR which resulted in reduction of kidney and biliary disease in both bpk and PCK mice models of ARPKD[425].	Phase 2 trial with frequent adverse events, including QT-prolongation. Study aiming to measure changes in htTKV was stopped 2 years ago[425,426].

Continued on following page

Table 45.9 Emerging Therapeutics in Cystic Kidney Diseases: Mechanisms, Preclinical Evidence, and Clinical Insights (Continued)

Categories	Mode of Action	ADPKD Preclinical Studies	Clinical Trials
Somatostatin Analogs	Lanreotide and Octreotide bind to SSTRs inhibiting AC activity and reduce cAMP production by maintaining intracellular Ca2+ levels[427].	-	Showed conflicting results in clinical trials. Aladin trials showed reduced TKV growth but no eGFR decline whereas DIPAK-1 trial showed no significant impact on eGFR decline[428,429]. Several clinical studies have shown that somatostatin analogs inhibit not only kidney cysts but also hepatic cyst growth [427].
Glucosylceramide Synthase (GCS) Inhibitors Venglustat AL01211	Designed to reduce the production of glucosylceramide (GL-1) and thus is expected to substantially reduce formation of glucosylceramide-based glycosphingolipids[430].	Reduced cyst growth and preserved kidney function in animal studies	Venglustat trial was stopped due to insufficient effectiveness (ClinicalTrials.gov ID NCT04908462) AL01211 is currently undergoing phase 1 clinical study ClinicalTrials.gov ID (NCT03523728)
Nrf2 Activators Bardoxolone Obacunone	Obacunone is a potent antioxidant that activates Nrf2 leading to suppression of lipid peroxidation and reduce cell proliferation by downregulating mTOR and MAPK signaling pathway[431].	In vitro obacunone significantly inhibited cyst formation and expansion of MDCK cysts and embryonic kidney cysts in a dose-dependent manner In vivo there was significant reduction in kidney cyst formation [431].	Unpublished PHOENIX study indicated kidney function improvement. FALCON trial aimed to assess safety and efficacy but was eventually terminated (Clinicaltrials.gov NCT03366337, NCT0391844)
Statin Therapy	Inhibits 3-hydroxy-3-methyl-glutaryl coenzyme A reductase reduces the farnesylation and activation of RAS guanosine triphosphate (GTP)-binding proteins that are important in numerous cellular functions, including the regulation of cellular proliferation[432]	Animal models of ADPKD have shown that statin treatment decreases cyst formation, preserves kidney blood flow, and mitigates interstitial inflammation[433]	Trials examining pravastatin's effect on TKV and combined with sodium citrate are still ongoing[206,434] The results of the pilot trial in adults with ADPKD, (presented at ASN 2024), did now show significant change in eGFR by using statins.

has been complicated by frequent adverse events including QT-prolongation.

Counseling, Pregnancy, and Reproductive Issues. Family planning is an important discussion with families with ADPKD.[46,221] The decisions revolve around the risk of the child having ADPKD (which is 50%), whether the couple would want genetic testing of the fetus, and what the clinical implications of this information would be in a pregnancy. Preimplantation genetic testing for monogenic disorders (PGT-M) is a potential solution for couples affected by ADPKD who aim to have children without passing on the disease. PGT-M screens IVF embryos for monogenic variants to identify and transfer unaffected embryos, reducing the risk of transmitting the disease to offspring. However, this approach presents challenges, such as high costs, specialized expertise requirements, and ethical dilemmas related to embryo selection and disposition.[222] This technique involves the screening of embryos created through in vitro fertilization (IVF) to identify ADPKD pathogenic variants before implantation.[223] Use of PGT-M for ADPKD poses challenges, however, primarily attributable to the nature of the *PKD1* gene situated in a region with multiple pseudogenes.[222] Overcoming these complexities requires specialized methodologies like karyomapping.[224] In vitro fertilization itself carries risks for the mother. In addition, it is important to discuss the risks to a mother with ADPKD with regard to progression of kidney function and pregnancy-associated complications. In women with ADPKD who have normal BP and kidney function, pregnancy typically proceeds favorably. However, they should be aware of a slightly elevated risk of developing pregnancy-induced hypertension and preeclampsia.[222] A history of multiple pregnancies (more than three) has been linked to a higher rate of GFR decline in ADPKD.[46,225] Similar to all women with CKD, pregnant women with ADPKD and reduced kidney function face an increased risk of early fetal loss, accelerated kidney function decline, and challenges in BP management. As a result, it is recommended to refer them to an obstetrician with expertise in managing high-risk pregnancies.[226] In the case of women with ADPKD of reproductive age, especially those with severe PLD, it is crucial to provide guidance regarding the potential exacerbation of their liver condition when exposed to estrogen.

ADPKD in Children

Diagnosis of ADPKD in children. Most cases of ADPKD typically manifest in adulthood after an asymptomatic phase through childhood. However, a minority of individuals experience early onset and rapidly progressive kidney disease. It is hypothesized that the early onset of ADPKD is inversely related to the level of polycystin function,[68] which results from rare genetic combinations including biallelic variants involving at least one hypomorphic *PKD1* or *PKD2* allele.[227,228] Additionally, the presence of an ADPKD allele combined with an allele from another cystic gene, such as TSC2 (in contiguous gene deletion syndrome), and *HNF-1β* can cause the early onset and progression of the disease.[68,112] The age spectrum of ADPKD thus includes typical adult manifestations, early onset disease (before 15 years, known as ADPKD_{EO}), and very early onset ADPKD (before 18 months, labeled as $\text{ADPKD}_{\text{VEO}}$).[229] Diagnostic criteria for $\text{ADPKD}_{\text{VEO}}$ include findings observed both in utero and during infancy. In utero, signs include oligohydramnios and hyperechoic enlarged kidneys (defined as kidney length >2 standard deviations). Between birth and 18 months, the diagnosis requires enlarged palpable kidneys in combination with at least one additional criterion: BP >95th percentile (or use of antihypertensive therapy) or eGFR <90 mL/min/1.73 m^2 or persistent and overt proteinuria.[71,230] Early onset ADPKD (ADPKD_{EO}), by contrast, applies to children between 18 months and 15 years of age) with diagnostic criteria requiring at least one of the following: enlarged palpable kidneys, elevated BP >95th percentile (or use of antihypertensive therapy), GFR <90 mL/min/1.73 m^2, or persistent and overt proteinuria.[71] Kidney US is the preferred method for screening children at risk for ADPKD, and the detection of one or more kidney cysts on a sonogram is suggestive of an ADPKD diagnosis in a child younger than the age of 15 years with a positive family history. Repeat US is warranted in an individual with positive family history but negative initial US findings.[40] In the absence of family history and presence of cysts, parents and grandparents should be screened for ADPKD.[108] While a negative US result does not rule out ADPKD, genetic testing using advanced sequencing techniques is advised for children with early onset or atypical ADPKD presentations or those with cystic kidneys but no family history.[108] Screening for ADPKD in children younger than 18 years of age is a subject of debate. Some argue that the potential adverse consequences of a positive diagnosis in young, asymptomatic individuals outweigh the benefits. In contrast, others suggest that early diagnosis provides an opportunity for optimal anticipatory care, such as BP control, and the future possibility of benefiting from new therapies. When appropriate, teenagers and informed younger children should be actively involved in discussions and decisions about screening.[40]

Manifestations of ADPKD in children. In children with ADPKD, the clinical manifestations often present challenges in diagnosis due to their variable nature: They may vary from mild or asymptomatic to severe disease that can mimic ARPKD (discussed later). These manifestations are generally nonspecific and include symptoms like generalized abdominal, flank, or back pain, but patients may also develop cyst infections or cyst bleeding.[229] Although symptoms like abdominal, flank, or back pain are reported in a considerable proportion (between 16% and 30%) of children with ADPKD, their occurrence is not significantly different from that in the general pediatric population.[231] One study showed that gross hematuria was observed in 10% to 14% of children before the age of 16, while polyuria, urinary frequency, and enuresis (reflecting reduced urine concentrating ability) were the most reported symptoms, affecting around 58% of individuals.[229,232,233] Approximately 20% of children with ADPKD exhibit mild proteinuria.[231] However, these numbers should be interpreted cautiously, especially in the absence of large comparative studies, as they can be similar to those found in the general population, and hematuria might be a consequence of UTIs.[234,235] Asymptomatic ADPKD remains the most common pattern. Approximately 20% to 40% of children showing signs of hypertension defined as BP >95th percentile for age, sex, and height, or ≥130/80 mm Hg in adolescents older than 13.[231,232,236] Notably, even though these hypertensive patients may be asymptomatic at the time of diagnosis, they are at an increased risk for cardiovascular events and often exhibit an elevated left ventricular mass index.[160,237] Children with ADPKD and elevated or borderline BP (>75th percentile) exhibited significantly higher rates of LVH compared with children with lower blood pressure.[237,238] Moreover, findings from a 24-hour BP monitoring study conducted in pediatric patients with ADPKD unveiled a notable prevalence of isolated nocturnal hypertension. This indicates that relying solely on office BP measurements may underestimate the actual occurrence of hypertension in this population. Hypertensive children diagnosed with ADPKD are likely to face a more rapid prospective decline in kidney function, kidney growth, and cyst development compared with their normotensive counterparts.[239,240] Extrarenal manifestations are rarely observed in young patients.[229,241] Hepatic manifestations are more common in adults than children and if present in the pediatric population, they typically do not cause symptoms.[242] Structural heart abnormalities are also uncommon in children with ADPKD.[243]

Children with $ADPKD_{VEO}$ experience an unusually rapid progression of the disease. These children are at risk of loss of kidney function at a younger age than those with non-VEO ADPKD.[244] Severe ADPKD can manifest in neonates with clinical findings resembling those seen in ARPKD. Neonates with ARPKD can display congenital hepatic fibrosis and characteristic ultrasound features like hepatomegaly, increased liver echogenicity, and dilation of peripheral intrahepatic ducts and the main bile ducts, which can help differentiate them from patients with ADPKD. However, only 40% of neonates with ARPKD have clinical evidence of liver involvement, so the absence of liver findings does not rule ARPKD.[40]

Management of ADPKD in Children

Management of ADPKD in children depends on the clinical manifestations: Asymptomatic patients may not require treatment before adulthood, while early symptomatic patients (such as those with hypertension) may require early treatment.[108] The use of tolvaptan in children with ADPKD is still under investigation.[245] A phase 3 clinical trial by Mekahli and colleagues[245] showed the effect of tolvaptan in slowing kidney volume growth and the decline in eGFR over the period of 12 months, although these findings were not statistically significant, and there were no reports of drug-induced hepatic injury.

Ambulatory BP monitoring is the preferred method for diagnosing high BP and assessing treatment effectiveness in children aged 5 years and older.[246] It provides a more accurate prediction of target organ damage compared with office measurements and can identify conditions like isolated nocturnal hypertension or nondipping patterns. If ambulatory BP monitoring is not available, routine office or home BP monitoring can be used as alternative methods.[247] In children with ADPKD and hypertension, RAAS blockade is the preferred drug class to control high BP because it slows kidney function decline and cardiovascular disease progression.[248,249] Studies suggest that targeting BP at the 50th percentile using RAAS inhibitors can help maintain kidney function and reduce LVM index.[239] While comparative studies in this specific population are limited, expert consensus supports the use of ACEI or ARBs due to their efficacy and generally favorable side effect profiles.[250,251]

While no randomized controlled trials specifically targeting lifestyle interventions in ADPKD children exist, general guidelines for pediatric populations and those with ESKF apply.[108] Obesity and metabolic disturbances play a significant role in the pathogenesis of ADPKD in the pediatric population. These prevalent issues further exacerbate glomerular hyperfiltration, leading to a faster decline in kidney function and an increase in kidney volume. Additionally, they heighten the risk of hypertension in obese children.[252,253] Maintaining a healthy weight is essential, as obesity can accelerate kidney function decline in ADPKD.[254] Salt intake in the pediatric population consuming a western diet exceeds the recommended amounts.[255] Therefore reducing salt intake is crucial, given its association with kidney growth and disease progression.[256] Although hydration is vital, the benefits of excessive water intake remain inconclusive.[257,258] Furthermore, a balanced diet that includes moderate protein is recommended to prevent malnutrition.[259,260]

Leuven Imaging Classification

A longitudinal study involving pediatric patients diagnosed with ADPKD younger than the age of 19 aimed to assess the utility of htTKV measured via a 3D US to discern rapidity of disease progression in children.[261] Applying MIC to the pediatric population led to underestimation of the disease severity, especially in patients younger than 10. Subsequent adjustments to the MIC model parameters resulted in an improved distribution of patients across severity scores; however, it still failed to adequately include the full range of pediatric ages.[261] Consequently, a predictive model was introduced, incorporating a formula based on htTKV (mL/m) = $A \times B^{(age\hat{}1.6)}$, with *A,* the starting htTKV at age 0 and *B,* the yearly htTKV % increase. This new model was able to better stratify the pediatric patients based on severity of the disease. Further validation using Mayo Clinic and CRISP data for the same age groups confirmed that the LIC model was better at sorting children into different risk levels compared with the MIC model.[261]

Prognosis of ADPKD in Children

Most children with ADPKD typically maintain adequate kidney function until their fourth decade of life. In some cases, glomerular hyperfiltration conceals the loss of GFR and is associated with a more significant increase in TKV, which leads to a faster GFR decline in the future.[249] In fact, hypertensive children with ADPKD are more likely to experience a decline in kidney function when compared with normotensive children with ADPKD.[239] A small number of symptomatic children are at risk for rapidly progressive disease that ultimately results in ESKF requiring KRT, ideally preemptive transplantation.[71]

Atypical ADPKD

ADPKD phenotypic spectrum entails a wide range of diseases with variable phenotypic and genotypic characteristics associated with monoallelic causes of cystic kidney disease (e.g., *PKD1, PKD2, GANAB, DNAJB11, IFT140, ALG5, ALG6, ALG8, ALG9, NEK8,* and monoallelic *PKHD1*).[63,262] The various genotypes are summarized in Table 45.4.

ADPKD-IFT140. IFT140, located on chromosome 16p13.3, is part of the IFT-A core complex, essential for retrograde protein trafficking within cilia. Dysfunction in IFT140 disrupts this process, causing ciliopathies such as cranioectodermal dysplasia and short-rib thoracic dysplasia 9 *(SRTD9)* in an autosomal recessive pattern.[59]

Monoallelic *IFT140* variants account for 1% of ADPKD cases.[62] Individuals with ADPKD-*IFT140* typically have asymmetrically enlarged kidneys with a few large, exophytic cysts contributing to TKV.[59,62] These patients typically have slow decline in GFR and lower risk of ESKF.[56,59]

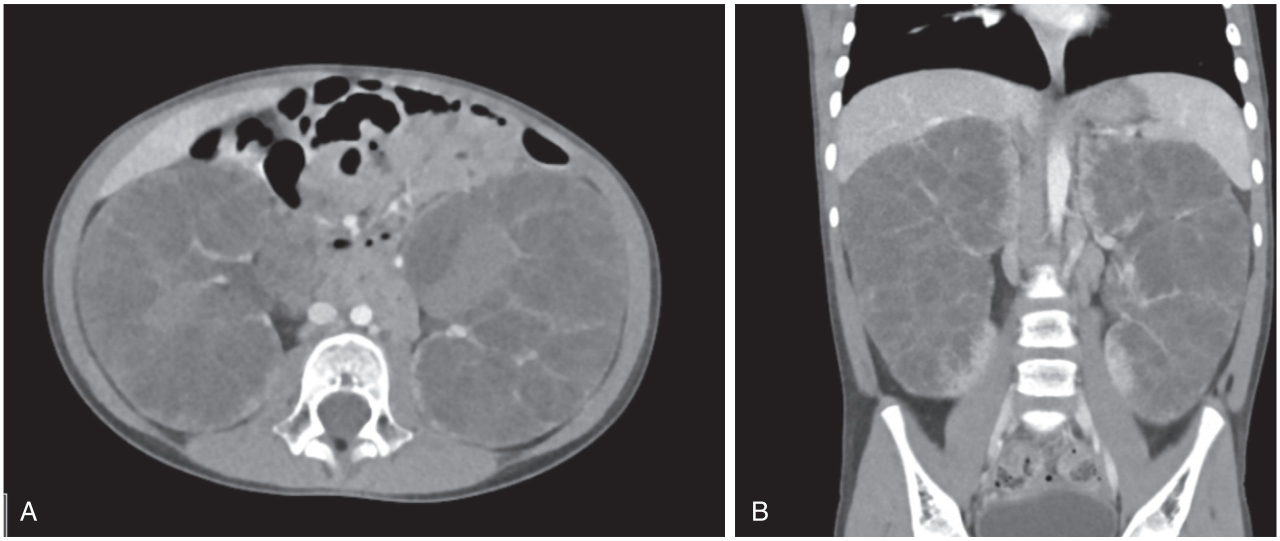


Fig. 45.14 Computed tomography scans (A, cross-sectional cut vs. B, coronal cut) of a 5-year-old female patient with NEK8 variant and eGFR of 25 mL/min/1.73 m^2, manifesting with early onset autosomal dominant polycystic kidney disease.

DNAJB11. Pathogenic variants in *DNAJB11* cause a mixed phenotype of ADPKD and autosomal dominant tubulointerstitial kidney disease (ADTKD).[263] Located in chromosome 3q27,[264] *DNAJB11* regulates ER protein folding, trafficking, and glycosylation. DNAJB11 functional disruption reduces PC1 expression in cystic kidneys, contributing to disease.[262,265,264] ADPKD-*DNAJB11* phenotype is distinct from *PKD1*-related disease. It features small kidney cysts, ESKF after the sixth decade, and significant interstitial fibrosis with milder cyst burden.[262]

GANAB. *GANAB* encodes the catalytic α subunit of the glucosidase II enzyme, while *PRKCSH* (ADPKD gene) encodes its β subunit. Together, these form an ER-associated N-linked glycan-processing enzyme implicated in the pathogenesis of ADPLD.[266,267] Pathogenic variants in *GANAB* affect the cell surface localization of PC1 and PC2, contributing to PKD and PLD phenotypes.[268] The ADPKD-GANAB phenotype is characterized by a smaller number of kidney cysts (10–15 cysts on average) and may have mild to severe PLD phenotype. Kidney function decline is slower than ADPKD-*PKD2*.

ALG5, ALG8, ALG9. These genes are involved in the N-linked glycosylation of proteins, crucial for their proper secretion and localization. Their disruption affects PC1 glycosylation and leads to kidney and liver cystic phenotypes. *ALG5* alters oligosaccharide synthesis, impacting PC1 glycosylation. ADPKD-*ALG5* typically presents with late-onset ESKF (after 60), kidney atrophy, and multiple cysts, but kidney function often remains normal until age 50. It may also be associated with PLD and colonic diverticulosis.[60,269] *ALG8* encodes an ER resident protein essential for PC1 glycosylation.[270] *ALG8* has milder kidney involvement compared with *PKD1* or *PKD2* variants. While liver cysts are less common,[271] kidney stones are more frequent.[272] Individuals with ALG8 pathogenic variants do not appear to have an increased risk of ESKF.[272] *ALG9* plays a role in kidney and biliary epithelial homeostasis.[273] ADPKD-*ALG9* presents with milder kidney cysts compared with *PKD1* or *PKD2*, but individuals older than 50 exhibit high cyst penetrance (88%).[274] Nephrolithiasis rates are similar to those with typical ADPKD.[274]

NEK8. Certain pathogenic variants in *NEK8* have been associated with early onset forms of PKD (Fig. 45.14). Monoallelic variants primarily cause kidney phenotypes, resembling ADPKD, while biallelic variants result in severe syndromic ciliopathies with extrarenal features.[275,276] Specific *NEK8* variants, like p.Arg45Trp and p.Lys157Gln, are linked to childhood ESKF, often requiring bilateral nephrectomies. These presentations overlap with ARPKD and ADPKDveo but align more closely with ADPKD inheritance and cyst localization.[110]

TUMOROUS SYNDROMES

TUBEROUS SCLEROSIS COMPLEX

Epidemiology, etiology, and pathogenesis of tuberous sclerosis complex

Tuberous sclerosis complex (TSC) is a genetic syndrome characterized by pathogenic variants in *TSC1* (chromosome 9q34) or *TSC2* (chromosome 16p13), affecting 1 in 6000 individuals. *TSC1* encodes hamartin, while *TSC2* encodes tuberin.[277] Together, they form a complex that regulates cell size and organ growth via inhibition of the mTOR pathway. Variants in either gene lead to constitutive mTOR activation, driving abnormal cellular proliferation and tumor development.[278,279]

Diagnosis and manifestations of tuberous sclerosis complex

TSC diagnosis requires specific major features (renal angiomyolipomas [AML], facial angiofibromas, nontraumatic ungual or periungual fibromas, hypomelanotic macules, shagreen patches, retinal nodular hamartomas, cerebral

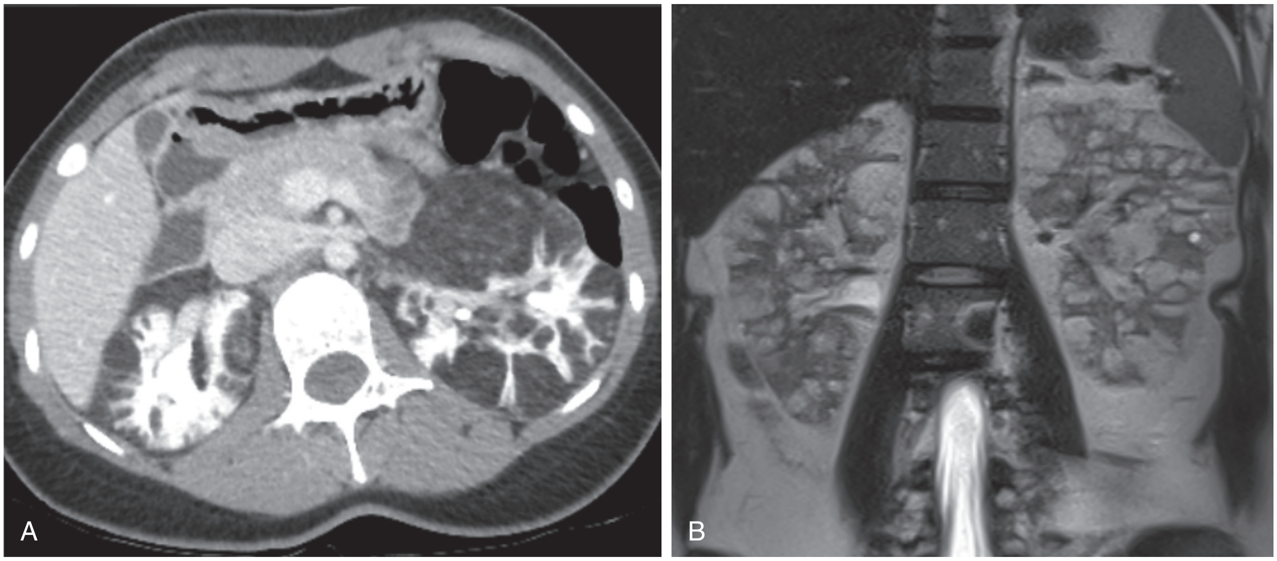


Fig. 45.15 Imaging representation of a 33-year-old woman diagnosed with tuberous sclerosis complex complex (eGFR at imaging of 60 mL/min/1.73 m^2) on computed tomography scan (A) and magnetic resonance imaging (B) (*arrow* showing angiomyolipomas).

cortical tubers, subependymal nodules, subependymal giant cell astrocytomas, benign cardiac rhabdomyomas, and pulmonary lymphangioleiomyomatosis) or minor features (renal cysts, nonrenal hamartomas, hamartomatous rectal polyps, retinal achromic patches, cerebral white matter radial migration tracts, bone cysts, gingival fibromas, and "confetti" skin lesions) (Fig. 45.15).[280] Abdominal MRI is preferred for evaluating AMLs and detecting lipid-poor lesions.[281,282] Additional imaging may reveal extrarenal findings like aortic aneurysms and neuroendocrine tumors.[283]

Kidney involvement includes AMLs, simple cysts, oncocytomas, clear cell RCCs, lymphangiomatous cysts, and, rarely, focal segmental glomerulosclerosis. AMLs consist of abnormal vessels, smooth muscle, and fat. Epithelioid AMLs, although rare, may undergo malignant transformation. Renal cysts occur in up to 32% of patients, often asymptomatic and without the need for routine surveillance unless associated with AMLs.[280,284]

TSC2/PKD1 contiguous gene syndrome (CGS), resulting from deletions of *TSC2* and *PKD1* on chromosome 16p13, causes early onset ADPKD with more severe renal manifestations (Fig. 45.16) including hypertension and ESKF by the second or third decade.[285,286] In contrast, non-CGS TSC patients typically have milder renal cystic disease.[287]

TSC is associated with two main manifestations: 1. Cystic kidney disease: Small, asymptomatic renal cysts are common, and often discovered incidentally. Surveillance focuses on AMLs due to their growth potential and RCC risk. TSC2/PKD1 CGS requires high suspicion in young patients with hypertension and early onset cystic disease[278,288]; 2. Glomerulocystic kidney disease: Rare and typically diagnosed in neonates, it presents with segmental lesions and enlarged kidneys with cysts.[278,288]

Disruptions in mTOR signaling and primary cilia deflection contribute to renal cystic diseases in TSC, which are second only to AMLs as common renal findings.[278,280,284,289,290]

Fig. 45.16 Imaging representation of a 2-year-old male patient with *TSC2/PD1* continuous gene syndrome (eGFR at imaging of 108 mL/min/1.73 m^2).

Management of tuberous sclerosis complex

AMLs often do not require immediate treatment. They warrant regular monitoring (every 1–3 years) with abdominal MRI. Interventions, such as selective arterial embolization, radiofrequency ablation, or cryoablation, are reserved for symptomatic AMLs, particularly those >4 cm due to bleeding risks.[277,280] Renal-sparing surgery is preferred to preserve kidney function. mTOR inhibitors like everolimus significantly reduce AML volume and are recommended for large or symptomatic AMLs.[291] The EXIST-2 trial demonstrated the efficacy of everolimus in reducing AML size, though long-term risks and benefits require further

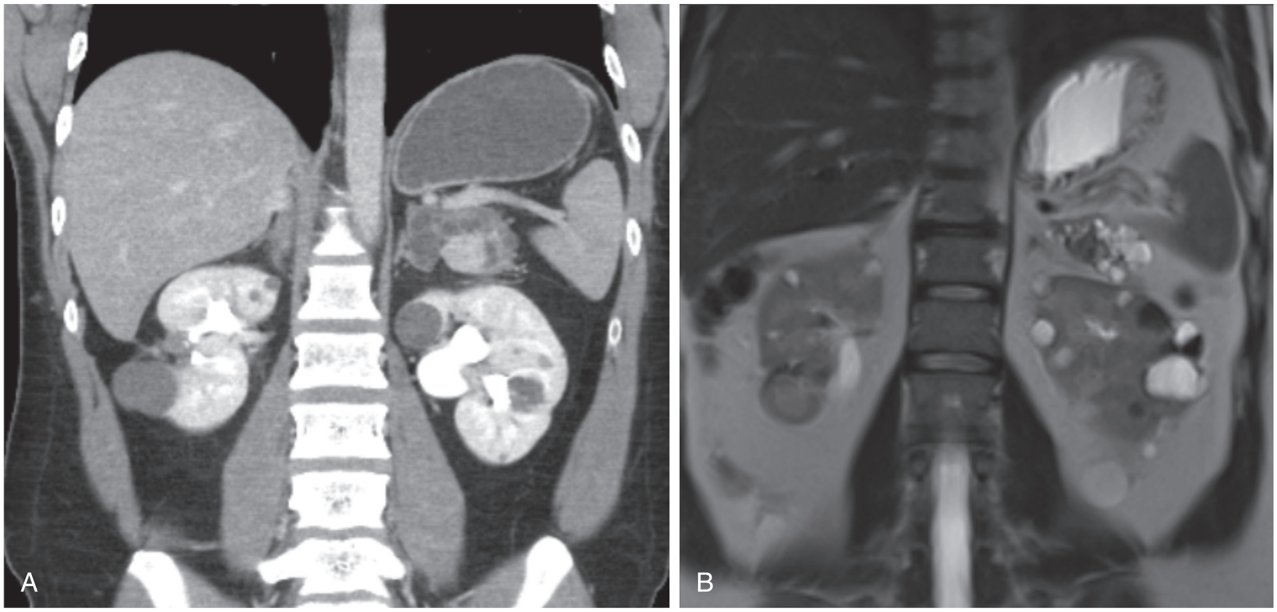


Fig. 45.17 Imaging representation of a 38-year-old male with Von Hippel-Lindau disease on computed tomography scan (A) and magnetic resonance imaging (B) (estimated glomerular filtration rate at imaging of 85 mL/min/1.73 m^2).

study.[292] For cystic disease, managing hypertension and monitoring for ESKF are priorities. In cases requiring kidney transplantation, bilateral nephrectomy may precede the procedure to mitigate hemorrhage and RCC risks.[277] While mTOR inhibitors show promise for AMLs, their efficacy in ADPKD-related cystic disease remains inconsistent. Adaptive mechanisms may explain paradoxical increases in cystic and AML volumes during prolonged treatment, as observed in some cases.[287,293,294]

VON HIPPEL-LINDAU SYNDROME

Etiology, epidemiology, and pathogenesis of Von Hippel-Lindau syndrome

Von Hippel-Lindau (VHL) syndrome is a rare genetic disorder caused by variants in the *VHL* tumor suppressor gene on chromosome 3p25-p26. It is characterized by benign and malignant growths in the eyes, brain, spinal cord, adrenal glands, and kidneys. Renal manifestations include benign cysts and clear cell carcinoma (RCC), with cysts present in 50% to 66% of patients, typically emerging in the third or fourth decade of life. These cysts are often asymptomatic and rarely lead to ESKF.[295,296]

The *VHL* gene encodes pVHL proteins (about 30 kDa and 19 kDa) that suppress RCC growth by degrading HIF-α subunits. Under normoxic conditions, pVHL targets HIF-Iα and HIF-2α for degradation. In hypoxia, stabilized HIF-α translocates to the nucleus, promoting angiogenesis, erythropoiesis, and mitogenesis.[295] Although HIF activation is necessary for VHL-associated tumors, it is not sufficient, as evidenced by Chuvash polycythemia (caused by homozygous *VHL* missense variant), which features elevated HIF but lacks typical VHL tumors.

Additional cellular functions of pVHL include apoptosis regulation, cilia maintenance, and extracellular matrix deposition. Loss of heterozygosity at the *VHL* locus in renal cysts from VHL patients suggests cyst formation is an early step in RCC pathogenesis. Renal cysts and RCC cells in VHL often exhibit reduced or rudimentary cilia.[295,296]

Diagnosis and manifestations of Von Hippel-Lindau syndrome

VHL is diagnosed through clinical criteria (e.g., central nervous system [CNS] or retinal hemangioblastomas, RCC, or family history) and confirmed via genetic testing, which has near-100% sensitivity.[297,298] Screening begins with abdominal ultrasound by age 8, advancing to MRI at age 16 if abnormalities are detected. Regular imaging (MRI or CT, every other year) is essential for monitoring visceral lesions in the kidneys, pancreas, adrenals, brain, and spine (Fig. 45.17). Renal cysts often precede RCC, which has a mean onset at age 35 and affects up to 70% of VHL patients by age 60. RCCs in VHL are typically clear cells, low-grade, and more likely to be multicentric and bilateral.[296,297] Metastatic RCC is the leading cause of death in VHL.[297,299] Early detection of asymptomatic RCCs can improve outcomes.

Management of VHL

Effective management of VHL syndrome requires early detection and close monitoring of complications including RCC and CNS lesions. Recommended surveillance includes annual physical and ophthalmologic examinations, blood or urinary catecholamine/metanephrine levels, and periodic imaging (ultrasound, MRI, or CT).[296] Renal lesion management aims to preserve renal function while minimizing invasive procedures. For tumors ≥3 cm, partial nephrectomy is the preferred approach based on the National Cancer Institute's "3-cm rule," as smaller tumors rarely metastasize.[297,299] Minimally invasive therapies, such as radiofrequency ablation and cryotherapy, are effective for smaller lesions and require regular follow-up. Targeted therapies including mTOR inhibitors (e.g., temsirolimus and everolimus) and tyrosine kinase inhibitors (e.g., sunitinib) have shown potential in treating VHL-associated tumors by inhibiting angiogenesis and tumor growth through VEGF, PDGF, and TGF-β signaling pathways.[296,298]

HYPERPARATHYROIDISM JAW TUMOR SYNDROME

Hyperparathyroidism-jaw tumor syndrome (HPT-JT) is a rare autosomal dominant disorder with incomplete penetrance, meaning not all individuals with the genetic variant will develop symptoms. This syndrome is primarily associated with germline variants in the *CDC73* gene and is characterized by parathyroid tumors, ossifying fibromas of the mandible and maxilla, uterine hyperplasia or neoplasms, and various renal abnormalities, both cyst and neoplastic.[300,301]

Primary hyperparathyroidism is the most common and often the initial clinical manifestation, while jaw tumors occur in about one-third of cases.[300,302] Renal involvement is less frequent, occurring in approximately 15% of individuals with HPT-JT. These renal manifestations range from benign cystic lesions to malignant tumors and may be unilateral or bilateral. Cystic kidneys are the most common finding, but patients may also develop hamartomas, Wilms tumors, or mixed epithelial-stromal tumors. The complex renal presentation often necessitates multiple abdominal surgeries during a patient's lifetime to address renal masses.[300,301,303] Proactive surveillance is critical for managing the diverse manifestations of HPT-JT. Recommended screening includes renal imaging (ultrasound, CT, or MRI) at diagnosis, with follow-up imaging every 5 years if no abnormalities are initially detected.[300,301]

BIRT-HOGG-DUBE SYNDROME

BHD syndrome is a rare autosomal dominant disorder caused by variants in the *FLCN* tumor suppressor gene on chromosome 17p11.2, which encodes folliculin.[304,305] Although the precise function of folliculin remains unclear, it likely regulates cell growth and survival through interactions with the mTOR signaling pathway.[306,307] The syndrome is characterized by cutaneous fibrofolliculomas, pulmonary and kidney cysts, recurrent spontaneous pneumothoraces, and various renal tumors.[308,309] Pulmonary involvement is a hallmark of BHD, with more than 80% of patients developing multiple bilateral pulmonary cysts, significantly increasing the risk of spontaneous pneumothorax. These cysts are thought to arise from dysregulated signaling pathways and impaired cell-cell adhesion.[308,310,311] Skin manifestations including fibrofolliculomas and trichodiscomas typically emerge in the third or fourth decade of life and are most commonly observed in Caucasian individuals.[312,313]

Renal manifestations are a major concern in BHD, with cysts occurring in 15% to 26% of patients and renal tumors in about 30%.[309] These renal tumors are frequently multifocal and bilateral, ranging from benign oncocytomas to malignant clear-cell renal carcinomas.[314,315]

Management involves regular renal surveillance using MRI, starting at age 20, with follow-up every 3 to 4 years.[316,317] Nephron-sparing surgeries are recommended for BHD-associated renal tumors to preserve kidney function. In cases of metastatic disease, mTOR inhibitors like everolimus have shown potential, though further research is needed to establish their efficacy.[308]

AUTOSOMAL DOMINANT TUBULOINTERSTITIAL KIDNEY DISEASE (PREVIOUSLY KNOWN AS MEDULLARY CYSTIC KIDNEY)

Epidemiology, Etiology, and Genetics of ADTKD

ADTKD is a rare autosomal dominant kidney disease characterized by slow progression of CKD and eventual ESKF. It manifests with impaired kidney function emerging in adolescence or early adulthood, with ESKF onset ranging from 20 to 70 years (median age 45–48). The disease affects both sexes equally and lacks geographic specificity, making prevalence estimation challenging.[318]

ADTKD is associated with variants in several genes, each contributing to unique clinical features:

a. **ADTKD-*UMOD*** is the most common subtype of ADTKD. It arises from pathogenic variants in the *UMOD* gene, which encodes uromodulin (previously known as Tamm-Horsfall protein). This subtype is characterized by gout, typically presenting during adolescence, followed by CKD that becomes apparent in adulthood.[318]
b. **ADTKD-*HNF1B*** results from variants in the *HNF1B* gene, a homeodomain-containing transcription factor. With more than 400 variants identified, de novo variants are observed in 30% to 50% of cases. ADTKD-*HNF1B* is associated with childhood genitourinary abnormalities, adulthood-onset gout or diabetes, and CKD developing by the third decade of life is associated with congenital anomalies of the kidney and urinary tract, including kidney cysts and dysplasia, often detected in childhood. CKD frequently begins in childhood or adolescence, and affected individuals are also at risk for early-onset diabetes mellitus (MODY5), hypomagnesemia, and gout in adulthood.[318] Patients may present with an array of urogenital abnormalities including atrophied vas deferens, structural uterine abnormalities, and epididymal cysts.[319-321] HNF1B nephropathy shares renal and diabetic features with 17q12 deletion syndrome but differs in its association with neurodevelopmental disorders (NDDs). While both conditions impair kidney function, 17q12 deletion syndrome exhibits a broader spectrum of NDDs including intellectual disability, developmental delays, and autism spectrum disorder, likely due to the involvement of other genes in the deleted region, such as *LHX1, PIGW,* and *PCGF2.* These genes play critical roles in neurodevelopment, and their deletion contributes to the expanded NDD phenotype observed in 17q12 deletion syndrome. Conversely, HNF1B variants predominantly result in diabetes and kidney disease/failure. The underlying mechanisms linking these genetic alterations to NDDs remain unclear but may involve haploinsufficiency of HNF1B and its interactions with other transcription factors like homeobox protein Hox-A1.[322-324]
c. **ADTKD-*MUC1*** is caused by pathogenic variants in the *MUC1* gene, which encodes mucin-1. Representing about 30% of ADTKD cases, ADTKD-*MUC1* is characterized by progressive CKD without other distinguishing characteristics such as gout, childhood anemia, hypotension, or hyperkalemia.[318]
d. **ADTKD-*REN*** is a less common subtype caused by variants in the *REN* gene, which encodes renin. Patients with variants in the signal or prepropeptide region of the *REN*

often present with low-normal BP, childhood anemia, progressive CKD, mild hyperkalemia, and hyperuricemia. In contrast, those with variants in the mature peptide of renin typically develop gout in their late teens, with progression to ESKF by around age 64.[318]

e. **ADTKD-*SEC61A1*** is linked to a missense variant in *SEC61A1,* which affects the α subunit of the ER membrane translocon SEC61. This subtype is associated with neutropenia and growth restriction.

Despite the variability in genetic etiology, ADTKD generally manifests with ESKF between the ages of 20 and 70, although renal decline may begin as early as ages 3 to 51, particularly in cases complicated by hyperuricemia. ADTKD may occasionally be misdiagnosed as ADPKD due to overlapping features, but it is distinguished by the absence of significant kidney enlargement and the presence of only a few cysts.[318,325-327]

Pathophysiology of ADTKD

The pathophysiology of ADTKD varies depending on the underlying genetic variant. In ADTKD-*MUC1,* variants in the *MUC1* gene result in an abnormal frameshift protein (MUC1fs), which accumulates in renal tubular cells, contributing to CKD.[328] ADTKD-*UMOD* is characterized by variants, often in cysteine residues, which lead to the retention of mutant uromodulin (mUMOD) in the ER of tubular epithelial cells. This ER stress triggers tubular cell death and impairs urinary excretion of wild-type uromodulin, resulting in defective urinary concentration and hyperuricemia.[318] In ADTKD-*REN,* nontruncating variants in the *REN* gene activate the unfolded protein response and disrupt renal development, contributing to ER stress.[329] ADTKD-*HNF1B* is linked to variants in the *HNF1B* homeobox B protein. Loss of *HNF1B* induces epithelial-mesenchymal transition, upregulating transforming growth factor-β (TGF- β) ligands and promoting kidney fibrosis. The variable expression of *HNF1B* in different tissues accounts for the diverse clinical features seen in *HNF1B*-related disorders.[330] Histologically, ADTKD is marked by interstitial fibrosis, tubular atrophy, thickened tubular basement membranes, and occasional tubular dilatation with microcysts. Immunofluorescence for immunoglobulins and complement is negative, while electron microscopy reveals mUMOD accumulation in the ER of thick ascending loop cells.[318]

Diagnostics, Clinical Manifestations, and Management of ADTKD

ADTKD should be suspected in patients presenting with CKD, a family history of autosomal dominant renal disease, bland urinary sediment, and absent or minimal proteinuria. In the absence of family history, the diagnosis may rely on histologic findings on renal biopsy, extrarenal features, or early onset hyperuricemia. However, genetic testing is essential to confirm the diagnosis, as kidney biopsy alone is insufficient.[318]

Diagnostic strategies vary by subtype. ADTKD-*MUC1* diagnosis requires detection of *MUC1* variants using indirect methods, as commercial clinical testing is unavailable. ADTKD-*UMOD* diagnosis is based on clinical presentation and genetic testing, which is widely accessible. ADTKD-*REN* diagnosis relies on genetic analysis of *REN,* as measuring renin and aldosterone levels is inadequate. Cost-effective NGS facilitates *HFN1B* variant diagnosis, while diagnostic criteria for ADTKD-*SEC61A1* are still evolving due to its rarity.[318,330-332]

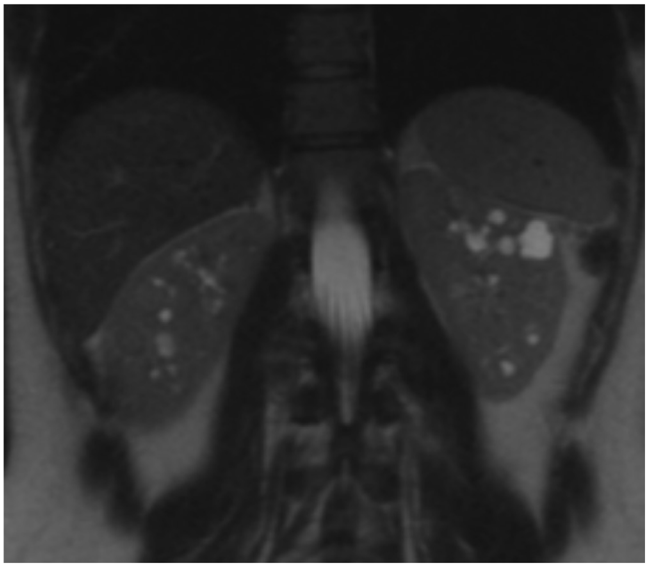


Fig. 45.18 Magnetic resonance imaging of a 32-year-old woman with *HNF1B*-associated kidney disease (estimated glomerular filtration rate at imaging of 90 mL/min/1.73 m²).

Clinical manifestations of ADTKD are gene specific. ADTKD-*UMOD,* historically called familial juvenile hyperuricemic nephropathy type 1, presents in adolescence with gout and hyperuricemia, progressing to ESKF by the fourth to seventh decades. ADTKD-*REN,* previously termed familial juvenile hyperuricemic nephropathy type 2, features hypoproliferative anemia, CKD, and hyperuricemia in adolescence, and ESKF in the fourth to sixth decades. ADTKD-*SEC61A1* is associated with dysplastic kidneys, CKD, anemia, neutropenia, and intrauterine growth retardation in some families. ADTKD-*HNF1B* (Fig. 45.18) has highly variable features including kidney cysts, glomerular tufts irregularities, and congenital abnormalities like horseshoe or solitary kidneys. A unique manifestation is early onset diabetes (MODY5), often preceded by renal abnormalities such as cystic kidney disease, which is the predominant feature in both children and adults.[320,321] More than one-third of individuals with *HNF1B* variants develop moderate to severe ESKF. Hypomagnesemia is a distinguishing feature in MODY5 patients with cystic kidney disease, helping differentiate it from ADPKD or ARPKD.[319,320]

Management of ADTKD is largely supportive, focusing on CKD complications like hypertension, anemia, metabolic bone disease, and electrolyte disturbances.[318] In ADTKD-*UMOD,* gout is managed with urate-lowering therapy including allopurinol or febuxostat, though cation is necessary in women of childbearing age due to teratogenic risks.[333,334] In ADTKD-*MUC1,* management is preventive and symptomatic, as no specific therapy exists, with kidney transplantation being the optimal option for ESKF.[334] For ADTKD-*REN,* hyperkalemia and mild hypotension can be addressed with fludrocortisone or a higher-sodium content diet, while low-sodium diets should be avoided to prevent acute kidney injury. ADTKD-*HNF1B* management includes monitoring liver function, hyperglycemia, hypomagnesemia, and hyperuricemia, with renal US for structural abnormalities. For ADTKD-*SEC61A1*, addressing congenital anemia and preventing infections are important to improve prognosis.[318,334]

PROGNOSIS OF ADTKD

While ADTKD typically follows a relatively benign course, it can progress rapidly to ESKF in some cases. Early and accurate diagnosis, coupled with timely supportive management, is essential to delay ESKF and mitigate complications, even though these measures do not target the underlying genetic cause. Kidney transplantation remains the optimal treatment for patients with advanced disease.[333]

X-LINKED DOMINANT DISEASES: OROFACIAL-DIGITAL SYNDROME TYPE 1

EPIDEMIOLOGY, ETIOLOGY, AND GENETICS OF OFD1 SYNDROME

Orofacial-digital syndrome type 1 (OFD1) is rare X-linked dominant ciliopathy characterized by a spectrum of congenital anomalies including cleft palate, tongue lobulation, cognitive impairment, and digital abnormalities. The syndrome primarily affects females, as it is embryonically lethal in males.[335,336]

Its inheritance pattern underscores its X-linked dominance, with clinical manifestations influenced by genetic heterogeneity and X-inactivation patterns.[335,336]

DIAGNOSTICS, CLINICAL MANIFESTATIONS, AND MANAGEMENT OF OFD1 SYNDROME

Genetic testing is essential for female patients lacking presenting with extrarenal manifestations such as cognitive impairment, cleft palate, misaligned teeth, and facial or digital malformations, particularly in the absence of family history of ADPKD. Regular kidney function monitoring and imaging are recommended to detect renal cystic involvement.[335,336]

OFD1 syndrome exhibits a broad clinical spectrum, with up to 50% of patients presenting with polycystic kidney disease, as well as cysts in the liver, pancreas, and ovaries (Fig. 45.19). This extensive cyst burden and kidney enlargement can mimic ADPKD. Renal abnormalities, often manifesting as bilateral PKD, become evident in approximately one-third of patients, typically during adulthood. Unlike the tubular cysts characteristic of ADPKD, the cysts in OFD1 syndrome are smaller and primarily glomerular in origin.[335,336]

Fig. 45.19 Magnetic resonance imaging of a 39-year-old woman with oral-facial-digital syndrome type 1 (estimated glomerular filtration rate at imaging of 52 mL/min/1.73 m^2).

AUTOSOMAL RECESSIVE DISEASES

AUTOSOMAL RECESSIVE POLYCYSTIC KIDNEY DISEASE

Epidemiology of ARPKD

ARPKD is an autosomal recessive genetic disorder that manifests as hepatorenal congenital fibrocystic syndrome, resulting in notable morbidity and mortality. More than 50% of individuals progress to ESKF by the first decade of life. Despite being one of the most common causes of inherited infantile cystic diseases, the estimated incidence of ARPKD only ranges between 1 in 20,000 to 50,000 live births. ARPKD was previously called infantile PKD; however, it can also be present in juvenile and adult populations.[337,338]

Etiology, genetics, and pathophysiology of ARPKD

ARPKD is primarily caused by variants in the *PKHD1* gene on chromosome 6 encoding fibrocystin. Fibrocystin can be found in three main locations: primary cilia of the collecting ducts and the thick ascending limb of the kidney, pancreatic islet cells, and hepatic bile duct epithelial cells.[338] Variants in the *PKHD1* can manifest at any point in life (prenatal period, infancy, childhood, or even adulthood). Variants in the *DZIP1L* gene have been reported as a rare cause of ARPKD. Both *PKHD1* and *DZIP1L*'s gene products localize to the cilia.[338,339] ARPKD patients with *CYS1* pathogenic variants have been identified.[340] Furthermore, the identification of a homozygous *CYS1* variant in an ARPKD patient, due to disrupted splicing, highlights the significance of the *CYS1* gene product for normal human collecting duct cell function. Notably, the pathogenic *CYS1* variant was initially overlooked in routine genetic analysis but was revealed by WGS.[340]

ARPKD primarily affects the kidney and the liver. As opposed to ADPKD, in which cystic dilation can be of various sizes and can affect any part of the nephron, ARPKD kidney enlargement happens secondary to microcysts and is typically confined to the collecting ducts. As such, renal collecting ducts undergo abnormal circumferential proliferation and accumulation of epithelial growth factor enriched fluid within those tubules. Hepatic findings are observed in half of the patients. They include ductal plate disruption, biliary duct proliferation, and ectasia along with periportal fibrosis. Hepatic fibrosis can potentially be present at birth.[341,342]

Diagnostics and clinical manifestations of ARPKD

ARPKD should be considered in individuals presenting with bilaterally enlarged, diffusely echogenic kidneys. Table 45.10 reflects a combination of clinical presentations and radiographic evidence that guide ARPKD diagnosis.[343]

Molecular genetic testing remains the gold standard for diagnosis. In the absence of genetic testing, clinical

Table 45.10 Diagnostic Criteria of ARPKD

ARPKD Diagnosis
Typical ARPKD renal findings on abdominal imaging
+ 1 or more of the following:
Imaging consistent with biliary duct ectasia
Clinical/laboratory signs of congenital hepatic fibrosis leading to portal hypertension, indicated by hepatosplenomegaly and/or esophageal varices
Hepatobiliary pathology demonstrating a characteristic developmental biliary ductal plate abnormality resulting in congestive heart failure
Absence of renal enlargement or characteristic imaging findings in both parents, as confirmed by high-resolution ultrasonography (HRUS) examination
Pathologic (biopsy or autopsy) or genetic diagnosis of ARPKD in an affected sibling

Fig. 45.20 Magnetic resonance imaging of a 22-year-old woman with autosomal recessive polycystic kidney disease (estimated glomerular filtration rate at imaging of 16 mL/min/1.73 m^2).

diagnosis is typically made through abdominal US, revealing large echogenic kidneys with poor corticomedullary differentiation and coexisting liver disease. US is most used in the perinatal and neonatal period due to its cost-effectiveness, safety, and ease of use. To note, renal US findings alone are not diagnostic; they are mainly useful to detect renal abnormalities. Biliary duct ectasia is commonly identified via magnetic resonance cholangiopancreatography (MRCP).[337,342,344] Fig. 45.20 represents an MRI of a patient with ARPKD.

ARPKD can present at different stages of life. These manifestations vary widely depending on age. As symptoms become apparent at a later stage, the degree of kidney involvement tends to be less severe, and kidneys may exhibit minimal cysts on US. ARPKD most commonly presents during the prenatal period and exhibits symptoms more severe than the ones encountered during childhood or adulthood. Severe ARPKD cases can be detected in utero after 24 weeks' gestation with markedly enlarged echogenic kidneys. Those patients typically show oligohydramnios and pulmonary hypoplasia and therefore have a poor prognosis.[342,344] ARPKD detected at the neonatal and infantile stage typically shows massively enlarged kidneys. Patients may additionally experience respiratory distress and can less commonly exhibit features of Potter syndrome. They can also present with hyponatremia that resolves with adequate kidney function and hypertension that develops during the first months of life in 80% of children and is often resistant to multidrug therapies. Infants often experience respiratory distress shortly after birth due to pulmonary hypoplasia, which is not prominent in adults. Respiratory distress is primarily caused by pulmonary hypoplasia, which is exacerbated by the presence of large kidneys and is associated with increased morbidity and mortality in the neonatal period. The long-term survival rate significantly improves to 80% upon starting respiratory support therapy and KRT.[342,344,346] When detected later during childhood, ARPKD patients commonly present with hepatic fibrosis and features of portal hypertension, splenomegaly, thrombocytopenia, and portosystemic varices, with severe cases potentially requiring liver transplantation. Kidney manifestations are less severe, but patients can eventually deteriorate into ESKF and require KRT.[342] Caroli syndrome, characterized by unobstructed dilatation of intrahepatic bile ducts, develops in 60% of patients with ARPKD. This increases the risk of recurrent bacterial ascending cholangitis with sepsis. Cholestasis associated with Caroli syndrome leads to malabsorption of fat-soluble vitamins (A, D, E, and K). Furthermore, the abnormal proliferation of biliary cells has been associated with the development of benign and malignant tumors including cholangiocarcinoma, particularly in adulthood.[347] ARPKD may rarely present in adulthood with features distinct from those seen in childhood. While both age groups exhibit liver involvement, adults typically present with symptoms such as hepatomegaly, portal hypertension, varices, and cholangitis, whereas pediatric cases often have milder liver manifestations.[348] Kidney manifestations in adults include renal enlargement, cysts, nephrocalcinosis, and ESKF, although less severe than in pediatric cases, which often present with early onset and rapidly progressive kidney disease leading to kidney insufficiency and the need for KRT. Proteinuria is present in both groups but tends to be milder in adults. Hypertension, a common feature in ARPKD, is typically less severe in adults compared with children, where it contributes significantly to the rapid progression of kidney disease.[348]

Management and prognosis of ARPKD

Management of ARPKD is supportive and depends on the clinical manifestations. As such, treatment starts with monitoring kidney, liver, respiratory function, BP measurements, and growth evaluation. Respiratory symptoms can further be managed by offering respiratory support therapy or mechanical ventilation. Unilateral or bilateral nephrectomy may be considered if massive kidney enlargements compress the diaphragm and lead to respiratory distress. KRT may be required as kidney function deteriorates.[337] Neonates born with anuria or oliguria may necessitate peritoneal dialysis or hemodialysis. Nephrectomy may be required to create space for peritoneal dialysis. Neonatal bilateral nephrectomy may be followed by significant hypotension; therefore BP must be closely monitored in the neonatal intensive care unit. In more severe cases with portal hypertension and varices, sclerotherapy and/or vascular banding may be required. Hepatotoxic medications such as acetaminophen should be used with caution to prevent further alterations in liver function.[337,342] It is also recommended to avoid nephrotoxic medications (NSAIDs and some antibiotics such as aminoglycosides) and sympathomimetics in cases of hypertension. As liver and kidney function deteriorate, dual organ transplant of liver and kidney will be required.[337] Genetic counseling should be considered in families with a history of ARPKD or individuals diagnosed with ARPKD as early detection can improve the quality of life of the patients.[342] Preimplantation genetic testing for monogenic (PGT-M) PKD may be an option for couples at risk of passing on this condition.[222] Given the success of tolvaptan in ADPKD, tolvaptan is currently being studied as a potential treatment in ARPKD patients aged between 28 days and 18 years to evaluate its effect on eGFR and TKV (Clinicaltrials.gov NCT04782258 and NCT04786574).

ARPKD outcome and prognosis depend on the degree of kidney and hepatic involvement, often reflecting the age of presentation. Neonates presenting with severe kidney disease have the highest mortality rate (30%) mainly due to respiratory dysfunction. Patients born with less severe manifestations are more likely to survive childhood but are at an increased risk for ESKF as they age.[342] In contrast, liver disease is typically milder during childhood and tends to worsen with age, possibly leading to portal hypertension secondary to chronic liver fibrosis.[342] For prenatally identified ARPKD cases with poor prognosis, options include pregnancy termination or providing comfort care for the neonate upon delivery. Serial US during pregnancy helps assess kidney size and amniotic fluid volume. If parents opt for full resuscitation, delivering at a center with a high-level neonatal intensive care unit is recommended. Cesarean delivery might be necessary due to complications related to massive kidney enlargement.[349]

CILIOPATHIES

Ciliopathies are a group of disorders caused by defects in the structure or function of cilia. Due to the widespread distribution and diverse roles of cilia, dysfunction can lead to a broad spectrum of clinical manifestations affecting multiple organ systems.[350] More than 950 cilia-related genes and their respective ciliary proteins have been identified.[351] More than 35 human genetic diseases, collectively known as “ciliopathies,” have been linked to pathogenic variants in over 180 established ciliopathy-associated proteins (Fig. 45.21).[352,353] These disorders can affect different organ systems including the kidneys, eyes, liver, brain, heart, and respiratory tract.[156] Table 45.11 summarizes the differences and similarities of various cystic diseases.

NEPHRONOPHTHISIS

Epidemiology and etiology of NPHP

NPHP is an autosomal recessive tubulointerstitial kidney disorder usually leading to kidney cysts and ESKF in children and adolescents. Occurring in 1 out of 50,000 births, NPHP accounts for 5% of ESKF cases in North American children.[354] It is a heterogeneous disorder involving more than 20 identified genes. It has been historically classified into three clinically different categories on the basis of the age of onset of ESKF: infantile (before the age of 4), juvenile (around the age of 13), and adolescent (median age of 19) NPHP. The juvenile form is the most common NPHP presentation.[355,356]

About 20% to 30% result from homogeneous deletions in NPHP genes, while 60% to 70% show no pathogenic variants in known genes. Patients with NPHP exhibit gene variants affecting the ciliary apparatus, with *NPHP1* gene variants accounting for at least 20% of cases. NPHP manifests with impaired urinary concentrating ability, bland urinalysis, anemia, chronic tubulointerstitial nephritis, and progression to ESKF, often before age 20. Infantile NPHP is linked to *INVS* (*NPHP2*) or *NPHP3* pathogenic variants, leading to ESKF by age 3.[356,357]

Pathophysiology of NPHP

In NPHP, nephrocystin involvement, especially in the primary cilia complex, centrosome, and actin, links kidneys and cholangiocytes, explaining kidney cysts and hepatic fibrosis. The juvenile and adolescent variants present with grossly normal or small kidneys, corticomedullary cysts, and histologic changes involving interstitial fibrosis and tubular damage. The infantile variant, related to *NPHP2* or *NPHP3* pathogenic variants, displays microcystic dilatations and distinct histologic patterns.[357,358] Kidney involvement is central, characterized by cortico-medullary cyst development, small kidneys, and fibrosis, with cysts originating in the collecting duct and remaining connected to the tubule.[359] Unlike ADPKD, where cysts separate from the tubule, NPHP manifests with overproliferation of tubular epithelium and disintegration of the tubular basement membrane, leading to impaired urine concentration.[360]

Diagnosis of NPHP

NPHP is suspected clinically and confirmed by genetic testing.[361] Genetic diagnosis of NPHP, though feasible in clinically suspected cases, remains complex and resource intensive.[362] Advances in genetic screening, notably through techniques like NGS, whole genome, or exome, have improved diagnostic rates, diagnosing up to 20% of genetic kidney diseases in adults and 46% to 63% of pediatric cases.[363] Genetic diagnosis aids prognosis and management, potentially avoiding the need for additional imaging, but necessitates genetic counseling.[364] Contrary to ADPKD, which has well-defined diagnostic criteria, NPHP lacks strict diagnostic guidelines unless a genetic diagnosis is achieved.[364] Kidney biopsy may show chronic tubulointerstitial changes but is not necessary to make the diagnosis.[111,365] Kidney US may initially reveal normal-size, hyperechoic kidneys with poor corticomedullary differentiation. However, in later stages of NPHP, kidneys typically become atrophied, smaller with increased echogenicity and notable cyst formation.[366]

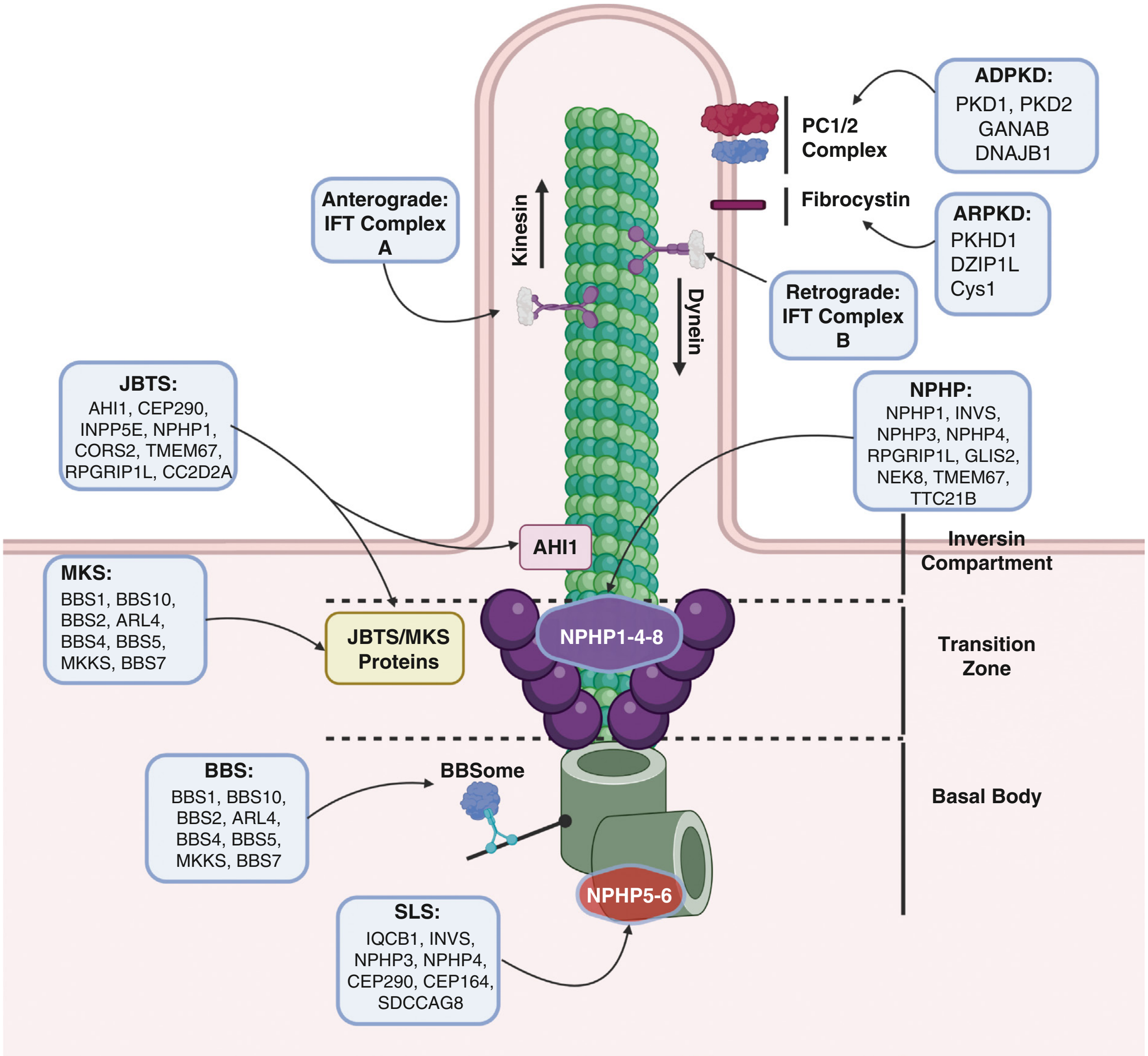


Fig. 45.21 Structure of the primary cilium including the distribution of cilioproteins and the disease and genes implicated in the ciliary diseases. This diagram illustrates the structure of the primary cilium, a sensory organelle implicated in the pathogenesis of polycystic kidney disease (PKD) and various ciliopathies. The primary cilium harbors distinct groups of cilioproteins, and intraflagellar transport (IFT) mechanisms play a crucial role in protein trafficking within the cilium. Two specific complexes, IFT Complex B with the kinesin 2 motor for anterograde movement and IFT Complex A with the dynein motor for retrograde movement, contribute to cilia formation. Proteins associated with Meckel and Joubert syndromes are involved in these transport processes. Additionally, nephronophthisis proteins (NPHP) form a complex at the transition zone, regulating cilium protein composition. Specific proteins like inversin (INVS), located in the proximal region (inversin segment) of the ciliary axoneme, have less clear functions. The Bardet-Biedl syndrome proteins, forming the BBSome, regulate membrane protein trafficking into the cilium. Variants in any of these proteins disrupt the normal composition, sensory functions, and signaling of the cilium, leading to pleiotropic phenotypes characteristic of diverse ciliopathies. Understanding these molecular intricacies provides insights into the pathogenesis of diseases associated with primary cilia dysfunction.

Clinical manifestation of NPHP

NPHP presents a spectrum of clinical manifestations affecting multiple organ systems. Clinical kidney symptoms include polyuria, polydipsia, nocturia, and microalbuminuria.[249,367] In addition to kidney manifestations, NPHP is associated with diverse ocular abnormalities such as retinitis pigmentosa, coloboma, and visual impairment.[364] Liver fibrosis, musculoskeletal abnormalities, reproductive system anomalies, and cardiovascular complications contribute to the heterogeneity of NPHP.[364] Chronic iron-resistant anemia, often out of proportion to the degree of kidney dysfunction, is another notable feature of NPHP.[354] NPHP-related ciliopathies include various autosomal recessive cystic kidney diseases including Senior-Loken, Joubert, Meckel-Gruber, and Sensenbrenner syndromes and asphyxiating thoracic dystrophy.[368]

Treatment of NPHP

No specific treatment exists for NPHP. Management mainly consists of supportive care, avoidance of unnecessary diuretics, and salt restriction in salt-wasting cases. Early stage management focuses on maintaining fluid balance, treating

Table 45.11 Similarities and Differences Between ADPKD and Other Cystic Kidney Diseases

Disease	Gene	MOI	Age at Presentation	Similarities With ADPKD	Differences With ADPKD
ARPKD	*PKHD1, DZIP1L, CYS1*	AR	Neonates, infancy, childhood	Kidney cyst development	Pulmonary hypoplasia and early onset KF & TKV decline over time Fibrosis (and not cysts) is predominant hepatic manifestation
Meckel Syndrome	*MKS1, TMEM216, TMEM67, CEP290, RPGRIP1L, CC2D2A, NPHP3, TCTN2, B9D1, B9D2, TMEM231*	AR	Neonates	Kidney cyst development	CNS malformations, occipital encephalocele, and polydactyly Liver and bile ducts involvement
OFD Syndrome Type 1	*OFD1*	XD	Neonates, infancy	Kidney cyst development in females	Cleft lip and palate, oral frenula, facial dysmorphisms and polydactyly
Joubert Syndrome	*INPP5E, TMEM216, AHI1, NPHP1, CEP290, TMEM67, RPGRIP1L, ARL13B, CC2D2AOFD1, KIF7, TCTN1, TMEM237, CEP41, TMEM138, C5ORF42, TCTN3, ZNF423, TMEM231, CSPP1, PDE6D*	AR	Infancy	Kidney cyst development	Cerebellar and brainstem malformations "molar tooth sign" Hypotonia, developmental delay, and abnormal eye movements
Alstrom Syndrome	*ALMS1*	AR	Infancy-childhood	Kidney cyst development	Cardiomyopathy, liver abnormalities, progressive vision, and hearing loss Renal cysts are not a defining feature
Senior-Loken Syndrome	*NPHP1, NPHP4, IQCB1*	AR	Infancy-childhood	Kidney cyst development	Retinal dystrophy, intellectual disability, cerebellar vermis hypoplasia, chronic malnutrition, and anemia
Tuberous Sclerosis Complex (TSC)	*TSC1, TSC2*	AD	Childhood	Kidney cyst development	Angiofibromas, rhabdomyoma, hamartomas, depigmented nevi, seizures
Hyperinsulinemic Hypoglycemia and Polycystic Kidney Disease (HIPKD)	*PMM2*	AR	Infancy, childhood	Kidney Cyst Development	ARPKD-like. Hepatomegaly, Respiratory distress Hyperinsulinemic hypoglycemia
Alagille Syndrome	*JAG1, NOTCH2*	AD	Infancy, childhood	Kidney Cyst Development	Facial appearance, heart defects, and eye abnormalities.
Nephronophthisis	*NPHP1, NPHP2/INVS, NPHP3, NPHP4, NPHP5/IQCB1, NPHP6/CEP290, NPHP7/GLIS2, NPHP8/RPGRIP1L, NPHP9/NEK8, NPHP10/SDCCAG8, NPHP11/TMEM67, NPHP12/TTC21B, NPHP13/WDR19, NPHP14/ZNF423, NPHP15/CEP164, NPHP16/ANKS6, NPHP17/IFT172, NPHP18/CEP63*	AR	Childhood-Adolescence	Kidney Cyst Development	Progressive tubulointerstitial nephritis, anemia, growth retardation

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Table 45.11 Similarities and Differences Between ADPKD and Other Cystic Kidney Diseases (Continued)

Disease	Gene	MOI	Age at Presentation	Similarities With ADPKD	Differences With ADPKD
Bardet-Biedl Syndrome	BBS1, BBS2, ARL6, BBS4, BBS5, MKKS, BBS7, TTC8, PTHB1, BBS10, TRIM32, BBS12, MKS1, CEP290, C2orf86, TMEM67, LZTFL1, BBIP1, IFT127	AR	Childhood-Adolescence	Kidney Cyst Development	Retinal dystrophy, cognitive impairment, obesity, polydactyly
Alport Syndrome	COL4A3, COL4A4, COL4A5	AD, AR, or XL	Childhood-Adolescence	Kidney Cyst Development	Hearing loss, retinal and eye abnormalities, thinning of the basement membrane and microhematuria
Hajdu-Cheney Syndrome	NOTCH2	AD	Adolescence	Kidney enlargement and Kidney Cyst Development	Skeletal abnormalities, coarse facial features, dental issues, joint hypermobility, and connective tissue disorders
HNF1B-related kidney disease	HNF1B	AD	Childhood-Adulthood	Kidney Cyst Development.	Early onset diabetes, hypomagnesemia, congenital urinary tract abnormalities
Urinary Stone Disease	CYP24A1 Deficiency SLC34A3 Variants HOGA1 Variants	AR (AD)	Infancy-Adulthood	Kidney Stones and Impaired renal function	Predominant stone formation and milder kidney cystic burden
Von Hippel-Lindau syndrome	VHL	AD	Adulthood	Kidney and possible Pancreatic Cyst Development	CNS and Retinal Hemangioblastomas, RCC, Pheochromocytoma
ADPKD	PKD1, PKD2 GANAB, DNJAJB11, ALG9, IFT140	AD	Adulthood	Progressive kidney and liver cyst development as well as kidney enlargement, and hypertension eventually leading to end-stage kidney failure. Patients may also develop hematuria, urinary tract infections, and nephrolithiasis. Extrarenal symptoms include cerebral aneurysms, polycystic liver disease, cardiomyopathies among others	
ADPLD	PRKCSH, SEC63, SEC61B, GANAB, ALG8, LRP5	AD	Adulthood	Overlap of kidney & liver cysts in most cases	Predominantly affects the liver with milder kidney involvement in some cases. NB: GANAB can be involved in both ADPKD and ADPLD
ADTKD	UMOD, MUC1, REN, SEC61A	AD	Adulthood	Kidney Cyst Development	No liver cysts and the decline in KF is not associated with an increase in TKV. Fibrosis (and not cysts) is the major kidney manifestation. Gout often encountered
Monoallelic PKHD1	PKHD1	AD	Adulthood	Overlap of kidney & liver cysts in most cases	Cyst development does not often lead to ESKF.
Birt-Hogg-Dubé Syndrome	FLCN	AD	Adulthood	Kidney Cyst Development	Cutaneous Fibrofolliculomas, pneumothorax, Renal tumors, lung cysts
Hereditary angiopathy with nephropathy, aneurysms, and muscle cramps (HANAC)	COL4A1	AD	Adulthood	Kidney Cyst Development and Cerebral Aneurysms	Peripheral muscle cramp, glomerular basement membrane abnormalities leading to hematuria

AD, Autosomal dominant; *AR,* autosomal recessive; *ESKF,* end-stage kidney failure; *XL,* X linked.

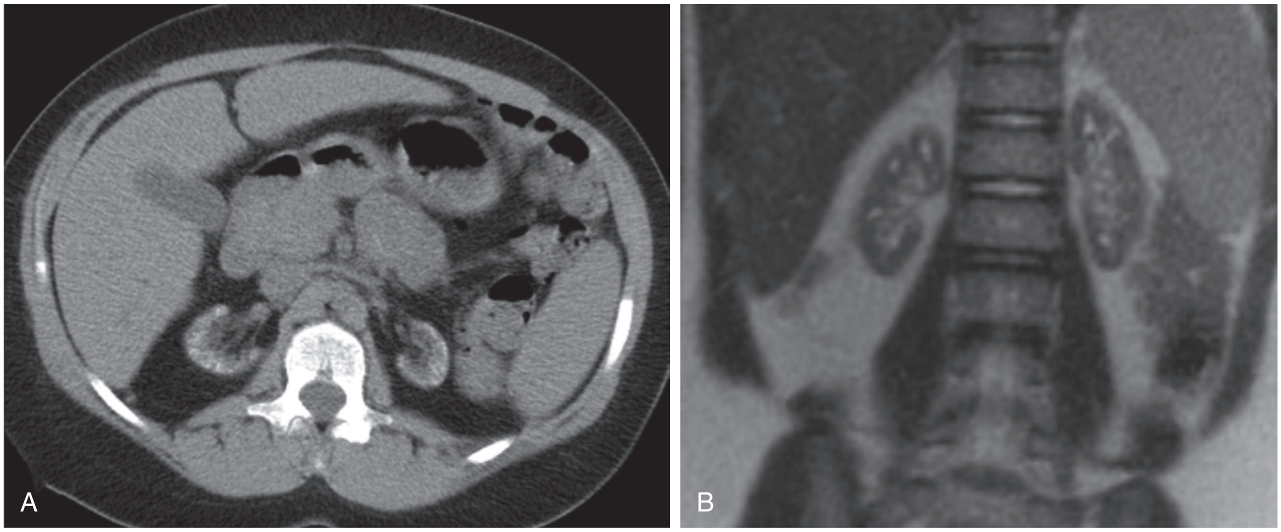

Fig. 45.22 Imaging representation of female patient with Joubert syndrome. (A) Computed tomography scan at 13 years of age with eGFR of 54.7 mL/min/1.73m^2. (B) Magnetic resonance imaging at 10 years of age with eGFR of 77.6 mL/min/1.73 m^2.

anemia, and promoting growth. Kidney transplantation is recommended for ESKF, with excellent outcomes and no recurrence of tubular injury in the transplanted kidney.[355,364]

JOUBERT SYNDROME

JS is a recessive neurodevelopmental disorder commonly presenting with a distinct cerebellar and brainstem malformation recognized as the "Molar Tooth Sign" on axial brain MRI, indicating cerebellar vermis aplasia or hypoplasia alongside abnormal superior cerebellar peduncles.[369] Most patients experience additional complications in the eyes, kidneys, liver, and skeletal structure. The syndrome can stem from pathogenic variants in more than 35 genes.[370,371] In around 50% to 92% of JS cases, biallelic pathogenic variants in known JS-related genes, or hemizygous pathogenic *OFD1* variants, can be identified.[354,370,372] Pathogenic variants in the *AHI1* gene are seen in 10% of infants with this condition,[373] and other genes involved are *NPHP1* and *MKS*.[374] Regarding kidney manifestations, fibrocystic kidney disease arises in 25% to 30% of individuals, often progressing to ESKF.[370] JS can appear in two variations: NPHP and cystic kidney disease characterized by corticomedullary cysts, kidney atrophy, and interstitial scarring resulting in small, scarred kidneys as observed pathologically. NPHP represents chronic tubulointerstitial nephropathy.[375,376] At onset, it displays signs of excessive urination and thirst. Some individuals might exhibit kidney features similar to ARPKD (mainly prenatal kidney enlargement and early hypertension).[375] Regular monitoring for CKD development and progression, yearly BP assessments, and checking laboratory tests (i.e., serum electrolytes, calcium, phosphorus, creatinine, BUN, and blood counts with hematocrit/hemoglobin levels) are crucial measures in all JS patients.[354,370] Imaging studies of a patient with Joubert syndrome can be seen in Fig. 45.22.

MECKEL SYNDROME

Meckel syndrome (MKS) is a rare and severe autosomal recessive genetic disorder primarily affecting fetal development with an estimated prevalence of around 2 cases per 100,000 births.[354] MKS has four key features: occipital encephalocele, large cystic–dysplastic kidneys, congenital hepatic fibrosis (or ductal plate malformation), and polydactyly. Bilateral renal cystic dysplasia and CNS defects are determined by a range of genetic factors, involving 11 identified genes. Beyond these primary features, a range of additional malformations has been observed, such as various brain abnormalities (like cerebellar hypoplasia, hydrocephalus, holoprosencephaly, and anencephaly), as well as eye, facial, heart, genital, and skeletal abnormalities.[354] The diagnosis of MKS is typically made through prenatal imaging during pregnancy or upon delivery. While MKS is usually linked with fatality before or around birth, rare instances of survival beyond infancy have been documented.[354]

SENIOR-LOKEN SYNDROME

SLS is a rare genetic ciliopathy characterized by juvenile nephronophthisis and progressive retinal degeneration leading to a renal-retinal syndrome, with CKD often necessitating kidney transplantation.[377] SLS is an autosomal recessive disorder, which has an estimated incidence of 1 in 100,000 individuals, with higher rates observed in families with consanguineous marriages.[378] It is caused by pathogenic variants mainly in the *NPHP1* gene and typically presents with nephronophthisis and retinal degeneration, progressing from polyuria to ESKF.[378,379] Although primarily considered a pediatric disease, adult-onset kidney disease cases have been reported, even though without *NPHP1* gene deletion.[380] As such, patients may present with decreased vision, pallor, abnormal kidney function tests, and chronic malnutrition with laboratory studies revealing anemia, and decreased GFR.[378] Imaging studies may reveal renal parenchymal changes and corticomedullary cysts.[378] The retinal phenotype in SLS patients varies widely, encompassing conditions such as Leber congenital amaurosis (LCA), retinitis pigmentosa (RP), cataracts, Coat disease, and keratoconus.[381] Notably, the *NPHP5* (IQCB1-p.R461*) pathogenic variant was identified in three unrelated families of Ashkenazi Jewish descent, potentially representing a founder variant in this ethnic group. Patients with *NPHP5* variants typically presented with LCA, while those with *NPHP4* manifested early RP.[377] Refractive errors in SLS patients typically include high hypermetropia, with variations observed depending on the genetic variant. Polydactyly and hepatic fibrosis may also be observed. Interestingly, some younger patients had normal kidney function, with kidney enlargement detected in US scans.[377]

BARDET-BIEDL SYNDROME

Bardet-Biedl syndrome (BBS) is a rare genetic syndrome commonly missed in childhood and diagnosed later in life. It can have varying prevalence in different regions. For instance, its prevalence is estimated at around 1:3700 in Faroe Islands, 1:18000 in Newfoundland, and around 1:13500 in specific Bedouin communities of the Middle East.[382] BBS is marked by significant genetic diversity, involving 24 loci and a wide-ranging mutational profile. It typically presents with rod-cone dystrophy, affecting almost all individuals, obesity (89% of cases), postaxial polydactyly (79% of patients), cognitive impairment (66% of cases), and kidney disease (52% of patients). Renal manifestations vary from urinary tract malformations to chronic glomerulonephritis. CKD has significant complications contributing to morbidity and mortality. Cystic kidneys, although uncommon, can also be detected in BBS and significantly contribute to CKD. As the disease progresses, ESKF may necessitate dialysis or transplantation, with favorable long-term outcomes reported post-transplantation.[382,383]

BBS can share phenotypic similarities with ADPKD, presenting a spectrum of clinical features that can resemble those seen in ADPKD. A study by Su and colleagues[384] has revealed that proteins associated with BBS, particularly BBS1 and BBS3, play important roles in orchestrating the transportation of PC1 to the primary cilia. This interplay between *PKD1* and *BBS* proteins elucidates a crucial aspect of their molecular interactions, shedding light on the shared pathophysiologic mechanisms underlying both BBS and ADPKD.[384] Furthermore, disruption of *BBS1* and *BBS3* can disrupt the normal trafficking of *PKD1* to the primary cilia, potentially contributing to the pathogenesis of both BBS and ADPKD.[384]

ALSTROM SYNDROME

Alstrom syndrome (AS) often overlaps with BBS. The incidence of this rare condition is uncertain but estimated to range between 1 in 500,000 to 1 in 1,000,000. Globally, around 1200 AS cases have been identified, equally affecting both sexes. It is an autosomal recessive disorder due to variants in the *ALMS1* gene, a ciliopathy gene involved in ciliary transport and cell cycle regulation.[385,386] AS is characterized by a spectrum of manifestations emerging from childhood including retinal dystrophy, neuronal hearing loss, and obesity.[387] Diagnosing AS can be challenging due to the variable presentation. Some symptoms may not emerge until later in life, contributing to underdiagnosis in early life. Signs of AS typically become more prominent as the child grows, often leading to a clearer clinical picture in late childhood.[388] Kidney manifestations in AS exhibit considerable variability and typically progress slowly, with symptoms often emerging in late childhood or adulthood.[385] ESKF may develop during late childhood in certain cases, while kidney cysts are also observed.[385] Histopathologic examination often reveals interstitial fibrosis and tubular hyalinization. However, comprehensive data on kidney dysfunction in AS are limited to sporadic case reports and a single large cohort study.[387,389,390] Despite the association of AS with common causes of ESKF, such as type 2 diabetes mellitus, hypertension, and urinary tract dysfunction, the precise influence of these factors on the onset and progression of ESKF in AS remains unclear.[389]

RENAL-HEPATIC-PANCREATIC DYSPLASIA-1 WITH NOVEL NPHP3 GENOTYPE

Renal-hepatic-pancreatic dysplasia-1 (RHPD1) is a rare sporadic and autosomal recessive disorder caused by biallelic *NPHP3* variants. Patients mainly present with bilateral polycystic kidneys, CNS anomalies (i.e., occipital encephalocele), and polydactyly.[391]

KIDNEY CYSTIC DISEASES WITHOUT FAMILY HISTORY

MEDULLARY SPONGE KIDNEY DISEASES

EPIDEMIOLOGY AND ETIOLOGY OF MEDULLARY SPONGE KIDNEY

Medullary sponge kidney (MSK), also know as precaliceal canaclicular ectasia or Lenarduzzi-Cacchi-Ricci disease, is a rare kidney disorder with a prevalence of approximately 1 in 5000, typically affecting individuals aged 20 and 30 but occasionally reported in neonates. MSK can be unilateral or bilateral, with 70% of cases involving both kidneys. Among patients with calcium-based urolithiasis, women are more likely than men to have MSK.[392-394]

While most cases are sporadic, about 5% demonstrate autosomal dominant inheritance with reduced penetrance and variable expressivity. MSK is also associated with syndromic conditions including Ehlers-Danlos syndrome, Wilms tumor, Beckwith-Wiedemann syndrome, PKD, and Caroli disease.[393] Some *PKHD1* variants, typically linked to ARPKD, have been associated with MSK.[395] Urinary proteomics analysis has revealed dysregulation in pathways involving kidney development, cell adhesion, and extracellular matrix organization, identifying *LAMA2* (laminin subunit alpha 2) as a promising diagnostic biomarker.[392,396,397] MSK may originate from disruptions in the interaction between the ureteric bud and mesenchyme during development, potentially linked to variants in *GDNF* and *RET* genes.[392,393]

PATHOPHYSIOLOGY OF MEDULLARY SPONGE KIDNEY

MSK is a benign congenital disorder characterized by malformations of the terminal collecting ducts in the renal medulla. These ducts become ectatic, forming small medullary cysts within the renal pyramids that do not extend into the cortex. The cyst, lined with cuboidal or flattened epithelium, gives the kidney a spongy appearance on gross examination.[398-400] Investigations have noted heightened medullary echogenicity and the presence of small liver cysts in parents carrying a single *PKHD1* variant, suggesting that MSK could result from heterozygous *PKHD1* variants.[395] Furthermore, studies in mice have suggested that heterozygous *PKHD1* variants (heterozygous *Pkhd1C642** mice) might induce cystic liver disease and tubule ectasia resembling MSK.[395,401]

DIAGNOSTICS AND CLINICAL MANIFESTATIONS OF MEDULLARY SPONGE KIDNEY

MSK should be suspected in patients with recurrent kidney stones, nephrocalcinosis, hematuria, or recurrent UTIs. Imaging plays an important role in diagnosis. While kidney US and plain KUB radiographs may suggest MSK, intravenous urography and contrast-enhanced CT urography are more specific, identifying the characteristic papillary brushlike appearance. However, intravenous urography is now rarely used in clinical practice.[402,403] For most individuals with recurrent calcium kidney stones, a definitive diagnosis of MSK may not be necessary, as the condition is generally benign and lacks specific treatments beyond stone prevention.[393,404]

MANAGEMENT AND PROGNOSIS OF MEDULLARY SPONGE KIDNEY

Management focuses on addressing complications. Adequate hydration and dietary modifications help prevent kidney stones, with 24-hour urine tests guiding treatment for hypercalciuria and hypocitraturia using thiazide diuretics and potassium citrate. Potassium citrate also mitigates long-term bone loss. While most stones pass spontaneously, surgical or lithotripsy interventions may be required in select cases.[394,405,406]

Prognosis is generally favorable, with normal kidney function maintained in most patients. However, approximately 10% may progress to ESKF due to recurrent severe infections or extensive stone formation.[394,402,403]

ACQUIRED CYSTIC KIDNEY DISEASES

EPIDEMIOLOGY AND PATHOPHYSIOLOGY OF ACQUIRED CYSTIC KIDNEY DISEASE

Acquired cystic kidney disease (ACD) encompasses various conditions including solitary multilocular cysts, hypokalemic cystic disease, cysts of advanced ESKF, hilar cysts, and perinephric pseudocysts. While ACD can occur in patients with relatively preserved renal function, its incidence increases as CKD progresses, particularly in those on dialysis for extended periods. Cyst prevalence correlates with dialysis duration, with cyst formation observed after an average of 49 months. Male sex and Black race are associated with higher disease prevalence.[407]

Cysts in ACD are confined to the kidneys and result from hyperplasia of proximal tubular epithelial cells, which secrete a plasma-like fluid. Compensatory hypertrophy driven by growth factors and proto-oncogene activation contributes to tubular hyperplasia and cyst development. Notably, proto-oncogenes implicated in ACD also play a role in RCC development.[408]

Diagnostics, clinical manifestation, and management of ACD

Diagnosis of ACD is often incidental, based on imaging in asymptomatic CKD patients. Confirmation typically requires three or more cysts in each kidney. US and CT are commonly employed for symptomatic patients presenting with hematuria, flank pain, or UTIs.[407,409] While most patients are asymptomatic, progressive cyst growth may cause complications such as hematuria, pain, and in rare cases, cyst rupture. ACD also increases RCC risk, though the incidence varies. Kidney transplantation often stabilizes or regresses cystic disease by reducing growth factor levels, improving outcomes compared with chronic dialysis, except in patients treated with cyclosporine, who have similar RCC risks to those on dialysis.[407]

RARE DISEASES AND SYNDROMES WITH KIDNEY CYSTS

Several rare genetic and metabolic syndromes are associated with kidney cyst formation, each presenting with distinct clinical features (Table 45.12A**).** These include Hajdu-Cheney syndrome, Alagille syndrome, and hyperinsulinemic hypoglycemia with polycystic kidney disease (HIPKD). Additionally, urinary stone diseases linked to variants in *HOGA1, SLC34A3,* or *CYP24A1* have also been associated with development of kidney cysts (see Table 45.12B).

RENAL CYSTIC DISEASES OF NONTUBULAR ORIGIN

Renal cystic diseases of nontubular origin encompass a diverse group of conditions affecting the kidneys and surrounding structures (see Table 45.12C). These include renal sinus cysts, perirenal lymphangiomas, subcapsular and perirenal urinomas, pyelocalyceal cysts, and conditions such as hypertrophic infundibular stenosis and megacalycosis. These disorders often arise from congenital abnormalities, trauma, or obstructions and are characterized by distinct imaging features and clinical presentations.

CYST-RELATED DYSPLASIAS AND NEOPLASMS

Cyst-related dysplasias and neoplasms include congenital anomalies and cystic tumors with distinct characteristics. Dysplastic conditions, such as cystic dysplasia of the kidney and multicystic dysplastic kidney (MCDK), arise from developmental defects, while neoplasms such as cystic renal cell carcinoma (RCC), multilocular cystic nephroma (MCN), and mixed epithelial and stromal tumors include both benign and malignant entities. Newer conditions, such as multiple unilateral subcapsular cortical hemorrhagic (MUCH) cysts and cystic partially differentiated nephroblastoma (CPDN), add to the spectrum of cystic kidney disorders. Table 45.12D and 45.12E summarizes these conditions, highlighting their etiology, clinical features, and management approaches. Imaging representation of RCC can be seen in Fig. 45.23. Kidney cancers are discussed further in Chapter 43.

Table 45.12 Comprehensive Overview of Diverse Kidney-Related Diseases: Etiology, Phenotypes, and Clinical Management

Disease or Syndrome	Etiology	Prevalence	Age at Presentation	Kidney Phenotype	Extrarenal Manifestation	Prognosis/Management
A. Rare Diseases and Syndromes Associated with Kidney Cysts						
Hajdu-Cheney syndrome	AD, *NOTCH2* gene variant. Can occur sporadically[436]	1/1,000,000[437]	Infancy-Childhood. Age-dependent progression[437]	Bilateral polycystic kidneys	Skeletal (Craniofacial and dental abnormalities, bone deformities, and acroosteolysis), neurologic complications, congenital heart defects, mitral and aortic valve abnormalities, respiratory infections, hearing loss, deep voice[436]	Morbidity and mortality increased due to bony fractures. Nonfavorable prognosis due to neurologic impairment (basilar invagination can occur in 50% of patients leading to central respiratory arrest)[438]
Alagille syndrome	AD, JAG1, NOTCH2 variant	1/30,000-70,000 live births[439] (increased to 1/30,000 with molecular diagnosis)[440]	Early childhood	Renal dysplasia, renal tubular acidosis, vesicular ureteric reflux, urinary obstruction,[441] small echogenic kidneys and kidney cysts,[442,443] glomerular mesangiolipidosis and proteinuria[444]	Liver (cholestasis, bile duct paucity, pruritus, xanthomas), cardiovascular (stenosis or hypoplasia of the branch pulmonary arteries, tetralogy of Fallot), vascular (intracranial bleeds, systemic vascular anomalies of the aorta), and skeletal (butterfly vertebrae, increased risk for fracture), and ocular abnormalities[439].	Significant morbidity and mortality due to the prevalence of cardiovascular abnormalities (6-year survival decrease to 40% compared with 5% of patients without intracardiac disease [445], 34% mortality due to noncardiac vascular anomalies) [446].

Table 45.12 Comprehensive Overview of Diverse Kidney-Related Diseases: Etiology, Phenotypes, and Clinical Management (Continued)

Disease or Syndrome	Etiology	Prevalence	Age at Presentation	Kidney Phenotype	Extrarenal Manifestation	Prognosis/Management
Hyperinsulinemic Hypoglycemia and Polycystic Kidney Disease (HIPKD)	AR, PMM2	1/50,000 for hyperinsulinemic hypoglycemia [447], 1/1,000,000 for HIPKD [448].	Infancy, childhood	Enlarged kidneys with multiple bilateral cysts	Hypoglycemia, cerebellar dysfunction, developmental delays, hepatic cysts, and abnormal fat distribution[449].	Variable disease severity, mainly based on case reports and genetic studies [450].
Disease or Syndrome	**Etiology**	**Prevalence**	**Age at Presentation**	**Kidney Phenotype**	**Extrarenal Manifestation**	**Prognosis/Management**
B. Urinary Stone Diseases						
HOGA1 variants and kidney cysts	HOGA1[451]	AR (AD)	Infancy-Adulthood	Stone formation, cystic burden, primary hyperoxaluria type 3 (PH3), higher urinary calcium excretion compared with PH1 and PH2 [452]		Variable disease severity, mainly based on case reports/series and genetic studies
Hypophosphatemic rickets with hypercalciuria	SLC34A3[453]			Stone formation, cystic burden	Rarely hepatic cysts, hypophosphatemia, along with hypercalciuria, osteoporosis, rickets, elevated 1,25-dihydroxyvitamin D levels	
CYP24A1 deficiency and kidney cysts	CYP24A1[454]			Stone formation, cystic burden, hypercalciuria	Rarely hepatic cysts, reduced 24-hydroxylase activity, increased serum 1,25-dihydroxycholecalciferol levels along with and without hypercalcemia	
Disease or Syndrome	**Etiology**	**Prevalence**	**Age at Presentation**	**Kidney Phenotype**	**Extrarenal Manifestation**	**Prognosis/Management**
C. Renal Cystic Diseases of Nontubular Origin						
Cystic diseases of the renal sinus[455-457]	Sporadic	Up to 1.5% of autopsy cases	Variable	Parapelvic or peripelvic cysts, kidney stones and mimic hydronephrosis, UTIs	Hypertension, abdominal pain, or prostatism	Unknown. Can be managed by retrograde intrarenal holmium-laser incision[458]

Continued on following page

Table 45.12 Comprehensive Overview of Diverse Kidney-Related Diseases: Etiology, Phenotypes, and Clinical Management (Continued)

Disease or Syndrome	Etiology	Prevalence	Age at Presentation	Kidney Phenotype	Extrarenal Manifestation	Prognosis/Management
Perirenal lymphangiomas (peripylic-pericalyceal lymphangiectasis)	Congenital disease (secondary to obstruction of regional lymph drainage during development) Identification of rare familial cases suggests a genetic inheritance pattern[459,460]	Difficult to estimate	Children	Hematuria, flank pain, abdominal pain, deterioration of kidney function over time if large enough to compress the kidney tissue[461,462] Nephromegaly, rarely perirenal location of the benign lymphatic tumor[463]	Commonly in the cervical neck and axilla, but may also be present in the retroperitoneum, mediastinum, mesentery, omentum, colon, and pelvis[461,462]	Incidental finding. For most asymptomatic patients, treatment may not be necessary. Less invasive interventions, such as percutaneous or laparoscopic aspiration and cyst marsupialization, should be considered for regional lymphangiomas. Nephrectomy may be needed in decompensated cases[459,463]
Subcapsular and perirenal urinomas (uriniferous pseudocysts)	Sporadic, develop secondary to obstruction, pregnancy, or iatrogenic		Variable	Encapsulated collections of extravasated urine in the kidney parenchyma, mainly the subcapsular or perirenal space (not true cyst) [464]. Possible hypertension, urinary peritonitis, renal atrophy, and kidney failure [465].		Can be complicated by abscesses, hydronephrosis, electrolyte imbalances, and rupture[465,466]. Worse prognosis with upper urinary tract obstruction. Possible spontaneous resolution but uncommon [467].

Table 45.12 Comprehensive Overview of Diverse Kidney-Related Diseases: Etiology, Phenotypes, and Clinical Management (Continued)

Disease or Syndrome	Etiology	Prevalence	Age at Presentation	Kidney Phenotype	Extrarenal Manifestation	Prognosis/Management
Pyelocalyceal Cysts (Pyelocalyceal Diverticula or Pyelogenic Cysts)		All age groups are affected with an incidence of 0.3% among adults [468,469].		Unilateral cystic enlargement of kidney pelvis or calyces. Urinary stasis and can be complicated by stones (10-40% of patients), infections, and less commonly transitional cell carcinoma[470,471].		Mostly symptomatic management. Surgery is considered only if conservative management of complications proves ineffective[468].
Renal hypertrophic infundibular stenosis and Megacalycosis				Obstruction to urine flow distal to the calyx, often linked to obstructions at the calyceal infundibulum or kidney pelvis[472]. Megacalycosis indicates a congenital increased volume and number of calyces without obstruction or abnormal renal function[473,474].		

Disease or Syndrome	Etiology	Prevalence	Age at Presentation	Kidney Phenotype	Extrarenal Manifestation	Prognosis/Management
D. Kidney Cyst–Related Dysplastic Diseases						

Continued on following page

Table 45.12 Comprehensive Overview of Diverse Kidney-Related Diseases: Etiology, Phenotypes, and Clinical Management (Continued)

Disease or Syndrome	Etiology	Prevalence	Age at Presentation	Kidney Phenotype	Extrarenal Manifestation	Prognosis/Management
Cystic dysplasia of the kidney	Intrinsic factors: early in utero, at the level of ureteric bud-mesenchyme interaction if it happens in a noncentral position, inducing a more cranial point of differentiation. Extrinsic factors: obstruction, when most nephrons are formed, leading to dilations and irregularities in branching[475]		in utero, neonatal, childhood	Cystic kidney dysplasia involves at least 1 kidney cyst, abnormal tissue development, primitive ducts, and presence of nonrenal tissues. Unlike most cystic kidney diseases, which feature normal or enlarged kidneys, cystic kidney dysplasia results usually in small kidneys[476]	It can occur as an isolated condition or alongside other urologic or genetic disorders such as branchio-oto-renal syndrome (caused by EYA1 variants) and renal-coloboma syndrome (caused by PAX2 variants, among others VACTERL, CHARGE)[361]	Depending on the severity, can lead to chronic kidney disease, kidney failure[361]
Multicystic dysplastic kidney (MCDK)	Familial (*PAX2*, *CCSAP*, *DSTYK*, *REN*, *AGT*, *TCF2/ HNF1β*) or sporadic[477-479]	1 per 1000 to 4000 live births Bilateral MCDK Is less prevalent with a reported incidence of 1 in 10,000.[480,481].	In utero–neonatal, childhood	Considered the severe form of kidney dysplasia with cysts Multiple kidney cysts of varying sizes, nonfunctional kidney tissue, and an abnormal urinary tract system. Can be unilateral (functionally solitary contralateral kidney, compensatory hypertrophy) or bilateral. Other anomalies included UPJ and UVJ obstruction. Horseshoe kidney and ureterocele are less common. In bilateral cases, MCDK may lead to Potter sequence, oligohydramnios, and be incompatible with life[482-484].	Cardiovascular disease, central nervous system, and gastrointestinal anomalies; omphalocele; and vertebral and skeletal malformations, and cryptorchidism[485,486].	Unilateral isolated MCDK with a normally functioning contralateral kidneys have an excellent prognosis. No hypertension, malignancy, or significant proteinuria is detected in the long follow-up [487]. Conservative management is appropriate for cases with a tendency to involute in the first 10-15 years of life with no increased risk for malignancy when compared to the general population[410,488]. Bilateral disease is incompatible with life[485].

Table 45.12 Comprehensive Overview of Diverse Kidney-Related Diseases: Etiology, Phenotypes, and Clinical Management (Continued)

Disease or Syndrome	Etiology	Prevalence	Age at Presentation	Kidney Phenotype	Extrarenal Manifestation	Prognosis/Management
Multiple unilateral subcapsular cortical hemorrhagic cysts of the kidney (MUCH)[489]	Unknown (potential genetic or developmental anomaly)	Unknown	Young	Predominantly on the left side. Flank pain or hematuria, with kidney cysts[489].		Prognosis is not well established given its rarity and limited research. However, several clinical and imaging markers can affect disease progression and prognosis. Hypertension or kidney function decline may develop over time, potentially impacting long-term prognosis. Current management is symptomatic[489].
Disease or Syndrome	**Etiology**	**Prevalence**	**Age at Presentation**	**Kidney Phenotype**	**Extrarenal Manifestation**	**Prognosis/Management**
E. Kidney Cyst–Related Neoplastic Diseases						

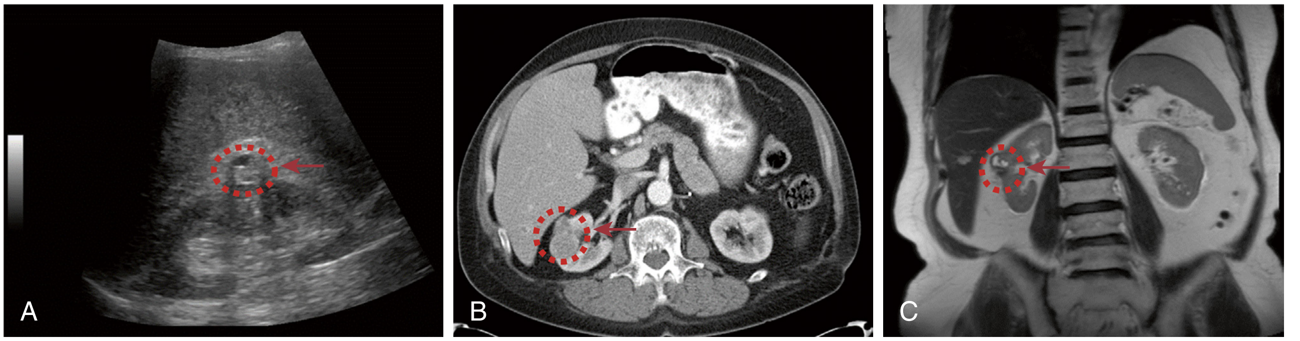

Fig. 45.23 Imaging representation of renal cell carcinoma in a 75-year-old male patient (eGFR = 49 mL/min/1.73 m^2) on ultrasound (A), transverse computed tomography (B), and coronal magnetic resonance imaging (C).

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